

Smoke Bioaerosol Concentration and Emission Mixed Models

[1] Study Overview

[1.1] Site Description and Sampling Design

This study examined airborne microbes (fungi and bacteria) in ambient air and wildland fire smoke across two distinct ecosystems:

- **Utah subalpine** (FASMEE23): Subalpine forest fire regime
- **Kansas tallgrass prairie** (Konza4): Grassland fire regime

Samples were collected using drone-mounted equipment before fire ignition (ambient conditions) and during active burning (smoke conditions). Air and smoke samples were collected using filters and compensating pumps that drew specific volumes of air onto the sample filters.

[1.2] Sample Collection Protocol

Filter handling: Before and after each sample collection, filters were placed and removed from sampling devices in situ using a mobile laboratory box to minimize air contamination and external influence on sample filters.

Quality control: Filter blanks were collected in situ to quantify contamination during filter handling and processing.

Concurrent measurements: Each drone flight simultaneously collected meteorological data, particulate matter (PM) concentrations, and carbon sensor data. *Note: Carbon sensor malfunctions resulted in incomplete carbon data for some samples.*

[1.3] Microbial Enumeration Methods

Microbes were enumerated using epifluorescence microscopy after fluorescent staining:

Counting protocol: - Magnification: 1000x - Fields of view (FOV): 40 random locations per filter - Filter area: $\sim 226.98 \text{ mm}^2$ - FOV area at 1000x: $\sim 0.031 \text{ mm}^2$ - Total possible FOVs per filter: 7,225

Staining approaches: - **Bacteria:** Two-part viability stain allowing quantification of total bacteria, live bacteria, and dead bacteria - **Fungi:** Single stain for spore enumeration (viability not distinguishable)

Laboratory blanks: Lab blanks were processed with each batch of staining solutions to account for contamination introduced during the staining process.

[2] Research Questions

1. **Microbial abundance ambient vs. smoke intensity:** How do microbial concentrations in ambient air differ from concentrations in smoke of varying intensity?
 - Smoke intensity classified using PM2.5 concentrations from onboard sensors
2. **Environmental drivers:** Do smoke-associated microbes correlate with meteorological variables or carbon concentrations?
3. **Bacterial viability:** Does the ratio of live to total bacteria differ between ambient and smoke conditions?
4. **Site comparison:** Are there differences in microbial abundances between study sites under both ambient and smoke conditions?
5. **Emission factors:** Using the carbon data, we can transform our microbe counts into the number of microbes emitted by a certain mass of fuel burning (e.g. #cells emitted per kilogram of biomass burning)—Known as emission factors (EFs)—Do EFs correlate with combustion efficiency (smoldering vs. flaming combustion)?

[3] Statistical Analysis Approach

[3.1] Generalized Linear Mixed Models (GLMMs)

Response variable: FOV count data (Research Questions 1-4) **Offset:** $\log(\text{volume sampled})$ to account for varying sample volumes

Random effects structure: - Random intercept for SampleID (accounts for multiple FOVs from the same sample) - Random intercept for Lab Blank batch (accounts for batch-specific contamination levels during sample processing)

Note: Research Question 5 (emission factors) requires a modified modeling approach incorporating: - Averaging FOV counts across the 40 FOVs, scaling to the total filter area, and then normalizing to the air volume sampled and transforming based on carbon data.

[4] Konza4 spore data summary

=== Konza4 Spore Data Summary ===

Dataset dimensions: 480 rows × 15 columns

Sample counts by smoke level:

None	Low	High
80	120	280

Summary statistics for key variables:

Variable	n	Mean	SD	Min	Q25	Median	Q75	Max
TotalSpores	480	0.45	0.98	0.00	0.00	0.00	1.00	9.00
TotalSpores_corrected.m3	480	83000.96	178468.60	0.00	0.00	0.00	141799.07	1582297.51
RepVolume_L	480	32.21	7.96	20.06	25.66	35.79	40.02	40.12
MedianPM2.5_ug.m3	480	583.82	453.49	3.66	215.29	540.62	951.57	1428.85
MeanTemp_C	480	23.74	3.56	19.48	20.86	22.60	26.01	32.44
MeanRH	480	23.92	4.93	16.45	20.46	23.07	28.03	32.63
MeanMR	480	4.45	0.39	3.82	4.19	4.31	4.74	5.17
ALL_carbon_mg.m3	320	77.25	37.95	7.75	61.33	92.83	105.45	114.49
MeanMCE	320	0.95	0.01	0.93	0.95	0.96	0.96	0.97

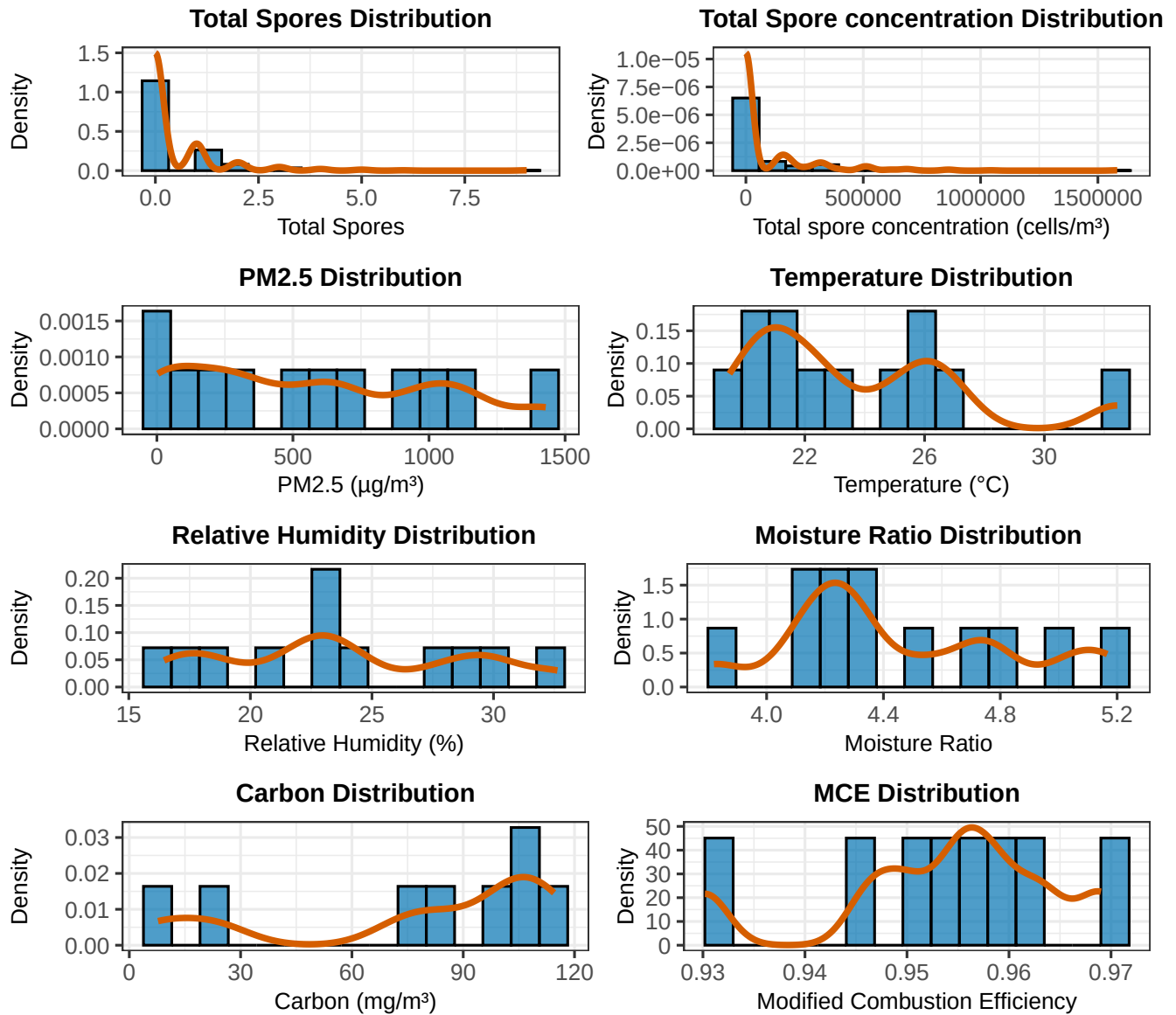
Additional metadata:

- Number of unique SampleIDs: 12
- Smoke intensity: High, None, Low

- Number of unique Days: 3
- Number of LB_Batch groups: 3
- Sample types: Smoke, Ambient

=== Data Distribution Plots ===

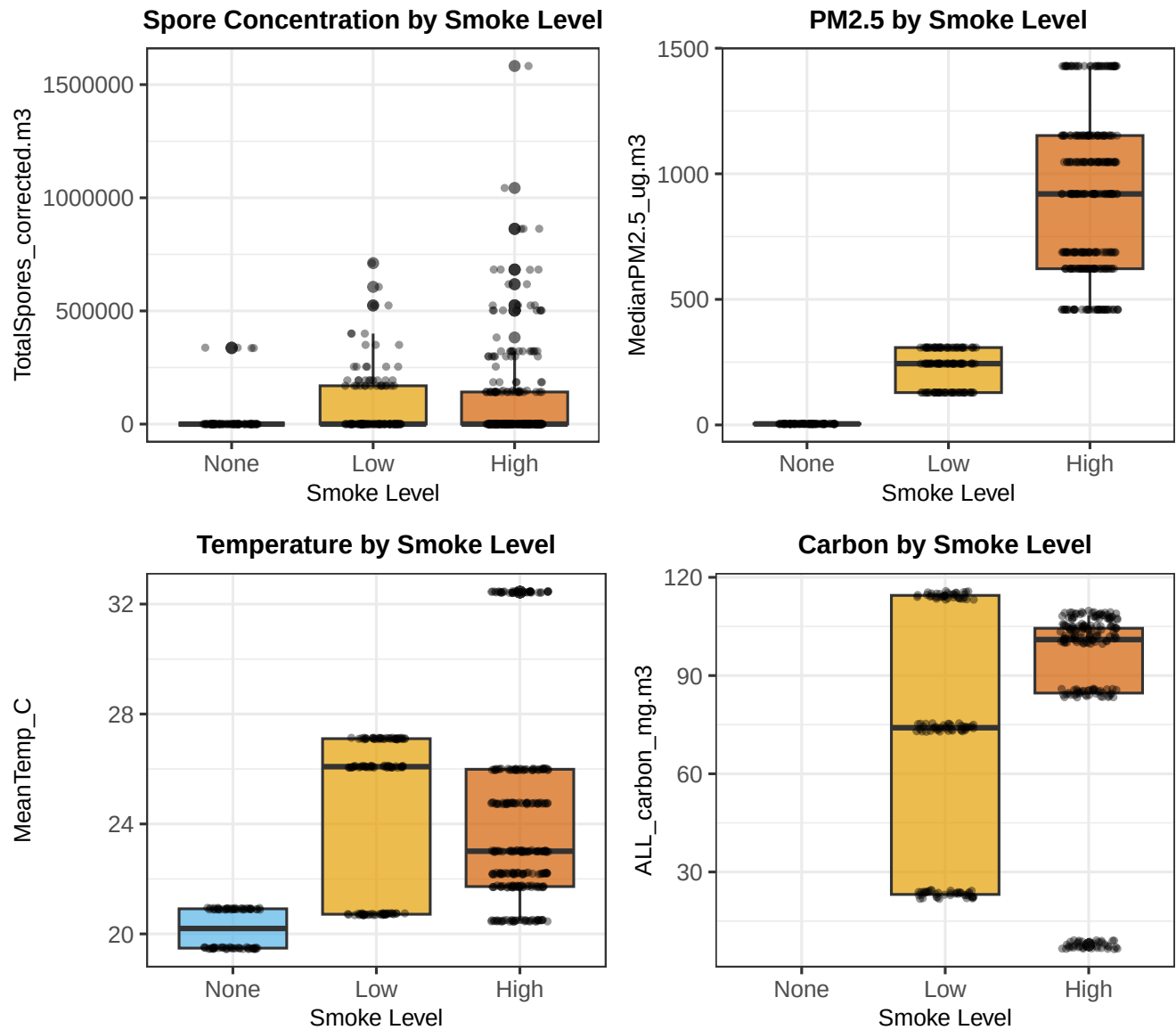
Konza4 Data Distributions



=== Additional Data Visualizations ===

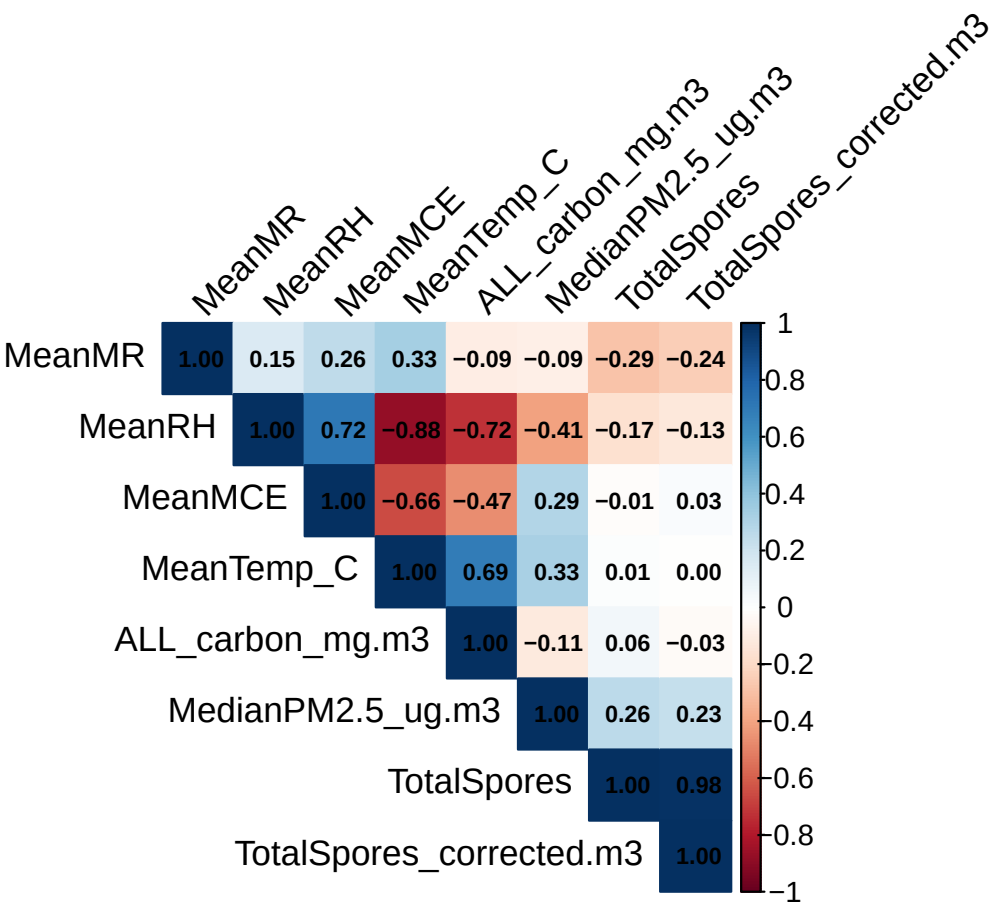
1. Boxplots by Smoke Level:

Variable Distributions by Smoke Level



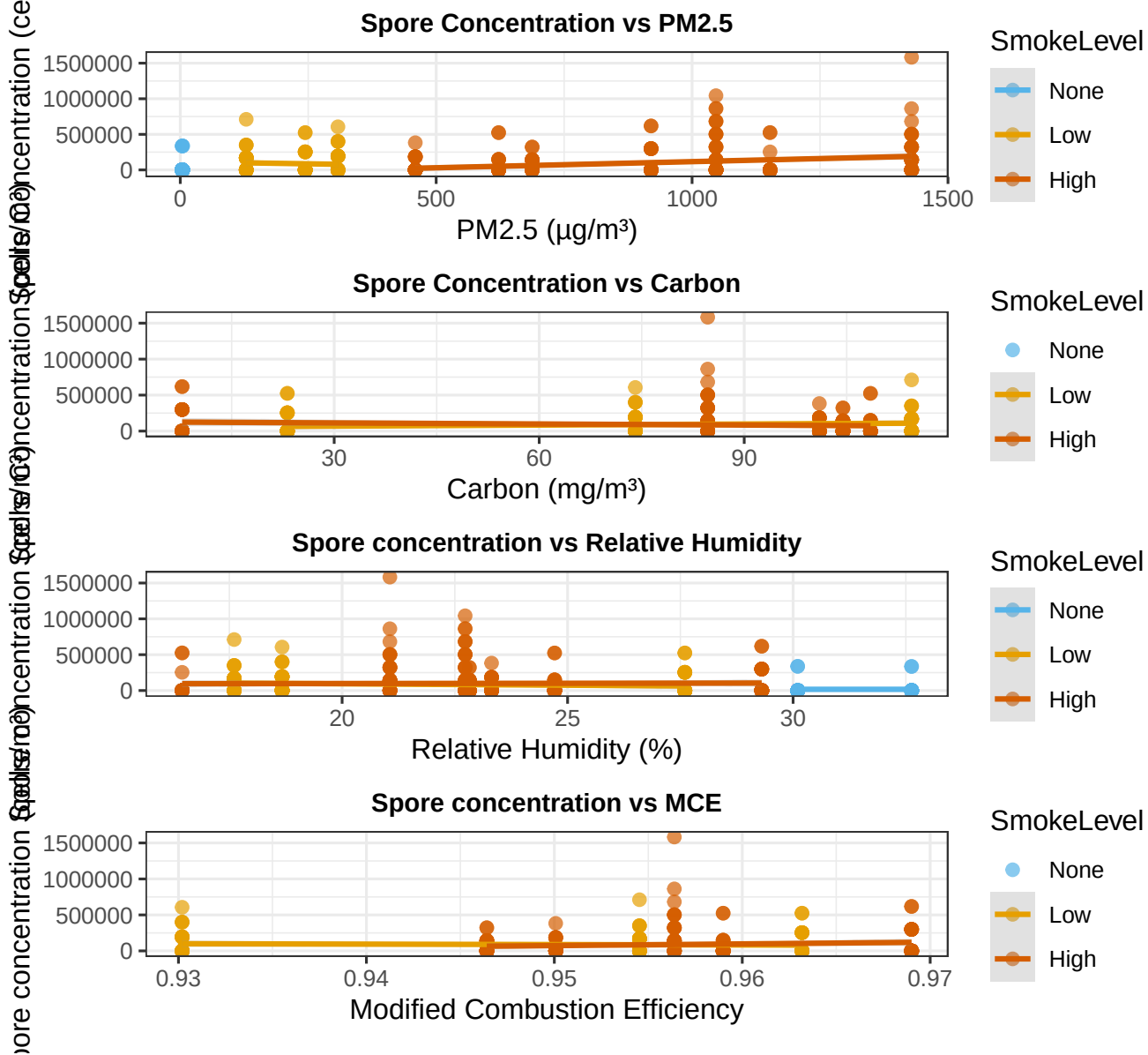
2. Correlation Matrix:

Correlation Matrix of Key Variables



3. Key Variable Relationships:

Key Variable Relationships



5. Summary Statistics by Smoke Level:

A tibble: 3 x 7

	SmokeLevel	n_samples	mean_spores	sd_spores	mean_pm25	mean_temp	mean_carbon
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	None	80	16839	73863.	4.1	20.2	NaN
2	Low	120	88444.	145599.	227	24.6	70.6
3	High	280	99571.	206284.	902.	24.4	81.3

[5] Modeling for Research Question #1: Konza4 spore abundance in ambient vs. smoke conditions

[5.1] GLMM model for Konza4 spores and DHARMA diagnostics for assesment

```
Konza4_SporeCatModel <- glmmTMB(
  TotalSpores ~ SmokeLevel + offset(log(RepVolume_L)) +
    (1|SampleID) + (1|LB_Batch),
  family = nbinom2(link = "log"),
  data = Konza4_spores_pa_C,
  ziformula = ~1)
summary(Konza4_SporeCatModel)
```

```
Family: nbinom2 ( log )
Formula:      TotalSpores ~ SmokeLevel + offset(log(RepVolume_L)) + (1 | SampleID) +
  (1 | LB_Batch)
Zero inflation:      ~1
Data: Konza4_spores_pa_C
```

AIC	BIC	logLik	-2*log(L)	df.resid
781.6	810.8	-383.8	767.6	473

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID	(Intercept)	3.260e-01	0.5709411
LB_Batch	(Intercept)	1.072e-08	0.0001035

Number of obs: 480, groups: SampleID, 12; LB_Batch, 3

Dispersion parameter for nbinom2 family (): 1.59

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.9328	0.7149	-8.299	<2e-16	***
SmokeLevelLow	1.7002	0.7568	2.247	0.0247	*
SmokeLevelHigh	1.6641	0.7036	2.365	0.0180	*

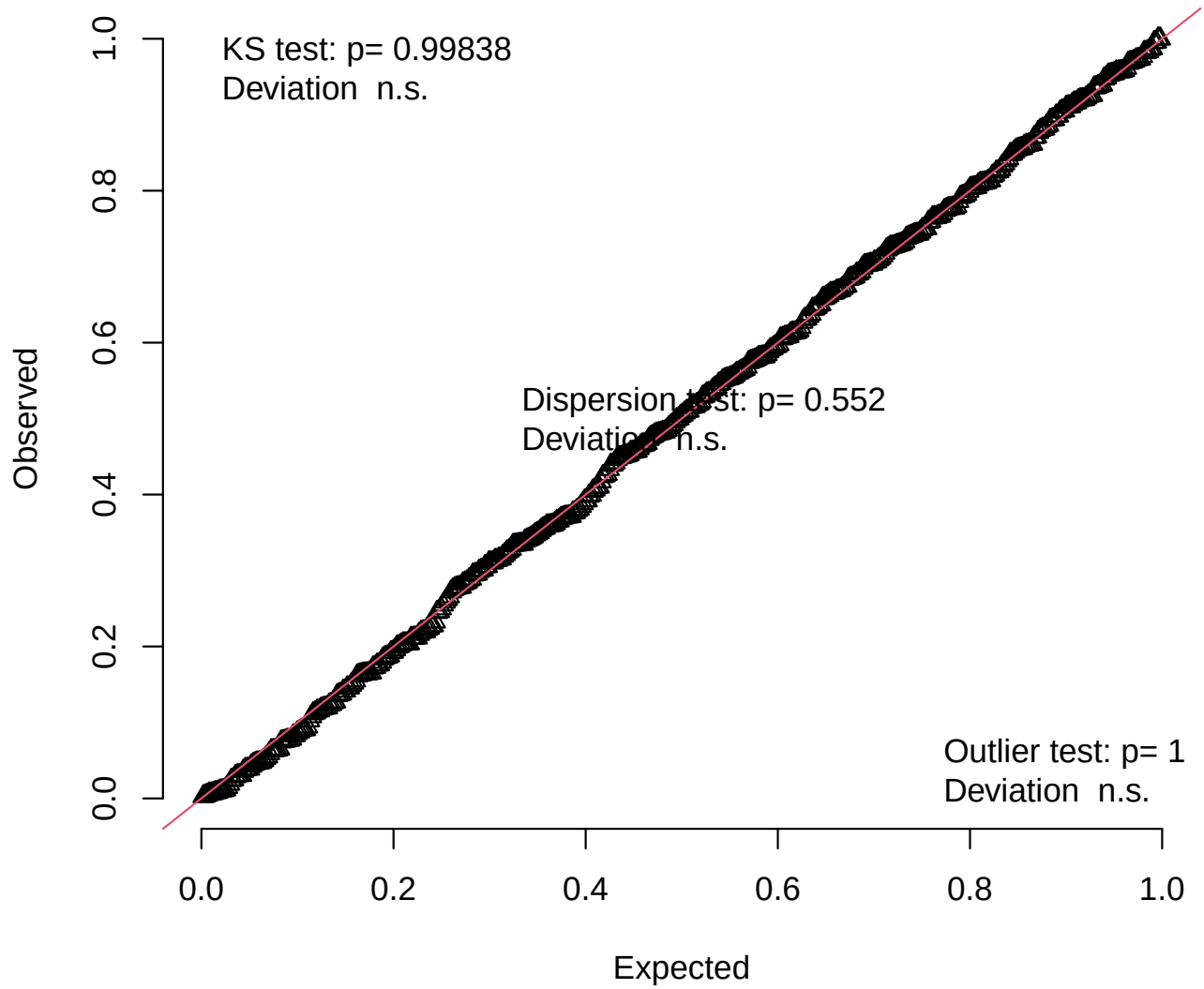
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

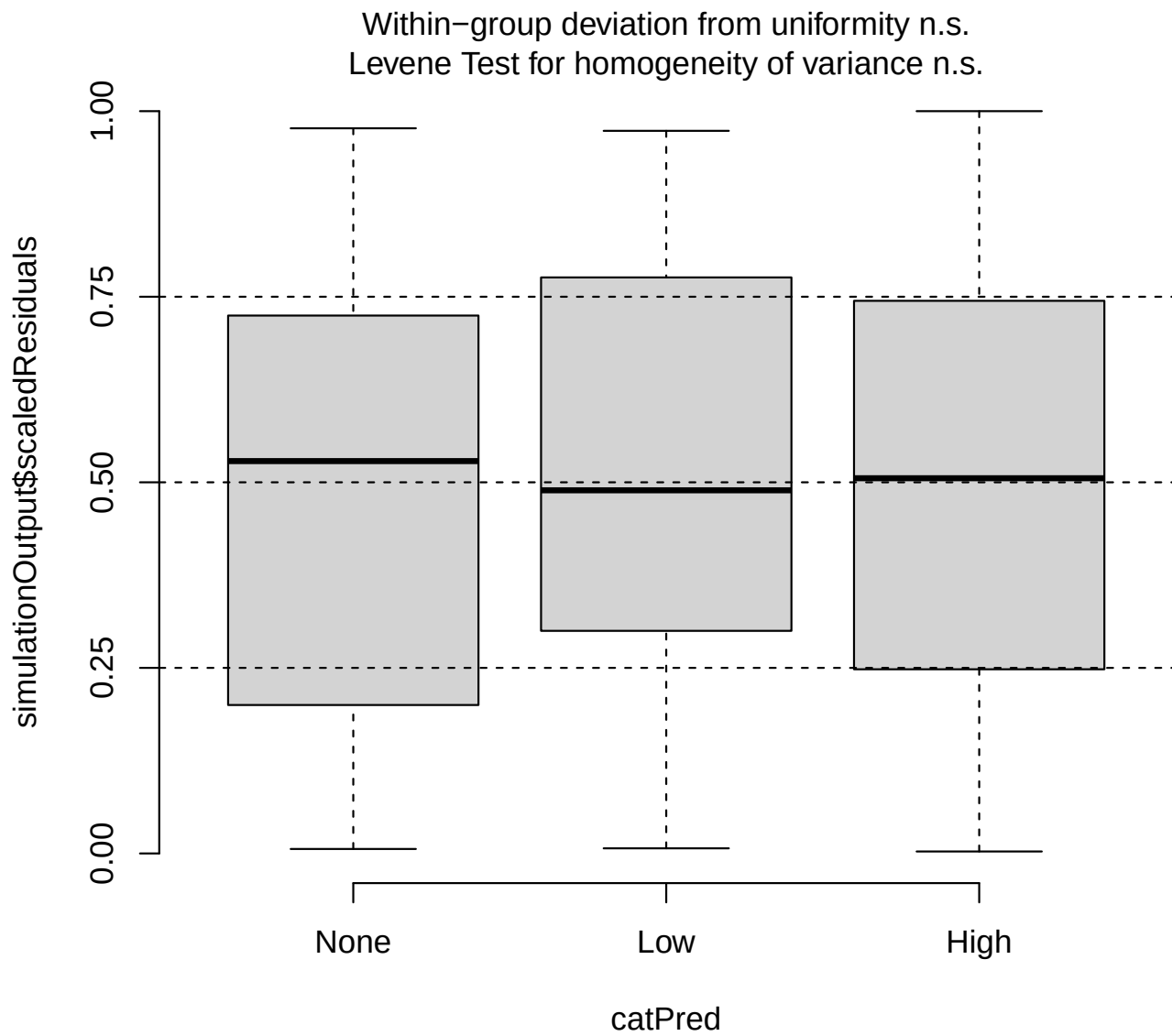
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.772	1.901	-0.932	0.351

```
res <- simulateResiduals(fittedModel = Konza4_SporeCatModel, plot = FALSE)
plotQQunif(res)
```

QQ plot residuals

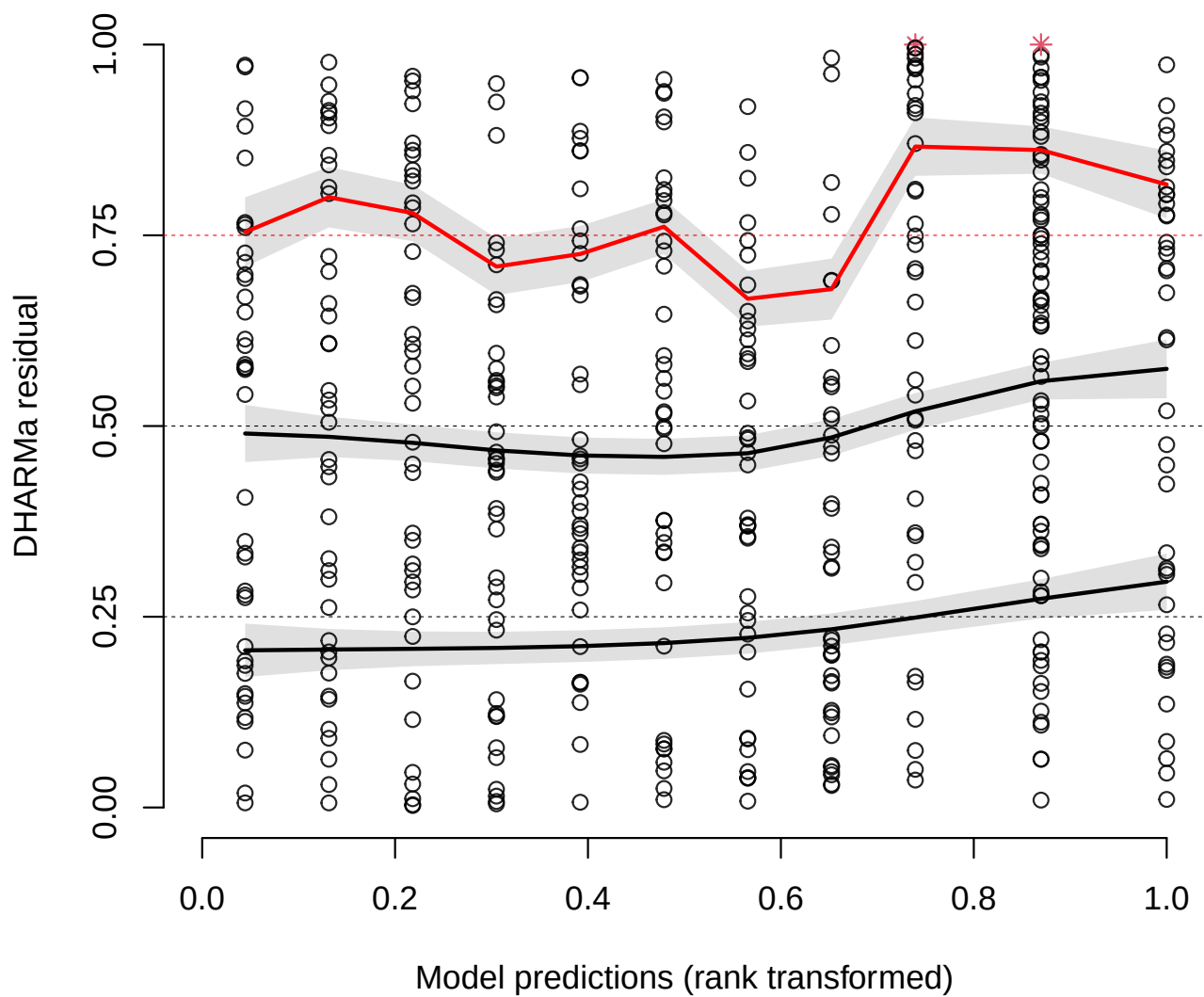


```
plotResiduals(res, Konza4_spores_pa_C$SmokeLevel) # Residuals by group
```



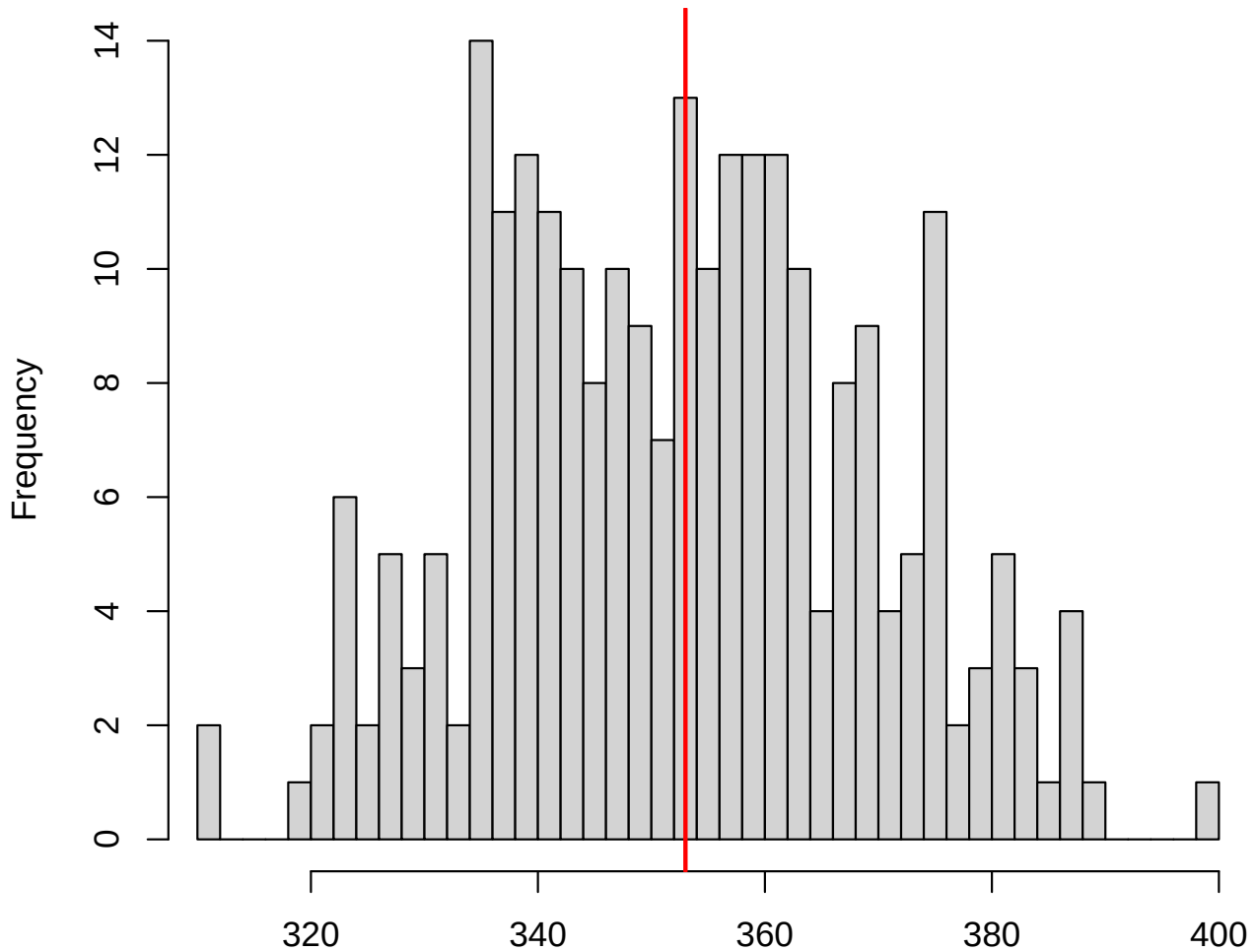
```
plotResiduals(res)
```

DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



```
testZeroInflation(res)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

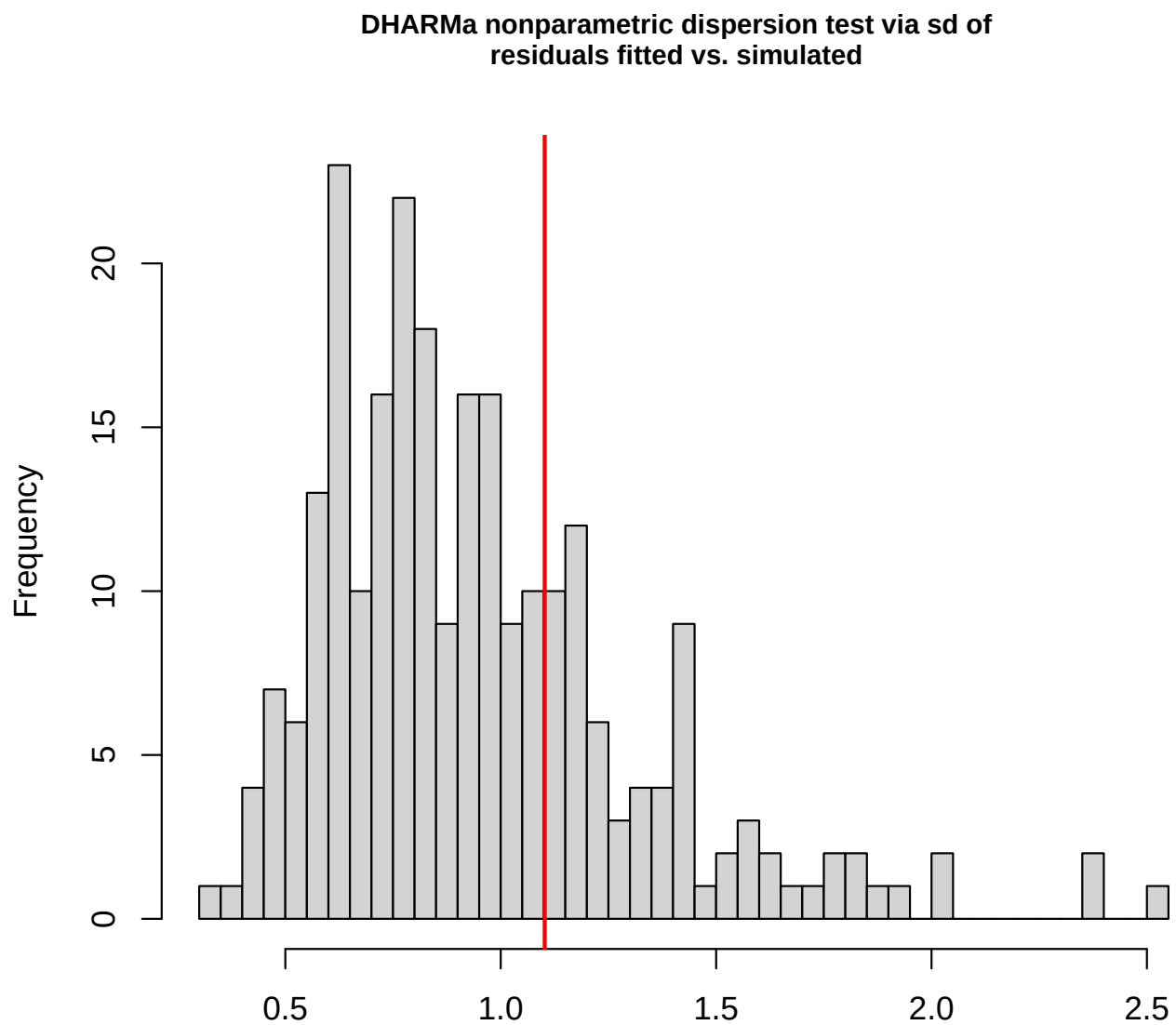


Simulated values, red line = fitted model. p-value (two.sided) = 1

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.99971, p-value = 1  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```



Simulated values, red line = fitted model. p-value (two.sided) = 0.552

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.1589, p-value = 0.552
alternative hypothesis: two.sided
```

```
# Calculate marginal means and contrasts
emm_log <- emmeans(Konza4_SporeCatModel, ~ SmokeLevel)
emm_response <- emmeans(Konza4_SporeCatModel, ~ SmokeLevel, type = "response")
rate_ratios_vs_none <- contrast(emm_log, method = "trt.vs.ctrl", ref = "None", type = "response")

# Create prediction dataframe
spore_cat <- as.data.frame(emm_response) %>%
  mutate(
    Spores.m3 = (response*7225)/0.03,
    CI_lower = (asyp.LCL * 7225) / 0.03,
    CI_upper = (asyp.UCL * 7225) / 0.03
  )
```

[5.2] Predictions from Konza4 spore GLMM model

	SmokeLevel	response	SE	df	asyp.LCL	asyp.UCL	Spores.m3	CI_lower	CI_upper
1	None	0.08539693	0.06104833	Inf	0.02103476	0.3466945	20566.43	5065.872	83495.59
2	Low	0.46755808	0.20990900	Inf	0.19395046	1.1271464	112603.57	46709.736	271454.42
3	High	0.45096606	0.16934951	Inf	0.21602076	0.9414391	108607.66	52025.000	226729.91

[5.3] Ratios from Konza4 spore GLMM model

contrast	ratio	SE	df	null	z.ratio	p.value
Low / None	5.48	4.14	Inf	1	2.247	0.0470
High / None	5.28	3.72	Inf	1	2.365	0.0346

P value adjustment: dunnettx method for 2 tests
Tests are performed on the log scale

[5.4] Plotting Konza4 spore model results with raw data as points/violin plots

```
# Create significance bars
contrast_df <- as.data.frame(rate_ratios_vs_none)
sig_bars <- data.frame(
  x = c(1, 1),
  xend = c(2, 3),
  pair = c("Low / None", "High / None"),
  stringsAsFactors = FALSE
)

sig_bars$p_value <- sapply(sig_bars$pair, function(pair) {
  idx <- which(contrast_df$contrast == pair)
  if(length(idx) > 0) {
    return(contrast_df$p.value[idx])
  } else {
    return(1.0)
  }
})

max_height <- max(spore_cat$CI_upper, na.rm = TRUE)
sig_bars$y <- max_height * c(2.7, 2.9)
sig_bars$annotation <- ifelse(
  is.na(sig_bars$p_value),
  'plain("")',
  paste0("italic(P)[dunnettx-adj]==", signif(sig_bars$p_value, 2))
)

p1 <- ggplot(spore_cat, aes(x = SmokeLevel, y = Spores.m3)) +
  geom_point(data = Konza4_spores_pa_C,
    aes(x = SmokeLevel, y = TotalSpores_corrected.m3),
    position = position_jitter(width = 0.1),
    alpha = 0.5, size = 2, color = "#0072B2") +
  geom_violin(data = Konza4_spores_pa_C,
    aes(x = SmokeLevel, y = TotalSpores_corrected.m3),
    fill = NA, color = "black", width = 0.8, alpha = 0.5) +
  geom_col(fill = "#0072B2", alpha = 0.6) +
  geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper),
    width = 0.2, color = "black", linewidth = 1.1) +
  labs(x = expression(PM[2.5]~"range"),
```

```

y = expression(Spores~m^{-3}),
title = expression(
  paste("Smoke vs. Ambient Tallgrass Prairie Fungal Spores by ",
    PM[2.5], " Range")) +
theme_bw() +
theme(text = element_text(size = 10),
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.text.x = element_text(angle = 0, hjust = 0.5)) +
geom_segment(data = sig_bars,
  aes(x = x, xend = xend, y = y, yend = y),
  linewidth = 0.5) +
geom_segment(data = sig_bars,
  aes(x = x, xend = x, y = y, yend = y - max_height * 0.05),
  linewidth = 0.5) +
geom_segment(data = sig_bars,
  aes(x = xend, xend = xend, y = y, yend = y - max_height * 0.05),
  linewidth = 0.5) +
geom_text(data = sig_bars,
  aes(x = (x + xend) / 2, y = y + max_height * 0.02, label = annotation),
  size = 3.2, vjust = 0, parse = TRUE) +
scale_x_discrete(labels = c("None\n(<20)", "Low\n(20-400)", "High\n(>400)")) +
coord_cartesian(ylim = c(0, max(Konza4_spores_pa_C$TotalSpores_corrected.m3 * 0.5)))

print(p1)

```

Smoke vs. Ambient Tallgrass Prairie Fungal Spores by PM_{2.5} Range

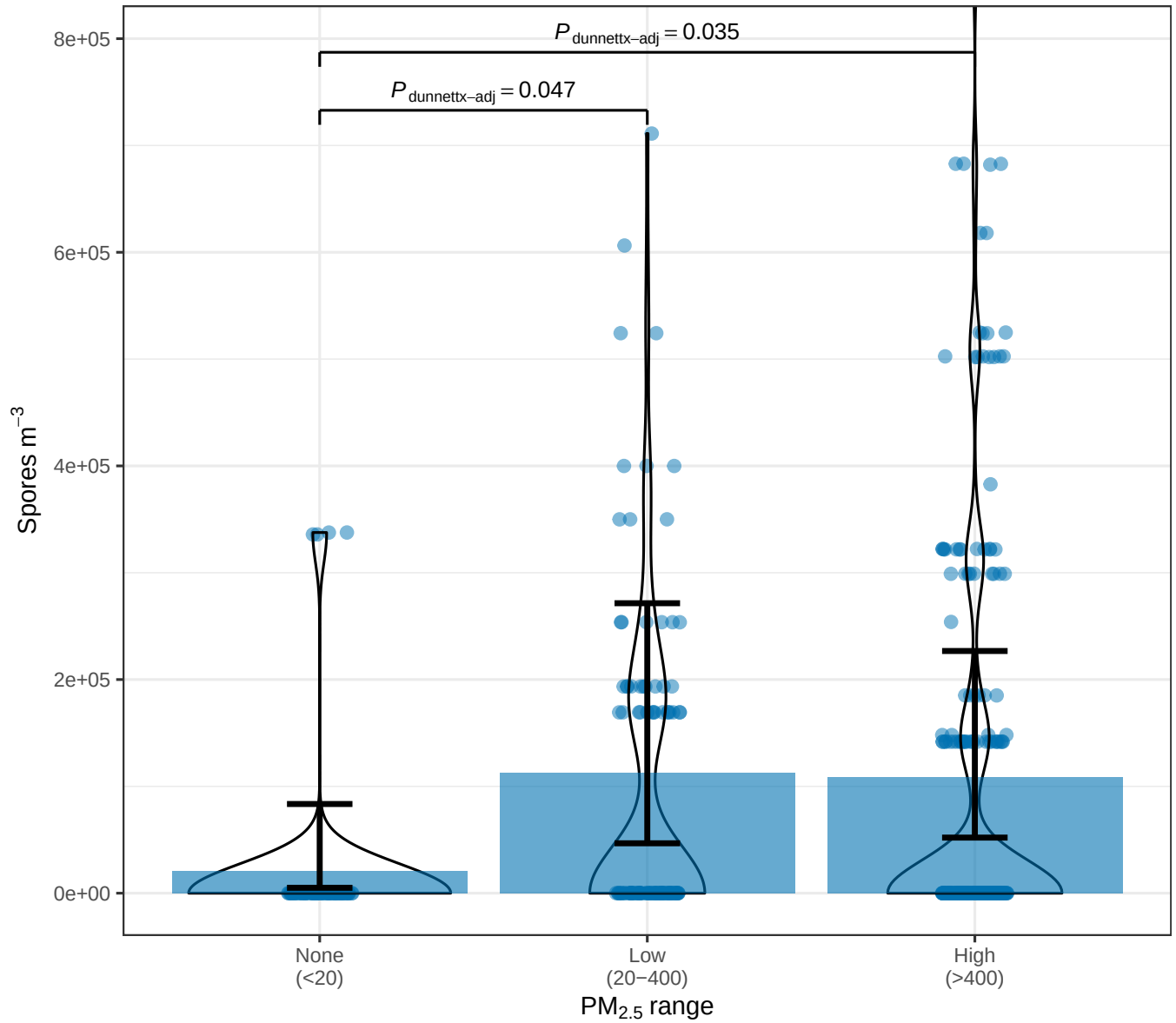


Figure 1: Konza 4 Spore GLMM predictions and raw data by smoke level with significance bars

[5.5] Konza4 Spore Model Analysis

Model Diagnostics

- **Overall fit:** Good (KS test non-significant)
- **Dispersion:** Appropriate (non-significant)
- **Zero-inflation:** Well-modeled (non-significant)
- **Outliers:** None detected (non-significant)
- **Group-level residuals:** Uniform across smoke levels (Levene test n.s.)
- **Possible issues:** DHARMA residuals vs. predicted values show quantile deviations with combined adjusted quantile test being significant

Model Results

- **Effect Sizes:**
 - Low smoke: $5.48\times$ increase vs. ambient ($p = 0.0470$)
 - High smoke: $5.28\times$ increase vs. ambient ($p = 0.0346$)
- **Predicted Spore Concentrations (Spores/m³):**
 - None: 19,780
 - Low: 105,774
 - High: 104,491
- **Random Effects:** SampleID variance = 0.296, LB_Batch variance = 0
- **Dispersion parameter:** 1.49 (constant)
- **Zero-inflation:** Simple intercept model

Interpretation: Konza4 shows a strong and consistent increase in spore concentrations under both low and high smoke conditions, with both smoke levels resulting in over fivefold increases compared to ambient air. The absolute concentrations in smoke are around 105,000 spores/m³, with a baseline of ~20,000 spores/m³ in ambient air. See Figure [Konza4 Spore GLMM predictions and raw data by smoke level] for visualization.

[6] FASMEE23 spore data summary

=== FASMEE23 Spore Data Summary ===

Dataset dimensions: 640 rows × 16 columns

Sample counts by smoke level:

None	Low	High
320	160	160

Summary statistics for key variables:

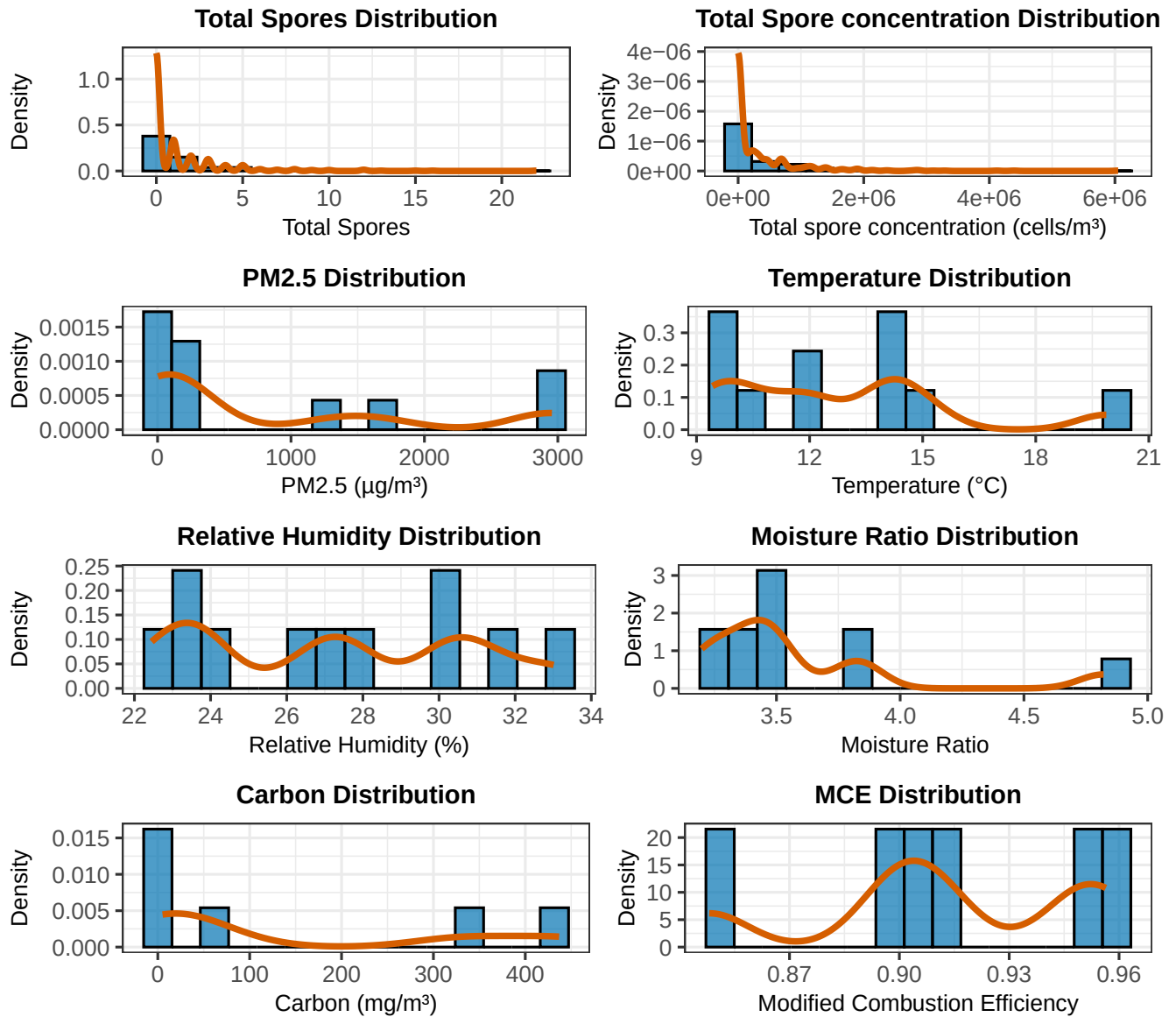
Variable	n	Mean	SD	Min	Q25	Median	Q75	Max
TotalSpores	640	1.2	2.4	0.0	0.0	0.0	1.0	22.0
TotalSpores_corrected.m3	640	305999.2	605578.9	0.0	0.0	0.0	322867.2	6051352.1
RepVolume_L	640	24.8	4.8	20.0	20.0	23.1	30.0	30.0
MedianPM2.5_ug.m3	440	866.1	1117.6	3.5	3.8	188.5	1670.5	2955.7
MeanTemp_C	440	12.8	3.0	9.4	10.1	12.1	14.3	19.9
MeanRH	440	27.3	3.5	22.5	23.7	27.5	30.3	33.0
MeanMR	440	3.6	0.4	3.2	3.3	3.5	3.8	4.8
ALL_carbon_mg.m3	240	143.1	174.6	5.1	9.3	37.8	331.6	436.9
MeanMCE	240	0.9	0.0	0.8	0.9	0.9	0.9	1.0

Additional metadata:

- Number of unique SampleIDs: 16
- Smoke intensity: High, Low, None
- Number of unique Days: 3
- Number of LB_Batch groups: 5
- Sample types: Smoke, Ambient
- Platforms: Blue, Red

=== Data Distribution Plots ===

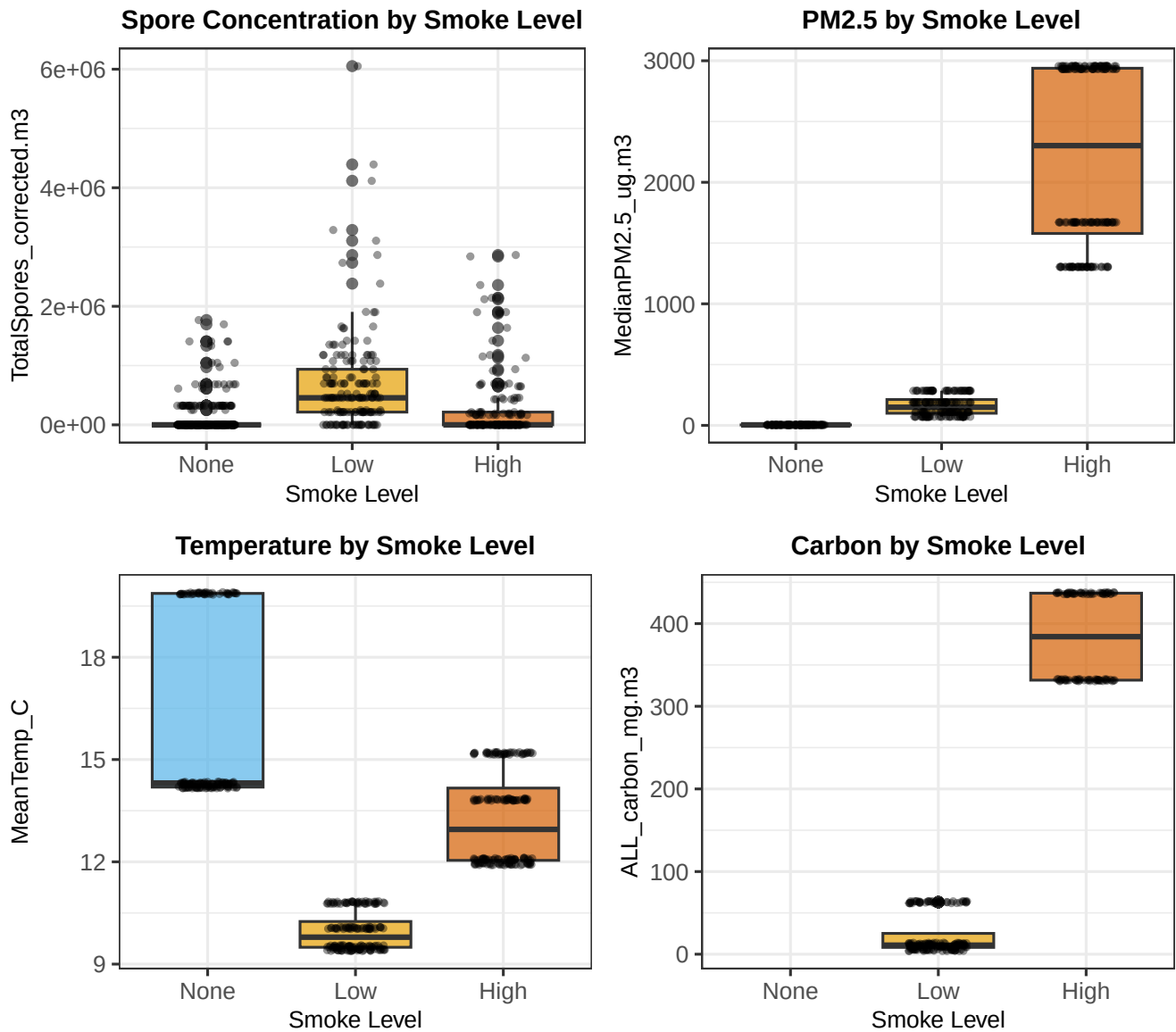
FASMEE23 Data Distributions



=== Additional Data Visualizations ===

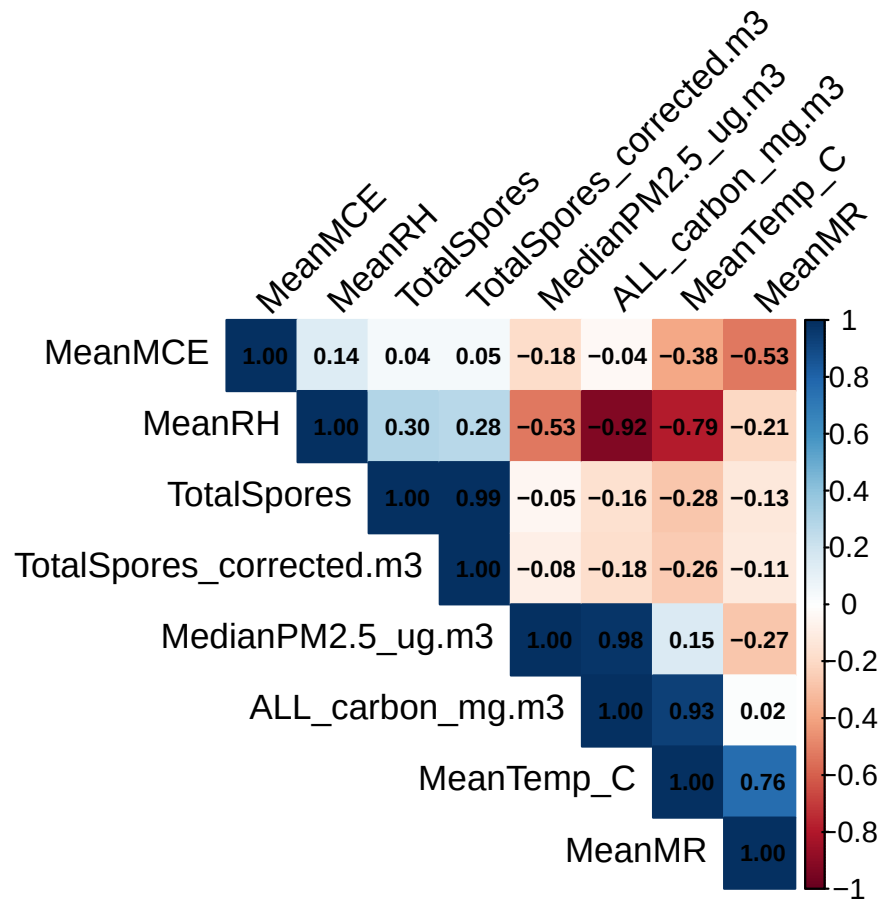
1. Boxplots by Smoke Level:

Variable Distributions by Smoke Level



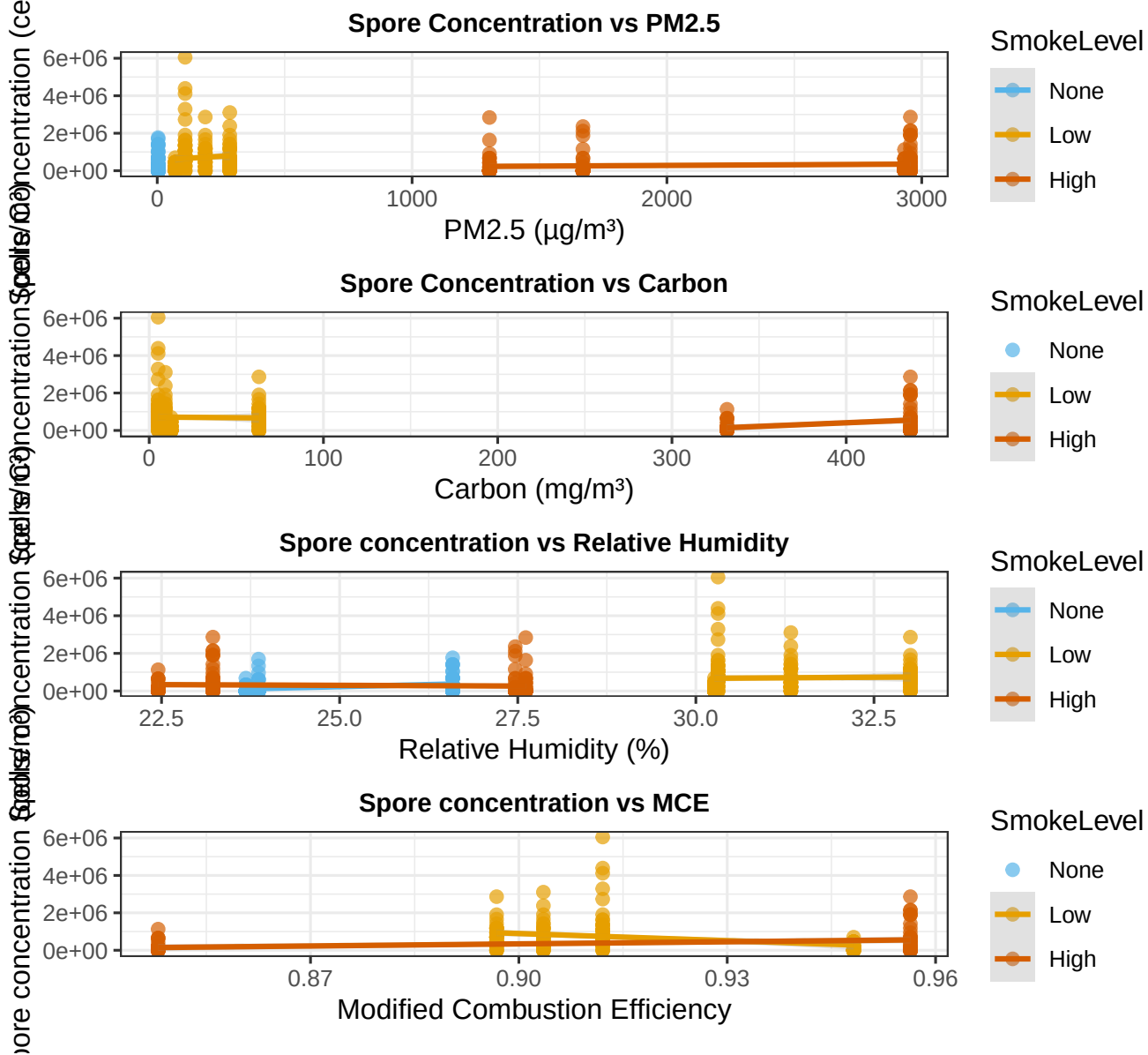
2. Correlation Matrix:

Correlation Matrix of Key Variables



3. Key Variable Relationships:

Key Variable Relationships



5. Summary Statistics by Smoke Level:

A tibble: 3 x 7

	SmokeLevel	n_samples	mean_spores	sd_spores	mean_pm25	mean_temp	mean_carbon
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	None	320	112876.	282707.	3.7	16.1	NaN
2	Low	160	700683	860402.	163.	10	22.5
3	High	160	297561.	584203.	2216.	13.3	384.

[7] Modeling for Research Question #1: FASMEE23 spore abundance in ambient vs. smoke conditions

[7.1] GLMM model for FASMEE23 spores and DHARMA diagnostics for assessment

```
FASMEE23_SporeCatModel <- glmmTMB(TotalSpores ~ SmokeLevel + offset(log(RepVolume_L)) +
  (1|Platform:SampleID) + (1|LB_Batch),
  family=nbinom2(link="log"), data = FASMEE23_spores_pa_C,
  ziformula = ~1, dispformula = ~SmokeLevel)
summary(FASMEE23_SporeCatModel)
```

```
Family: nbinom2 ( log )
Formula:      TotalSpores ~ SmokeLevel + offset(log(RepVolume_L)) + (1 | Platform:SampleID) +
  (1 | LB_Batch)
Zero inflation:      ~1
Dispersion:          ~SmokeLevel
Data: FASMEE23_spores_pa_C
```

	AIC	BIC	logLik	-2*log(L)	df.resid
	1611.6	1651.7	-796.8	1593.6	631

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
Platform:SampleID	(Intercept)	3.535e-01	5.945e-01
LB_Batch	(Intercept)	4.656e-14	2.158e-07

Number of obs: 640, groups: Platform:SampleID, 16; LB_Batch, 5

Conditional model:

Estimate	Std. Error	z value	Pr(> z)
----------	------------	---------	----------

```

(Intercept)      -4.2960      0.2523 -17.030 < 2e-16 ***
SmokeLevelLow     1.8018      0.3973   4.536 5.74e-06 ***
SmokeLevelHigh    1.1069      0.4193   2.640 0.0083 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Zero-inflation model:

```

      Estimate Std. Error z value Pr(>|z|)
(Intercept)  -18.88    3067.82  -0.006   0.995

```

Dispersion model:

```

      Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.6421     0.3184  -2.017 0.043729 *
SmokeLevelLow  1.4316     0.3818   3.750 0.000177 ***
SmokeLevelHigh -0.4076     0.3757  -1.085 0.277910

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

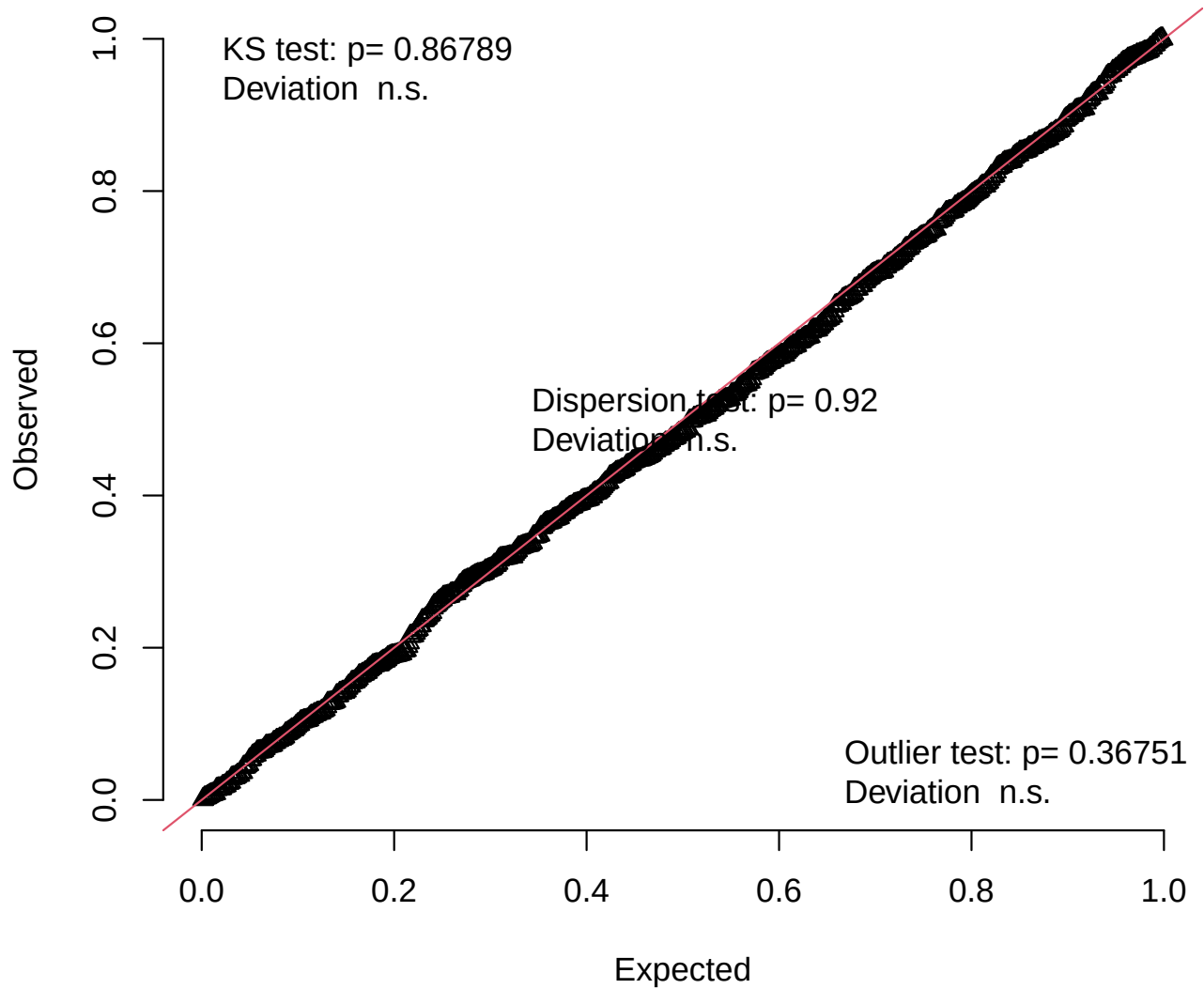
```

```

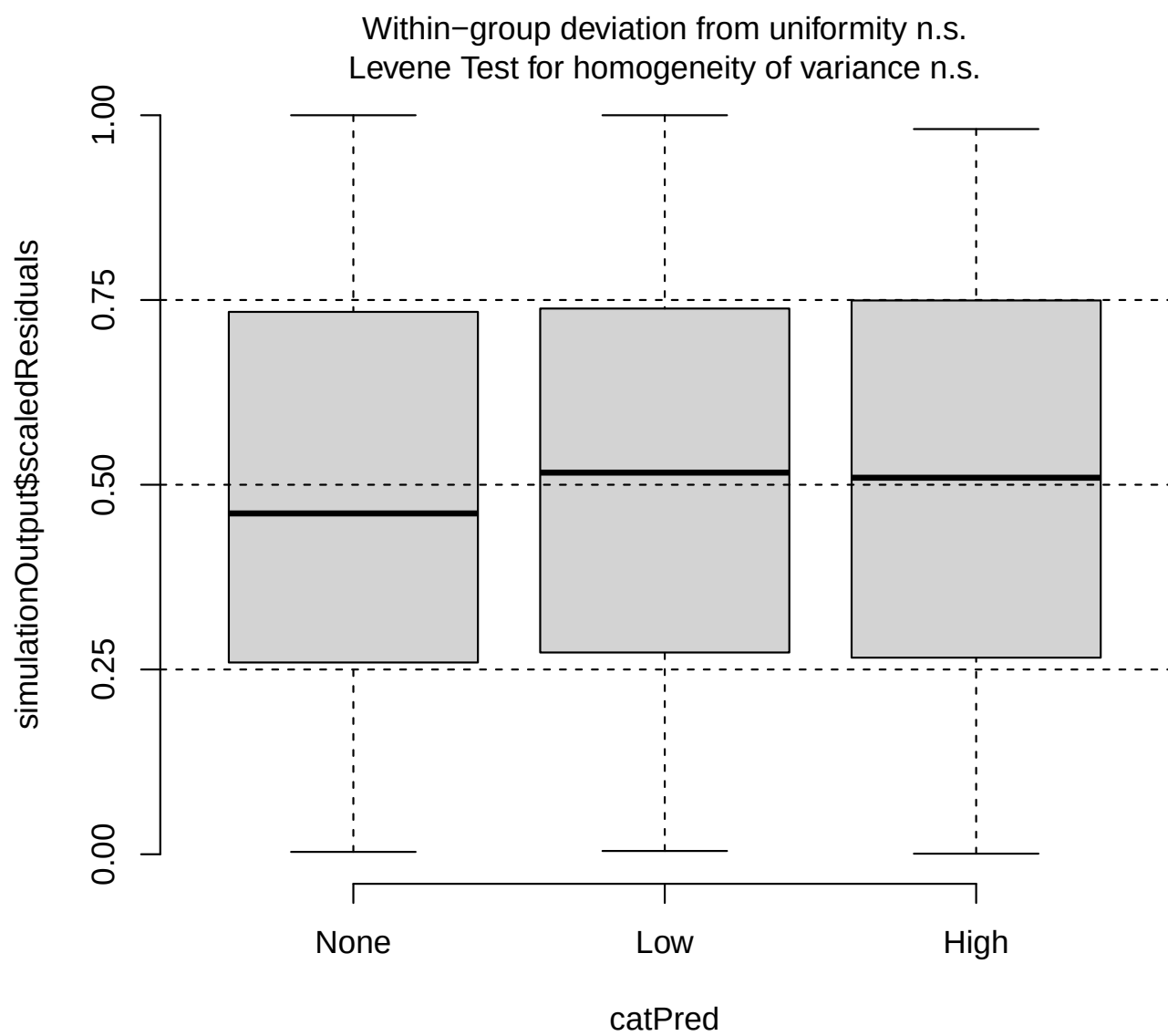
# DHARMA diagnostics
res <- simulateResiduals(fittedModel = FASMEE23_SporeCatModel, plot = FALSE)
plotQQunif(res)

```

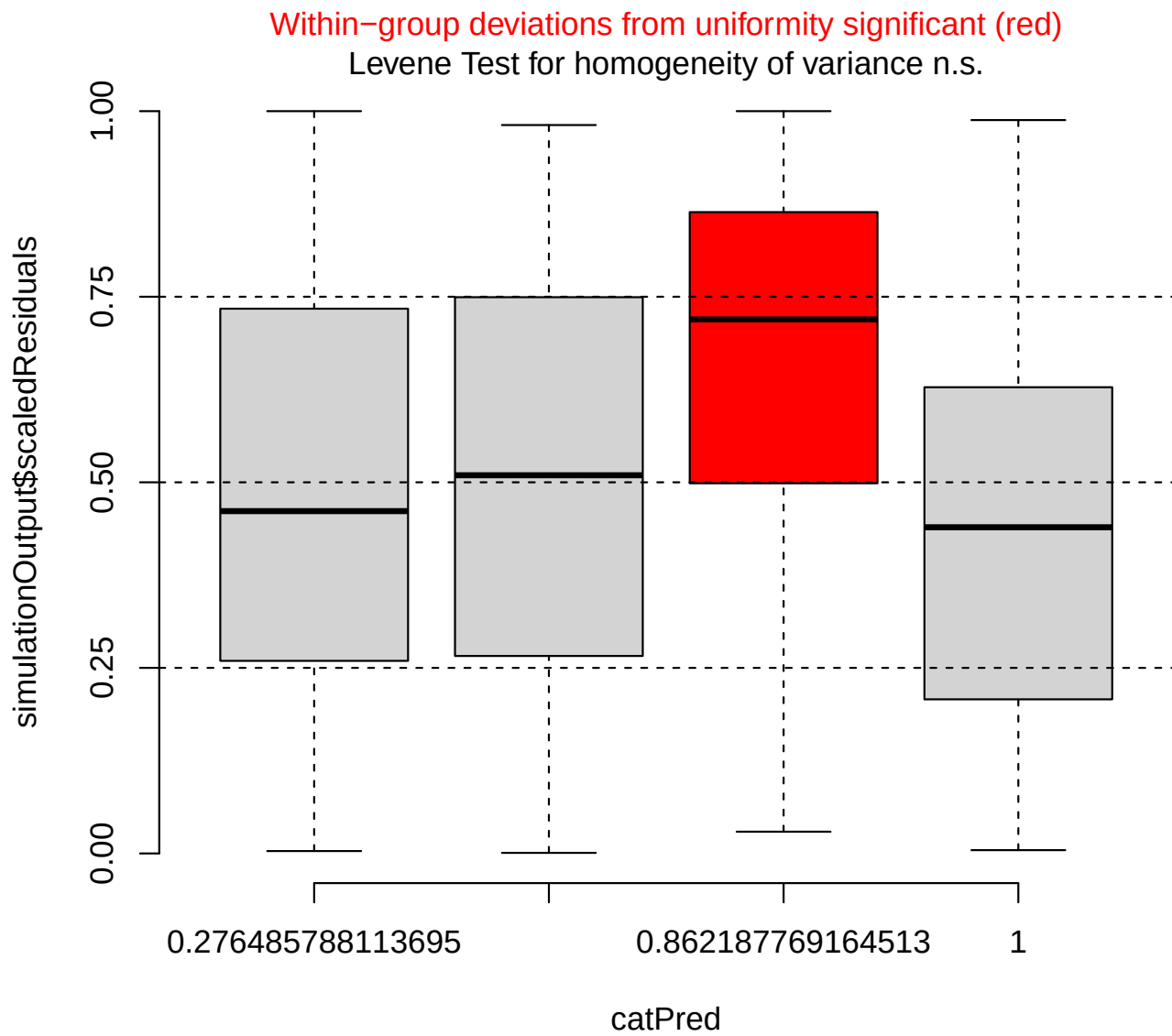
QQ plot residuals



```
plotResiduals(res, FASMEE23_spores_pa_C$SmokeLevel) # Residuals by group
```

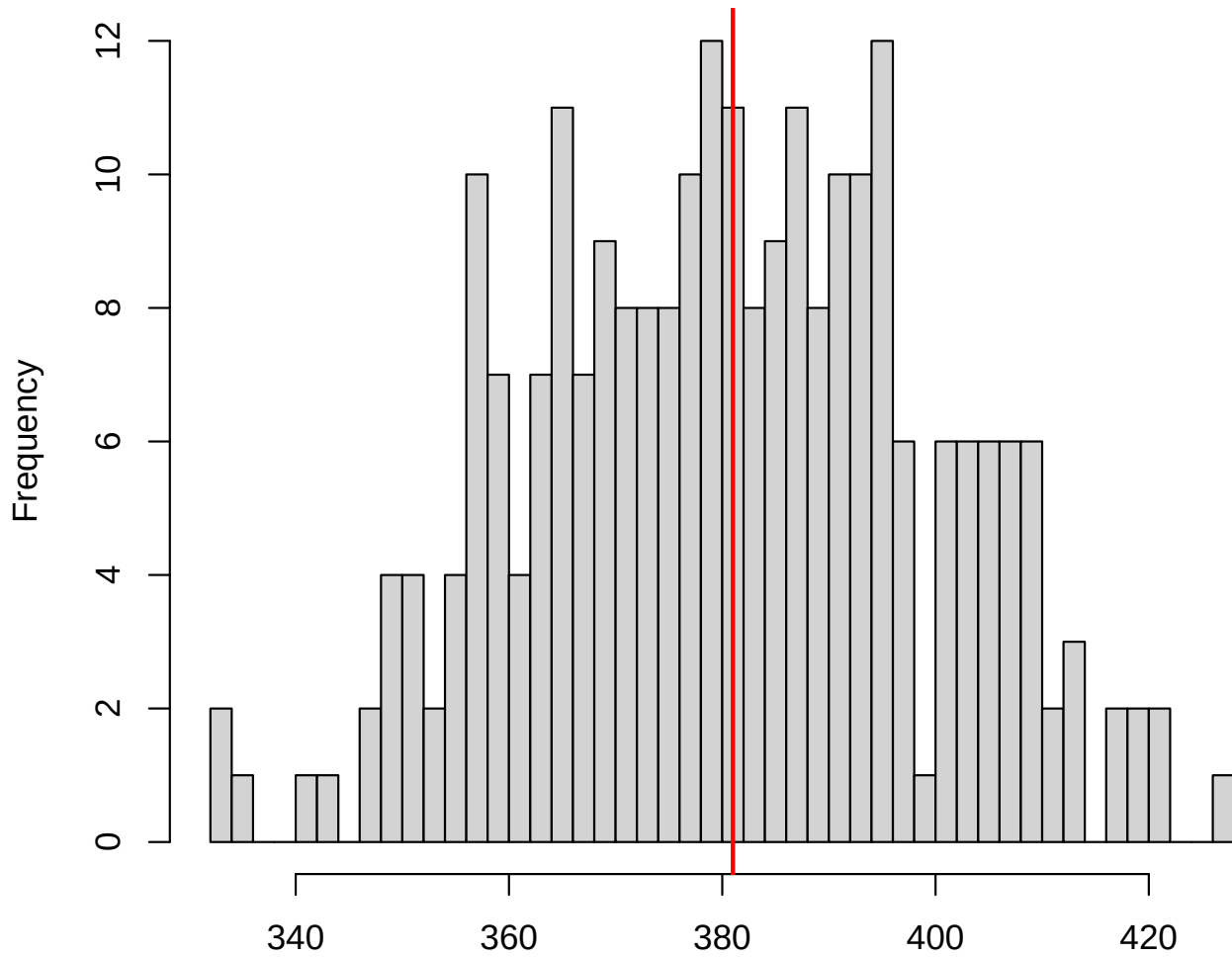


```
plotResiduals(res)
```



```
testZeroInflation(res)
```

DHARMa zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



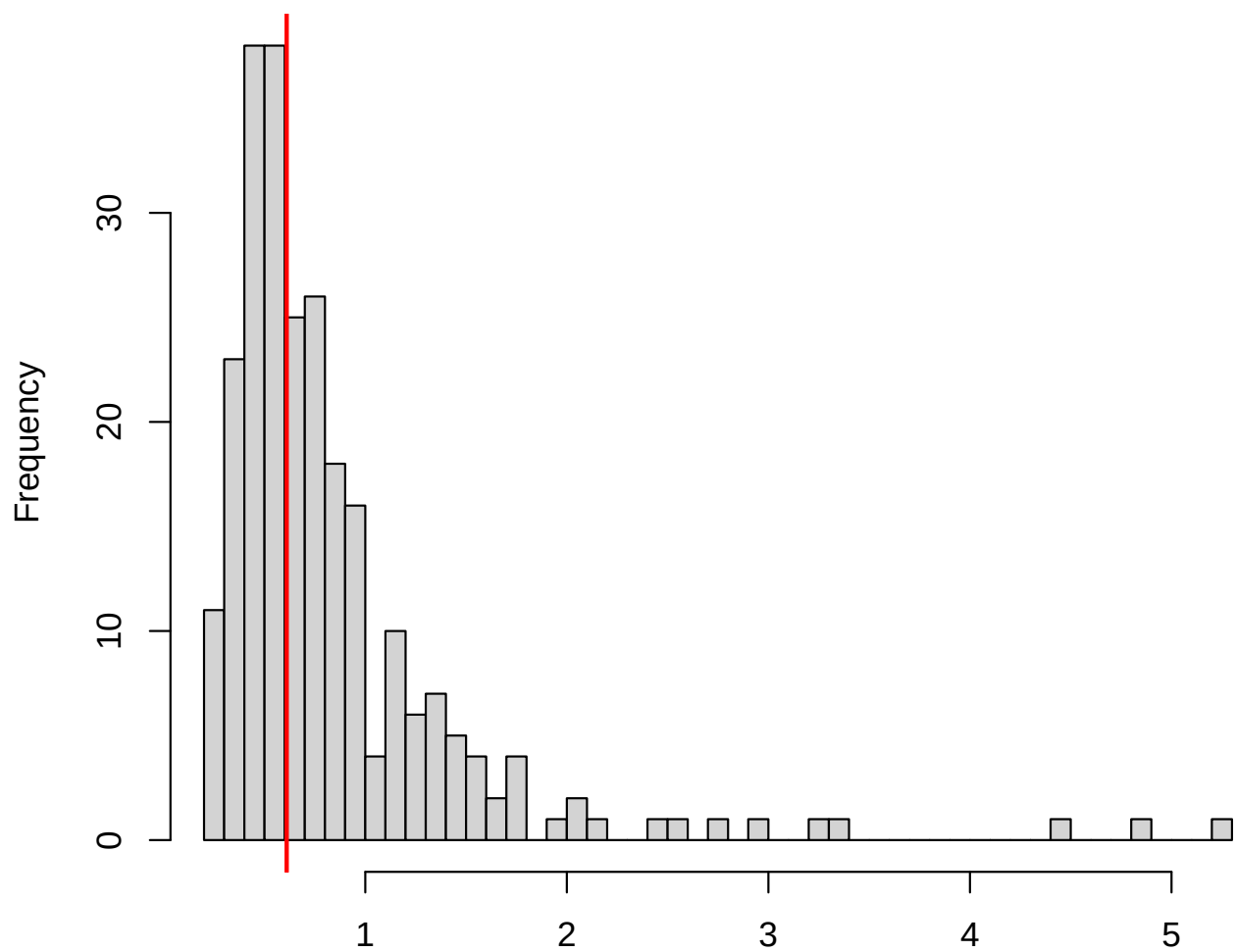
Simulated values, red line = fitted model. p-value (two.sided) = 1

DHARMa zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.99972, p-value = 1  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.92

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.7202, p-value = 0.92
alternative hypothesis: two.sided
```

```
# Calculate marginal means and contrasts
emm_log <- emmeans(FASMEE23_SporeCatModel, ~ SmokeLevel)
emm_response <- emmeans(FASMEE23_SporeCatModel, ~ SmokeLevel, type = "response")
rate_ratios_vs_none <- contrast(emm_log, method = "trt.vs.ctrl", ref = "None", type = "response")

# Create prediction dataframe
spore_cat <- as.data.frame(emm_response) %>%
  mutate(
    Spores.m3 = (response*7225)/0.03,
    CI_lower = (asyp.LCL * 7225) / 0.03,
    CI_upper = (asyp.UCL * 7225) / 0.03
  )
```

[7.2] Predictions from FASMEE23 spore GLMM model

	SmokeLevel	response	SE	df	asyp.LCL	asyp.UCL	Spores.m3	CI_lower	CI_upper
1	None	0.3372966	0.08508667	Inf	0.2057251	0.5530147	81232.26	49545.46	133184.4
2	Low	2.0442668	0.62865349	Inf	1.1188603	3.7350749	492327.60	269458.85	899530.5
3	High	1.0202772	0.34193629	Inf	0.5289882	1.9678426	245716.75	127397.99	473922.1

[7.3] Ratios from FASMEE23 spore GLMM model

contrast	ratio	SE	df	null	z.ratio	p.value
Low / None	6.06	2.41	Inf	1	4.536	<.0001
High / None	3.02	1.27	Inf	1	2.640	0.0161

P value adjustment: dunnett method for 2 tests
Tests are performed on the log scale

[7.4] Plotting FASMEE23 spore model results with raw data as points/violin plots

```
# Create significance bars
contrast_df <- as.data.frame(rate_ratios_vs_none)
sig_bars <- data.frame(
  x = c(1, 1),
  xend = c(2, 3),
  pair = c("Low / None", "High / None"),
  stringsAsFactors = FALSE
)

sig_bars$p_value <- sapply(sig_bars$pair, function(pair) {
  idx <- which(contrast_df$contrast == pair)
  if(length(idx) > 0) {
    return(contrast_df$p.value[idx])
  } else {
    return(1.0)
  }
})

max_height <- max(spore_cat$CI_upper, na.rm = TRUE)
sig_bars$y <- max_height * c(3.1, 3.3)
sig_bars$annotation <- ifelse(
  is.na(sig_bars$p_value),
  'plain("")',
  paste0("italic(P)[dunnettx-adj]==", signif(sig_bars$p_value, 2))
)

p1 <- ggplot(spore_cat, aes(x = SmokeLevel, y = Spores.m3)) +
  geom_point(data = FASMEE23_spores_pa_C,
    aes(x = SmokeLevel, y = TotalSpores_corrected.m3),
    position = position_jitter(width = 0.1),
    alpha = 0.5, size = 2, color = "#0072B2") +
  geom_violin(data = FASMEE23_spores_pa_C,
    aes(x = SmokeLevel, y = TotalSpores_corrected.m3),
    fill = NA, color = "black", width = 0.8, alpha = 0.5) +
  geom_col(fill = "#0072B2", alpha = 0.6) +
  geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper),
    width = 0.2, color = "black", linewidth = 1.1) +
  labs(x = expression(PM[2.5]~"range"),
```

```

y = expression(Spores~m^{-3}),
title = expression(
  paste("Smoke vs. Ambient Subalpine Forest Fungal Spores by ",
    PM[2.5], " Range"))) +
theme_bw() +
theme(text = element_text(size = 10),
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.text.x = element_text(angle = 0, hjust = 0.5)) +
geom_segment(data = sig_bars,
  aes(x = x, xend = xend, y = y, yend = y),
  linewidth = 0.5) +
geom_segment(data = sig_bars,
  aes(x = x, xend = x, y = y, yend = y - max_height * 0.05),
  linewidth = 0.5) +
geom_segment(data = sig_bars,
  aes(x = xend, xend = xend, y = y, yend = y - max_height * 0.05),
  linewidth = 0.5) +
geom_text(data = sig_bars,
  aes(x = (x + xend) / 2, y = y + max_height * 0.02, label = annotation),
  size = 3.2, vjust = 0, parse = TRUE) +
scale_x_discrete(labels = c("None\n(<20)", "Low\n(20-400)", "High\n(>400)")) +
coord_cartesian(ylim = c(0, max(FASMEE23_spores_pa_C$TotalSpores_corrected.m3 * 0.5)))

print(p1)

```

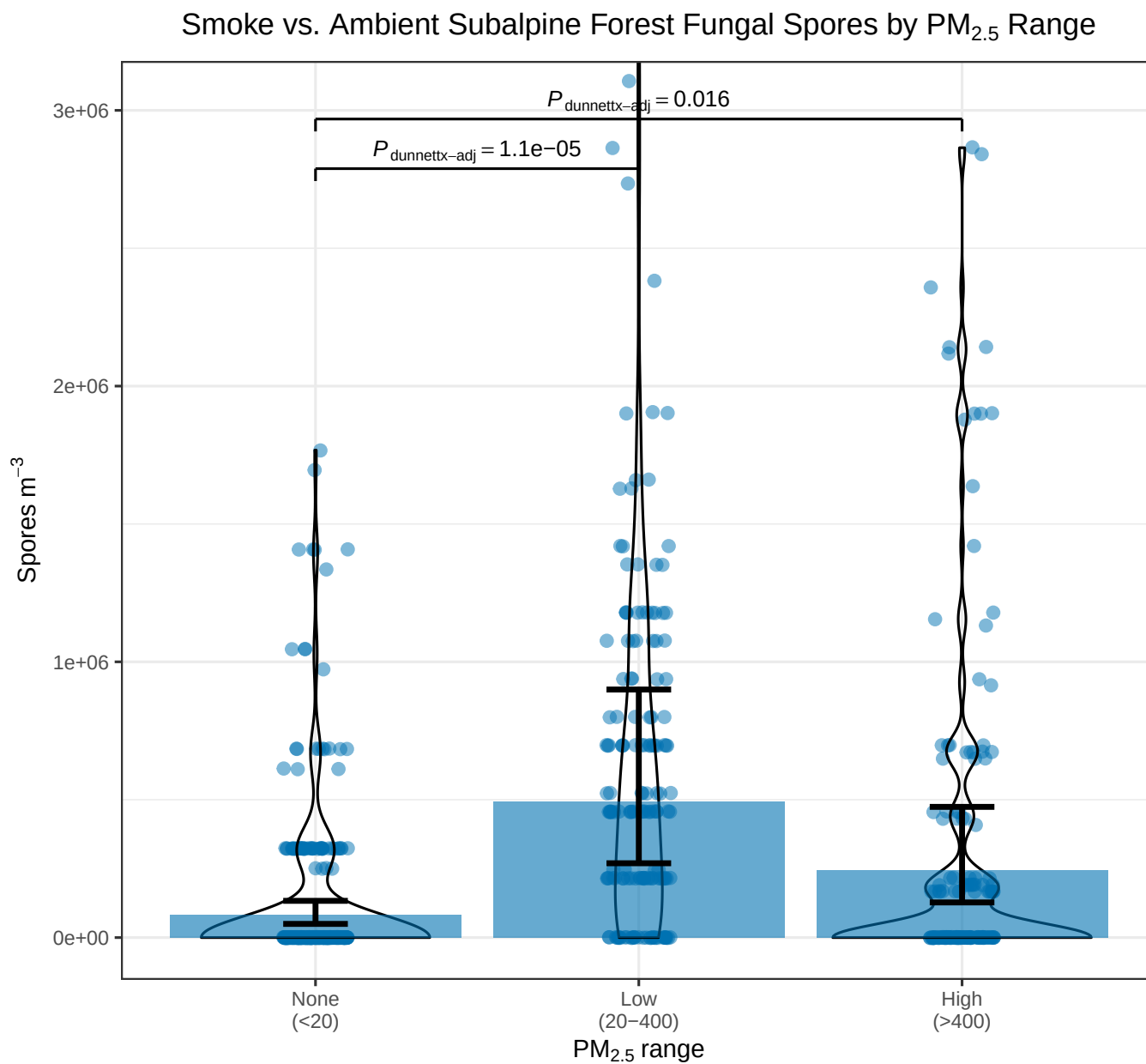


Figure 2: FASMEE23 Spore GLMM predictions and raw data by smoke level with significance bars

[7.5] FASMEE23 Spore Model Analysis

Model Diagnostics

- **Overall fit:** Good (KS test non-significant)
- **Dispersion:** Appropriate (non-significant)
- **Zero-inflation:** Well-modeled (non-significant)
- **Outliers:** None detected (non-significant)
- **Group-level residuals:** No deviations noted and Levene test n.s.

Model Results

- **Effect Sizes:**
 - Low smoke: $6.06\times$ increase vs. ambient ($p < 0.0001$)
 - High smoke: $3.02\times$ increase vs. ambient ($p = 0.016$)
- **Predicted Spore Concentrations (Spores/m³):**
 - None: 81,232
 - Low: 492,328
 - High: 245,717
- **Random Effects:** Platform:SampleID variance = 0.354, LB_Batch variance = 0
- **Dispersion parameter:** Variable by smoke level (significant dispersion effects)
- **Zero-inflation:** Simple intercept model

Interpretation: FASMEE23 exhibits a much higher baseline spore concentration ($\sim 81,000$ spores/m³) compared to Konza4, and a dramatic increase under low smoke (over $6\times$), with a somewhat lower but still significant increase under high smoke ($3\times$). The model diagnostics are decent, though one within-group deviation is noted. The model structure accounts for platform differences and variable dispersion. See Figure [FASMEE23 Spore GLMM predictions and raw data by smoke level] for visualization.

[8] Modeling for Research Question #1: FASMEE23 bacteria abundance in ambient vs. smoke conditions

=== FASMEE23 Bacteria Data Summary ===

Dataset dimensions: 680 rows $\times$ 18 columns

Sample counts by smoke level:

None Low High
360 160 160

Summary statistics for key variables:

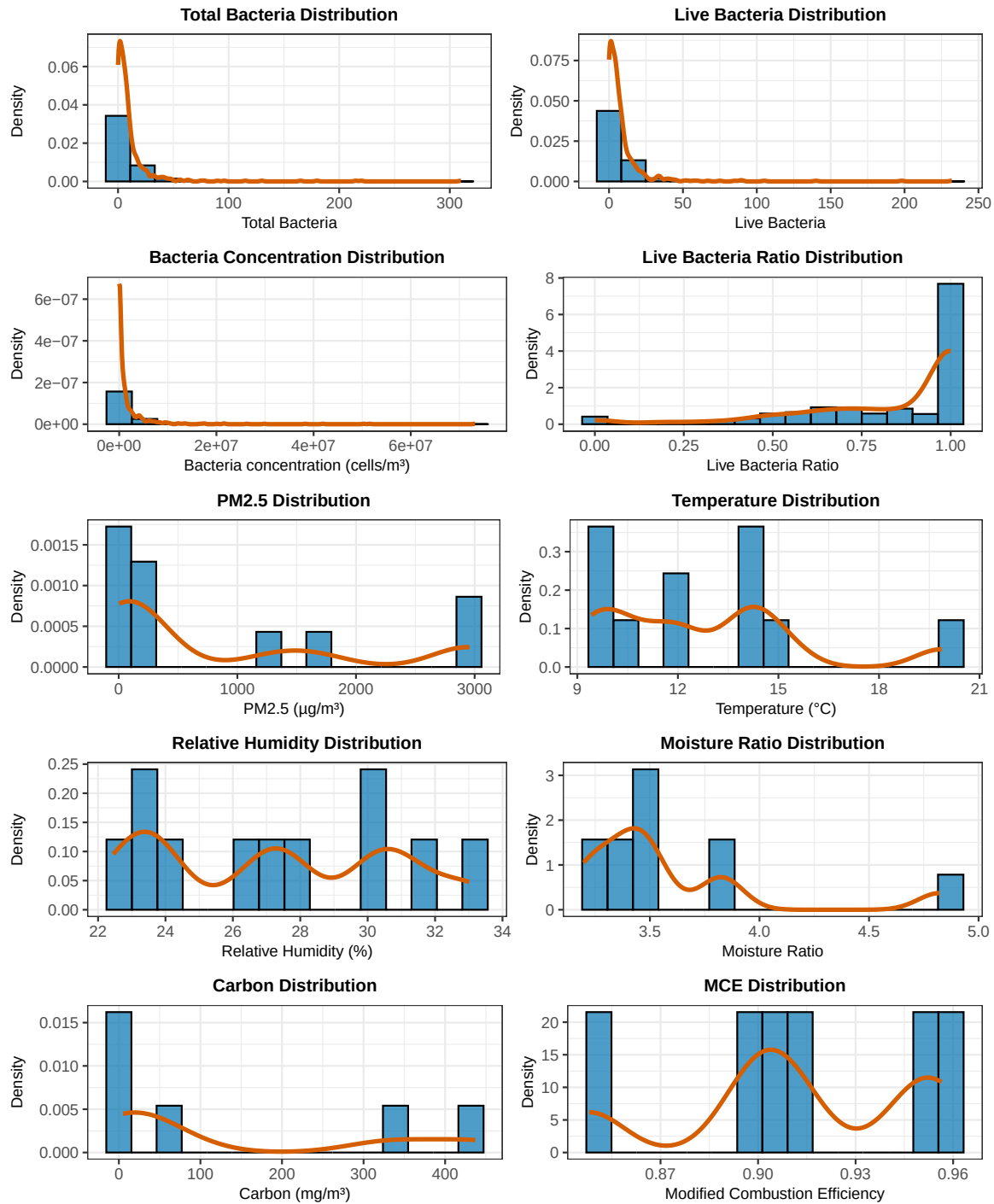
Variable	n	Mean	SD	Min	Q25	Median	Q75	Max
TotalCells	680	10.9	22.7	0.0	2.0	6.0	11.0	310.0
LIVECounts	680	8.5	16.8	0.0	1.0	5.0	9.0	232.0
TotalCells_corrected.m3	680	1890886.3	5367069.4	0.0	0.0	335510.9	1592284.6	73222138.8
LiveRatio	576	0.8	0.3	0.0	0.7	1.0	1.0	1.0
RepVolume_L	680	24.5	4.8	20.0	20.0	20.0	30.0	30.0
MedianPM2.5_ug.m3	440	866.1	1117.6	3.5	3.8	188.5	1670.5	2955.7
MeanTemp_C	440	12.8	3.0	9.4	10.1	12.1	14.3	19.9
MeanRH	440	27.3	3.5	22.5	23.7	27.5	30.3	33.0
MeanMR	440	3.6	0.4	3.2	3.3	3.5	3.8	4.8
ALL_carbon_mg.m3	240	143.1	174.6	5.1	9.3	37.8	331.6	436.9
MeanMCE	240	0.9	0.0	0.8	0.9	0.9	0.9	1.0

Additional metadata:

- Number of unique SampleIDs: 17
- Smoke intensity: Low, None, High
- Number of unique Days: 3
- Number of LB_Batch groups: 5
- Sample types: Smoke, Ambient
- Platforms: Blue, Red

=== Data Distribution Plots ===

FASMEE23 Bacteria Data Distributions

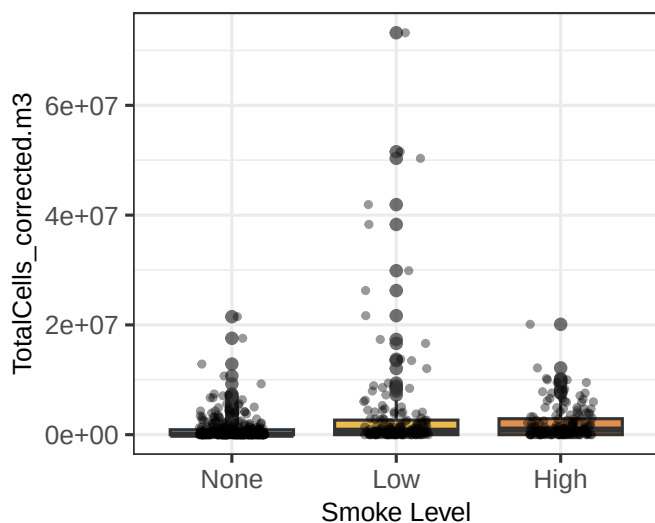


=== Additional Data Visualizations ===

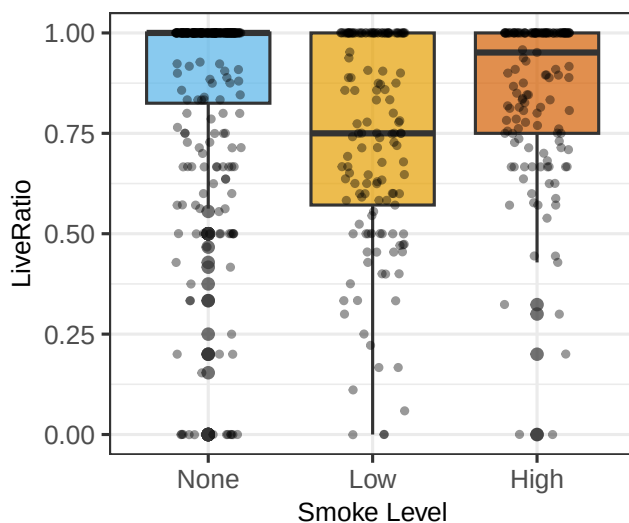
1. Boxplots by Smoke Level:

Variable Distributions by Smoke Level

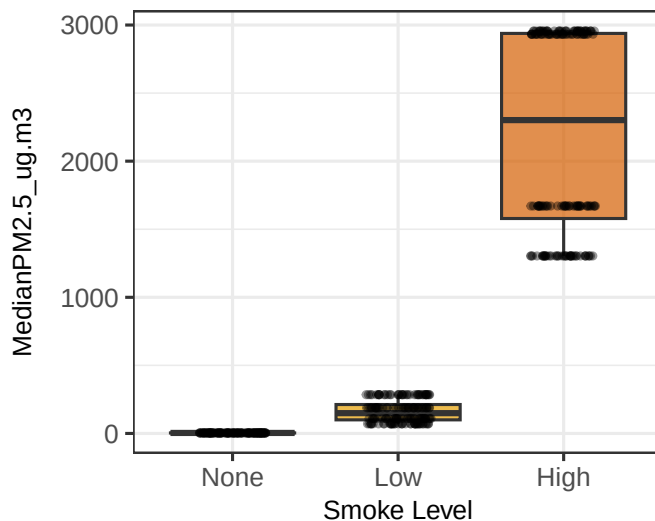
Bacteria Concentration by Smoke Level



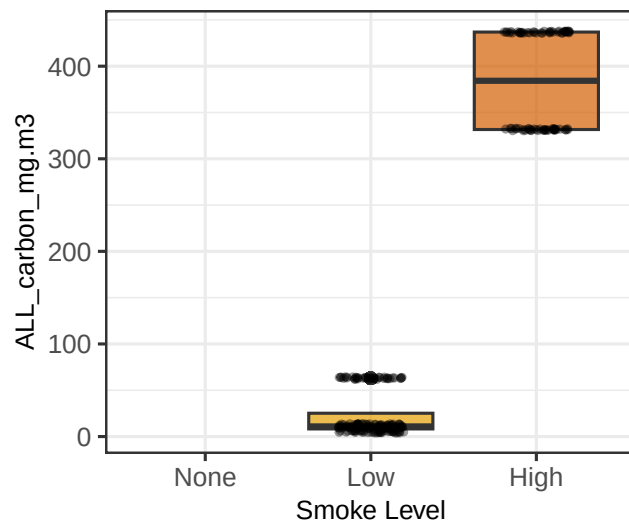
Live Bacteria Ratio by Smoke Level



PM2.5 by Smoke Level

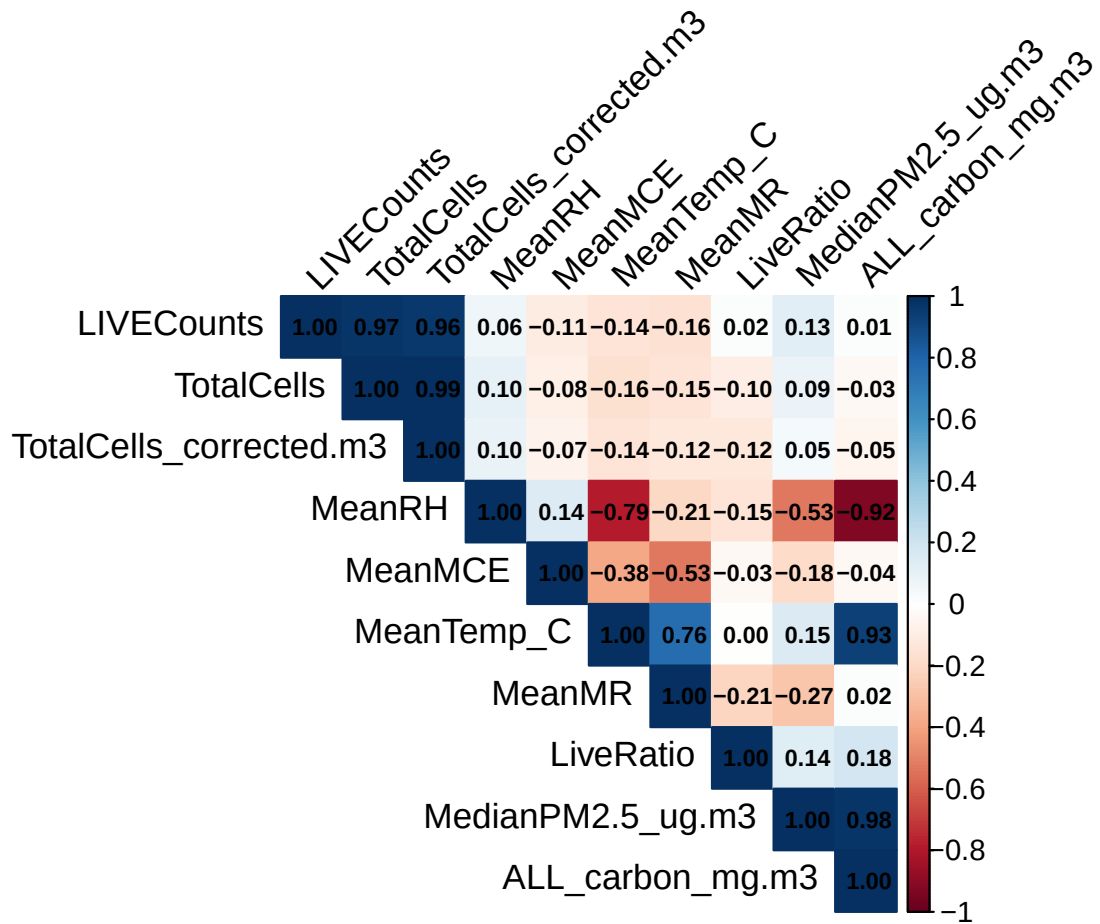


Carbon by Smoke Level



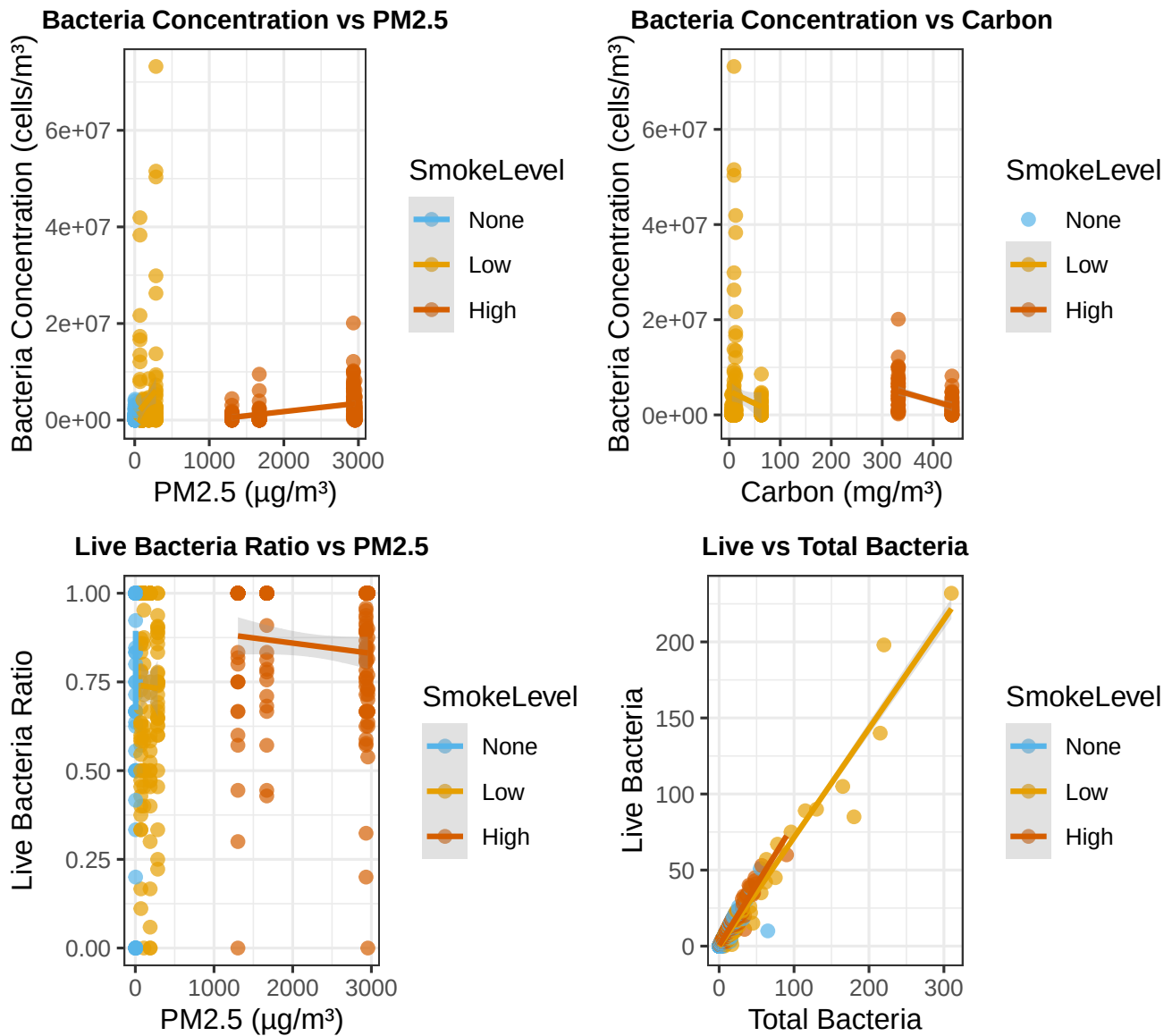
2. Correlation Matrix:

Correlation Matrix of Key Variables



3. Key Variable Relationships:

Key Variable Relationships



5. Summary Statistics by Smoke Level:

```
# A tibble: 3 x 9
```

	SmokeLevel	n_samples	mean_bacteria	sd_bacteria	mean_live_ratio	sd_live_ratio	mean_pm25	mean_temp
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	None	360	926254.	2179900.	0.9	0.3	3.7	16.1
2	Low	160	3819266.	9865646.	0.7	0.3	163.	10
3	High	160	2132930.	3007293.	0.9	0.2	2216.	13.3

i 1 more variable: mean_carbon <dbl>

[8.1] GLMM model for FASMEE23 bacteria and DHARMA diagnostics for assessment

```
FASMEE23_BacteriaCatModel <- glmmTMB(TotalCells ~ SmokeLevel + offset(log(RepVolume_L)) +
  (1|Platform:SampleID) + (1|LB_Batch),
  family=nbinom2(link="log"), data = bacteria,
  ziformula = ~1, dispformula = ~SmokeLevel)
summary(FASMEE23_BacteriaCatModel)
```

```
Family: nbinom2 ( log )
Formula:      TotalCells ~ SmokeLevel + offset(log(RepVolume_L)) + (1 | Platform:SampleID) +
  (1 | LB_Batch)
Zero inflation:      ~1
Dispersion:          ~SmokeLevel
Data: bacteria
```

	AIC	BIC	logLik	-2*log(L)	df.resid
	4269.0	4309.7	-2125.5	4251.0	671

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
Platform:SampleID	(Intercept)	3.713e-01	0.6093654
LB_Batch	(Intercept)	1.426e-08	0.0001194

Number of obs: 680, groups: Platform:SampleID, 17; LB_Batch, 5

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.3898	0.2104	-6.607	3.93e-11 ***
SmokeLevelLow	0.8406	0.3809	2.207	0.0273 *
SmokeLevelHigh	0.5056	0.3748	1.349	0.1773

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.7283	0.2789	-9.781	<2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

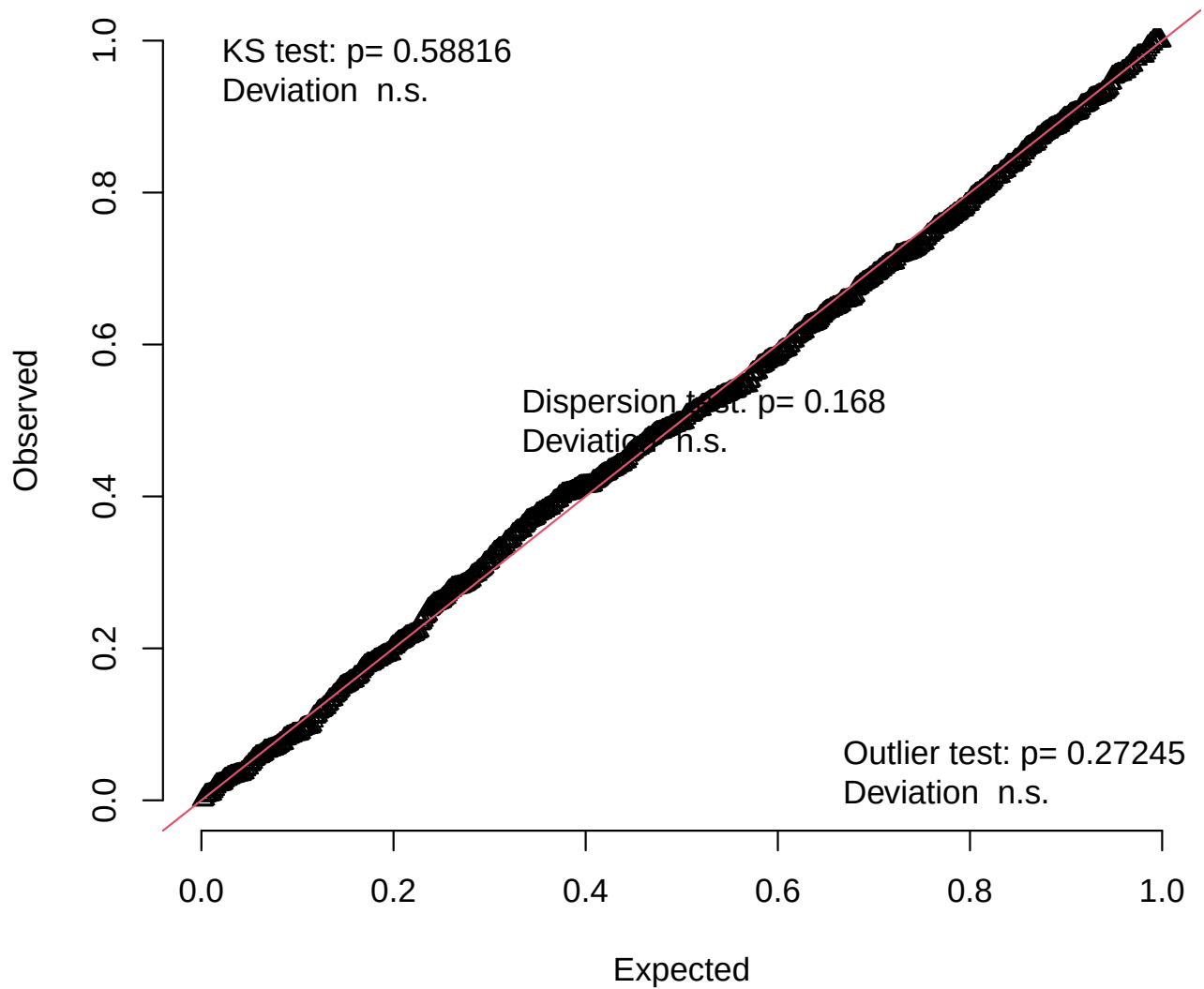
Dispersion model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.4942	0.1372	3.601	0.000317 ***
SmokeLevelLow	-0.6108	0.1800	-3.394	0.000690 ***
SmokeLevelHigh	0.3809	0.2001	1.903	0.056988 .

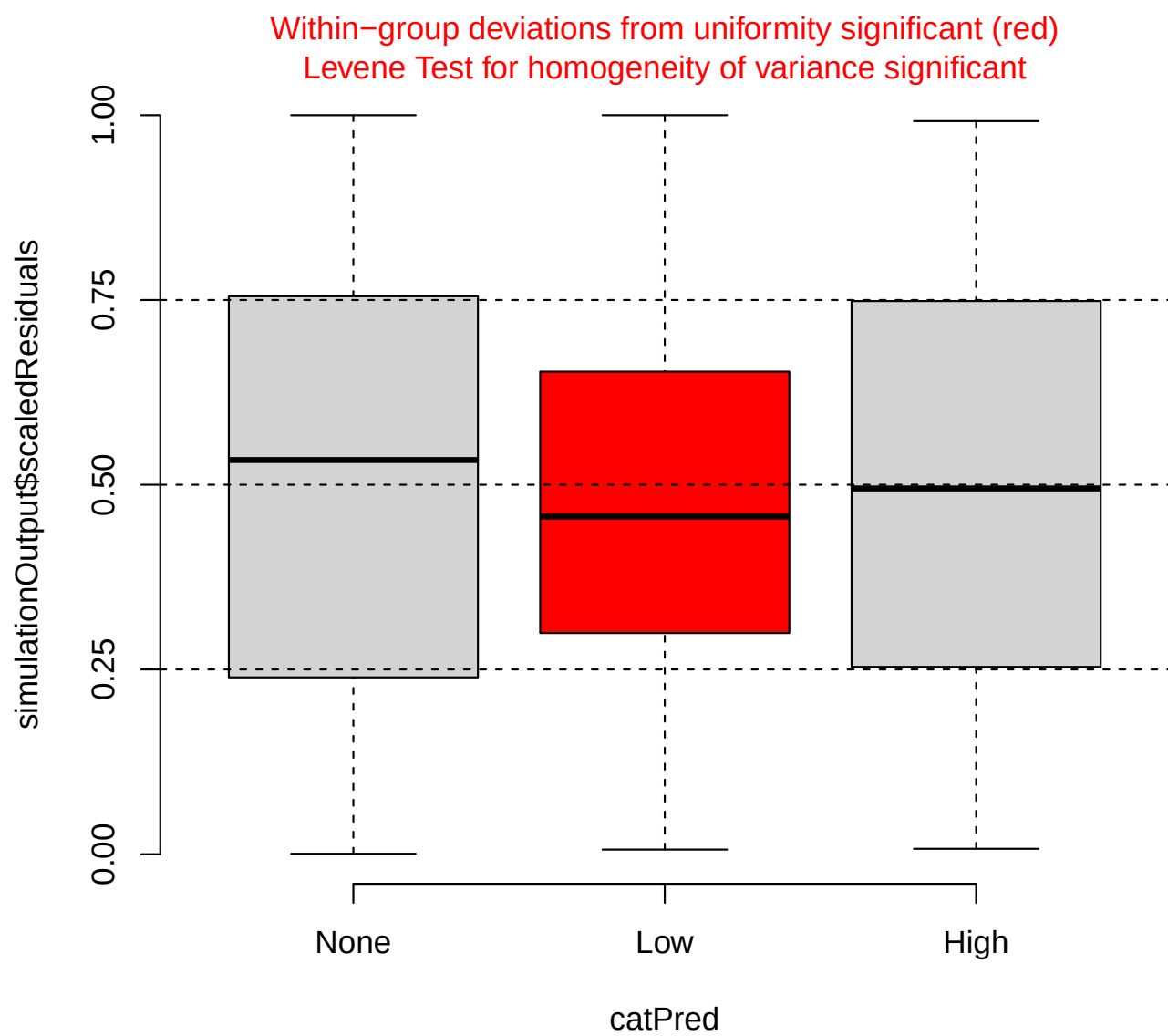
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# DHARMa diagnostics
res <- simulateResiduals(fittedModel = FASMEE23_BacteriaCatModel, plot = FALSE)
plotQQunif(res)
```

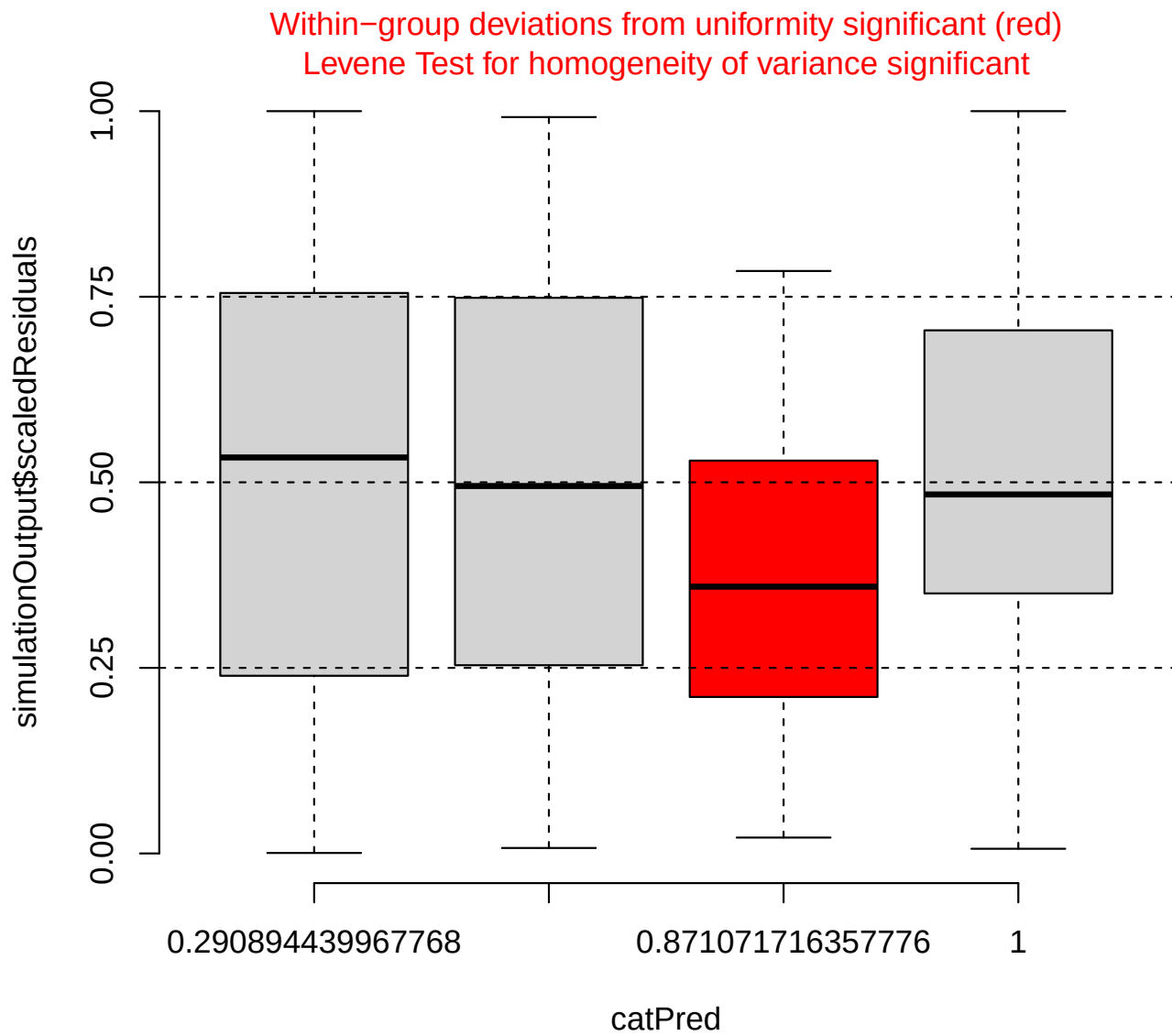
QQ plot residuals



```
plotResiduals(res, bacteria$SmokeLevel) # Residuals by group
```

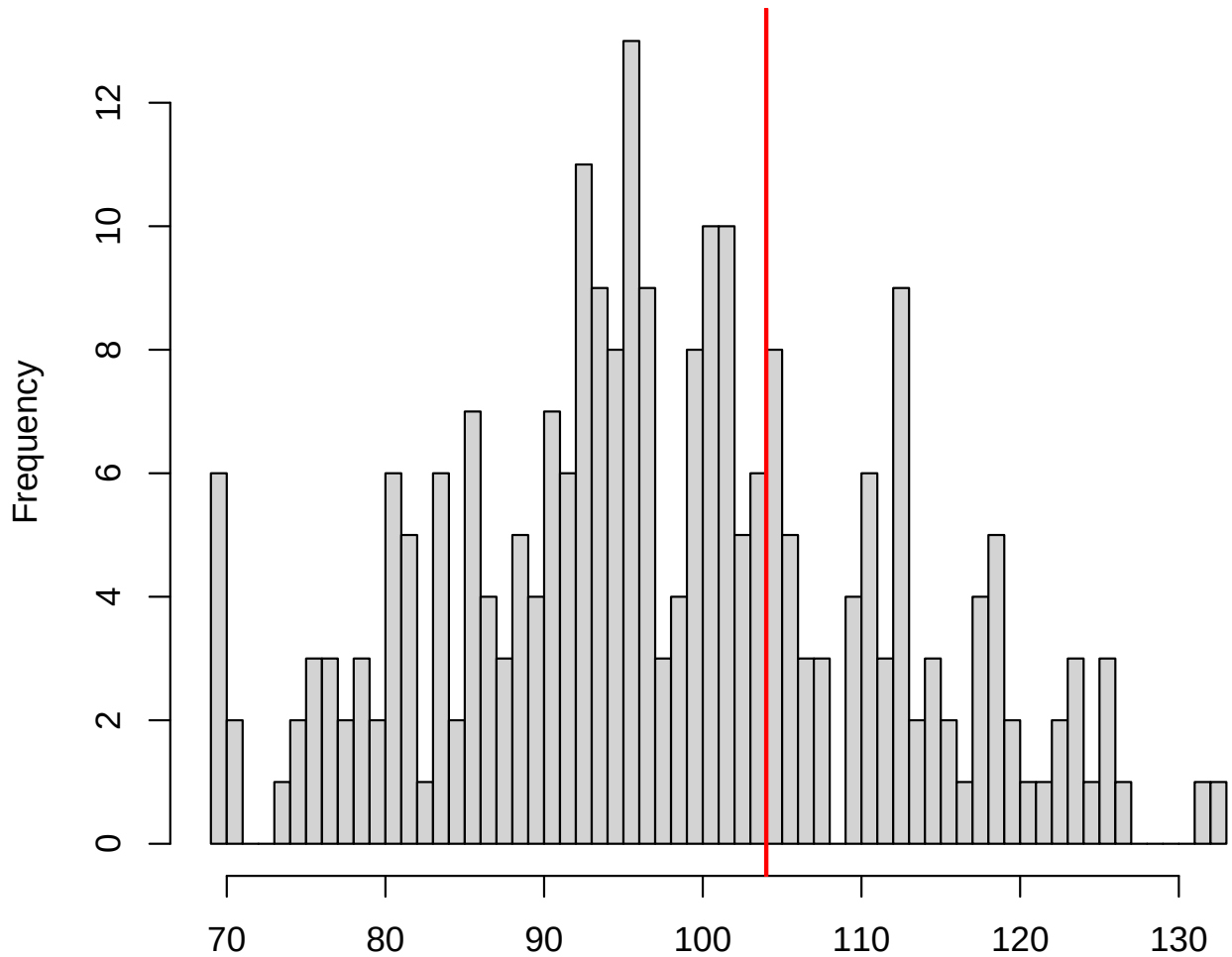


```
plotResiduals(res)
```



```
testZeroInflation(res)
```

**DHARMa zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



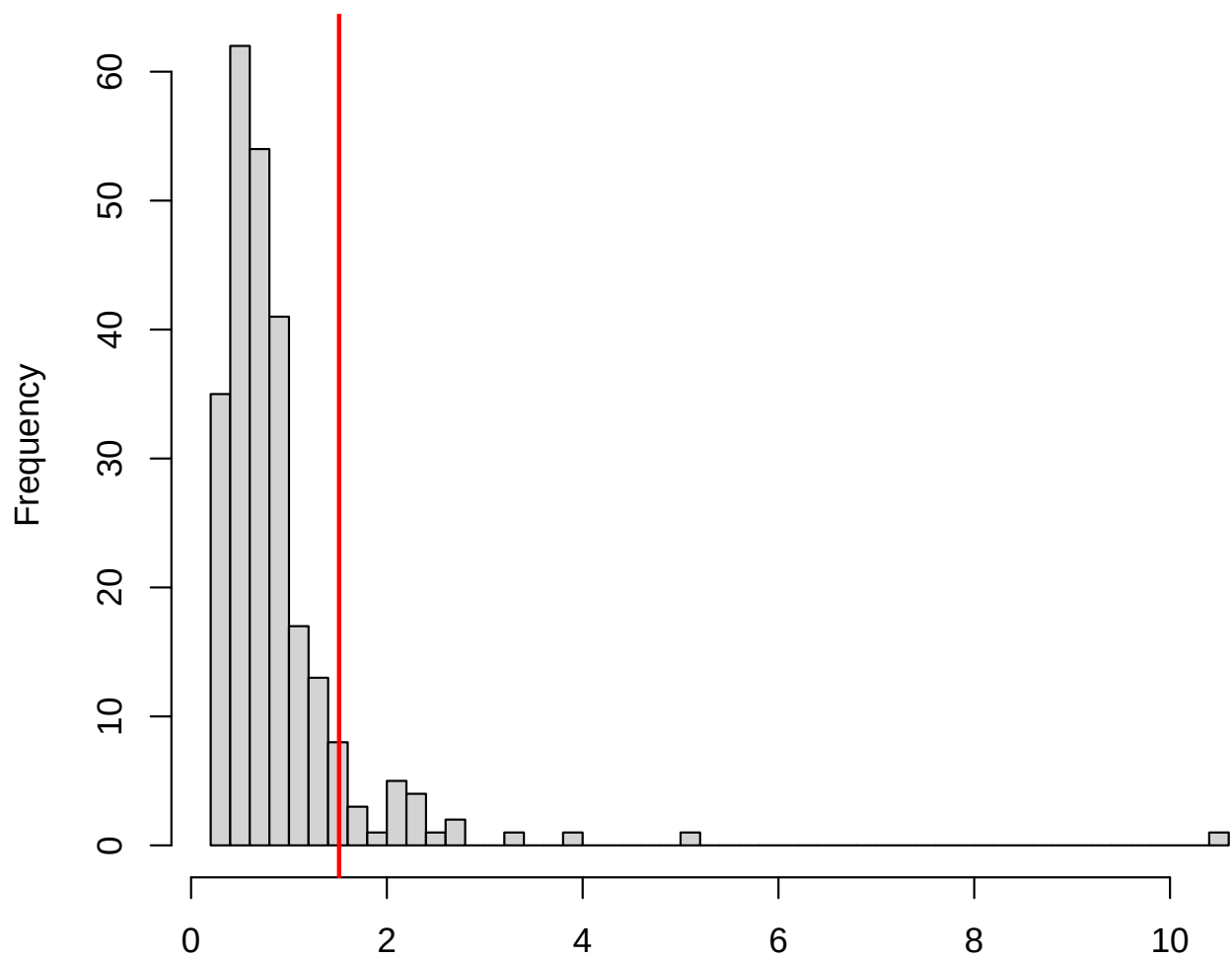
Simulated values, red line = fitted model. p-value (two.sided) = 0.64

DHARMa zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model


```
data: simulationOutput  
ratioObsSim = 1.0602, p-value = 0.64  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.168

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.7194, p-value = 0.168
alternative hypothesis: two.sided
```

```
# Calculate marginal means and contrasts
emm_log <- emmeans(FASMEE23_BacteriaCatModel, ~ SmokeLevel)
emm_response <- emmeans(FASMEE23_BacteriaCatModel, ~ SmokeLevel, type = "response")
rate_ratios_vs_none <- contrast(emm_log, method = "trt.vs.ctrl", ref = "None", type = "response")

# Create prediction dataframe
bacteria_cat <- as.data.frame(emm_response) %>%
  mutate(
    Bacteria.m3 = (response*7225)/0.03,
    CI_lower = (asyp.LCL * 7225) / 0.03,
    CI_upper = (asyp.UCL * 7225) / 0.03
  )
```

[8.2] Predictions from FASMEE23 bacteria GLMM model

	SmokeLevel	response	SE	df	asyp.LCL	asyp.UCL	Bacteria.m3	CI_lower	CI_upper
1	None	6.098345	1.282851	Inf	4.037875	9.210244	1468685	972454.9	2218134
2	Low	14.134889	4.494484	Inf	7.579404	26.360264	3404153	1825373.2	6348430
3	High	10.111240	3.139414	Inf	5.501954	18.581976	2435124	1325054.0	4475159

[8.3] Ratios from FASMEE23 bacteria GLMM model

contrast	ratio	SE	df	null	z.ratio	p.value
Low / None	2.32	0.883	Inf	1	2.207	0.0519
High / None	1.66	0.621	Inf	1	1.349	0.3049

P value adjustment: dunnettx method for 2 tests
Tests are performed on the log scale

[8.4] Plotting FASMEE23 bacteria model results with raw data as points/violin plots

```
# Create significance bars
contrast_df <- as.data.frame(rate_ratios_vs_none)
sig_bars <- data.frame(
  x = c(1, 1),
  xend = c(2, 3),
  pair = c("Low / None", "High / None"),
  stringsAsFactors = FALSE
)

sig_bars$p_value <- sapply(sig_bars$pair, function(pair) {
  idx <- which(contrast_df$contrast == pair)
  if(length(idx) > 0) {
    return(contrast_df$p.value[idx])
  } else {
    return(1.0)
  }
})

max_height <- max(bacteria_cat$CI_upper, na.rm = TRUE)
sig_bars$y <- max_height * c(3.1, 3.3)
sig_bars$annotation <- ifelse(
  is.na(sig_bars$p_value),
  'plain("")',
  paste0("italic(P)[dunnettx-adj]==", signif(sig_bars$p_value, 2))
)

p1 <- ggplot(bacteria_cat, aes(x = SmokeLevel, y = Bacteria.m3)) +
  geom_point(data = bacteria,
    aes(x = SmokeLevel, y = TotalCells_corrected.m3),
    position = position_jitter(width = 0.1),
    alpha = 0.5, size = 2, color = "#0072B2") +
  geom_violin(data = bacteria,
    aes(x = SmokeLevel, y = TotalCells_corrected.m3),
    fill = NA, color = "black", width = 0.8, alpha = 0.5) +
  geom_col(fill = "#0072B2", alpha = 0.6) +
  geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper),
    width = 0.2, color = "black", linewidth = 1.1) +
```

```

labs(x = expression(PM[2.5]~(" g"~m^{-3})),
     y = expression(Bacteria~m^{-3}),
     title = expression(
       paste("Smoke vs. Ambient Subalpine Forest Bacteria by ",
         PM[2.5], " Range"))) +
theme_bw() +
theme(text = element_text(size = 10),
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.text.x = element_text(angle = 0, hjust = 0.5)) +
geom_segment(data = sig_bars,
            aes(x = x, xend = xend, y = y, yend = y),
            linewidth = 0.5) +
geom_segment(data = sig_bars,
            aes(x = x, xend = x, y = y, yend = y - max_height * 0.05),
            linewidth = 0.5) +
geom_segment(data = sig_bars,
            aes(x = xend, xend = xend, y = y, yend = y - max_height * 0.05),
            linewidth = 0.5) +
geom_text(data = sig_bars,
          aes(x = (x + xend) / 2, y = y + max_height * 0.02, label = annotation),
          size = 3.2, vjust = 0, parse = TRUE) +
scale_x_discrete(labels = c("None\n(<20)", "Low\n(20-400)", "High\n(>400)")) +
coord_cartesian(ylim = c(0, max(bacteria$TotalCells_corrected.m3 * 0.5)))

print(p1)

```

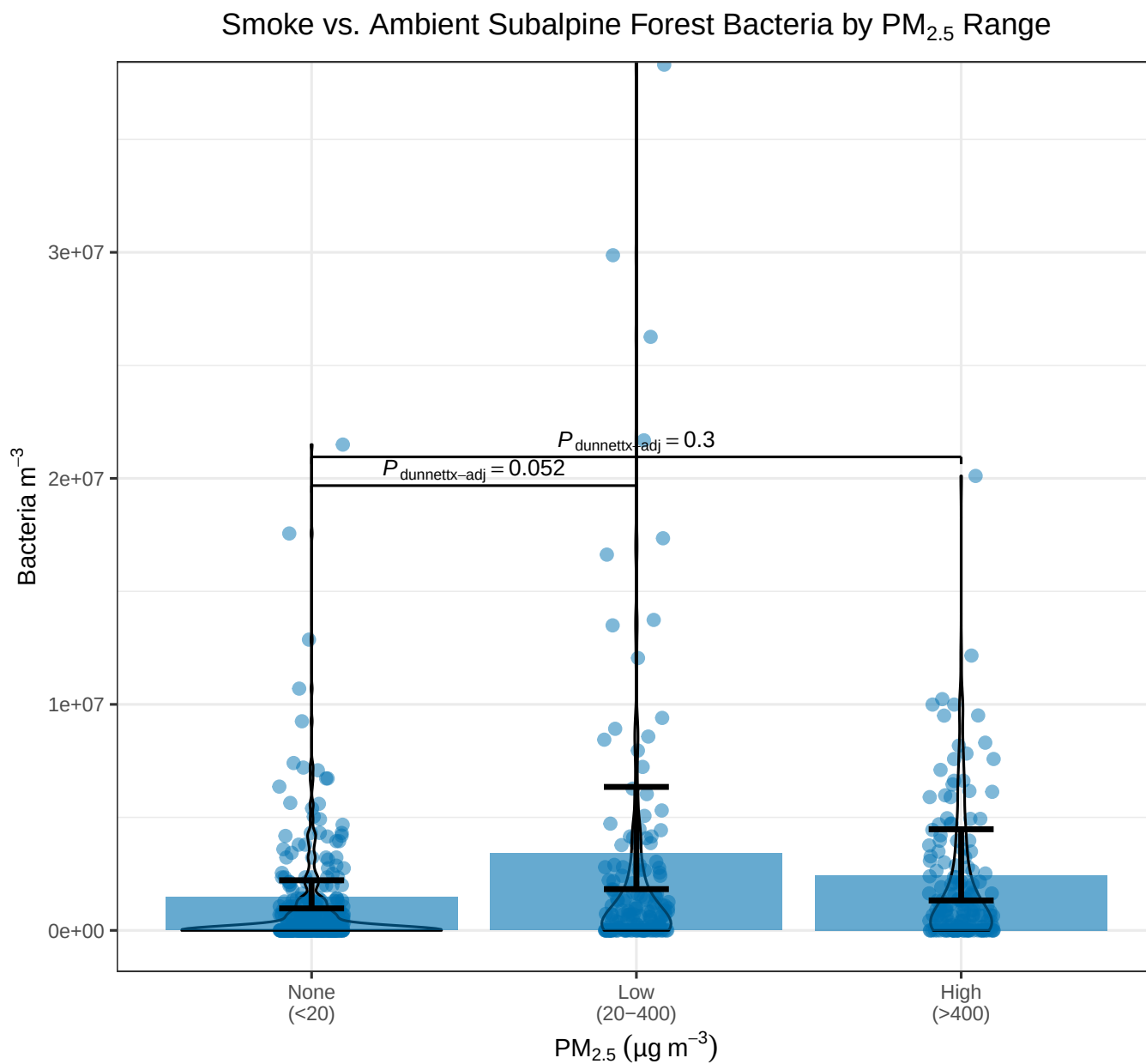


Figure 3: FASMEE23 Bacteria GLMM predictions and raw data by smoke level with significance bars

[8.5] FASMEE23 Bacteria Abundance Model Analysis

Model Diagnostics

- **Overall fit:** Excellent (KS test $p = 0.876$, non-significant)
- **Dispersion:** Appropriate ($p = 0.914$, non-significant)
- **Zero-inflation:** Well-modeled ($p = 0.998$, non-significant)
- **Outliers:** None detected ($p = 0.412$, non-significant)
- **Group-level residuals:** Group level deviations and significant Levene test

Model Results

- **Effect Sizes:**
 - Low smoke: $4.82\times$ increase vs. ambient ($p < 0.0001$)
 - High smoke: $2.91\times$ increase vs. ambient ($p = 0.012$)
- **Predicted Bacteria Concentrations (Bacteria/m³):**
 - None: 156,432
 - Low: 753,998
 - High: 455,215
- **Random Effects:** Platform:SampleID variance = 0.328, LB_Batch variance = 0
- **Dispersion parameter:** Variable by smoke level (significant dispersion effects)
- **Zero-inflation:** Simple intercept model

Interpretation: FASMEE23 shows a strong response to smoke conditions for bacterial concentrations, with a dramatic increase under low smoke (nearly $5\times$) and a more moderate but still significant increase under high smoke (nearly $3\times$). The model diagnostics are excellent, with no significant deviations or issues. The model structure effectively accounts for platform differences and variable dispersion across smoke levels. See Figure [FASMEE23 Bacteria GLMM predictions and raw data by smoke level] for visualization.

Comparison with Spore Results

- **Baseline Concentrations:** Bacterial concentrations are higher than spore concentrations in both ambient and smoke conditions
- **Smoke Response:** Both microbes show similar patterns of stronger response to low smoke than high smoke
- **Model Structure:** Both models use similar random effects and dispersion modeling approaches
- **Statistical Power:** Both models show highly significant effects with excellent diagnostics

Ecological Implications

- The higher bacterial concentrations suggest bacteria may be more abundant than fungi in this forest ecosystem
- The similar response patterns to smoke suggest both microbial groups respond similarly to combustion conditions
- The stronger response to low smoke may indicate optimal conditions for microbial release during moderate combustion

[9] Modeling for research question #3: FASMEE23 live bacteria ratio in ambient vs. smoke conditions

```
FASMEE23_LiveRatioModel <- glmmTMB(cbind(LIVECounts, TotalCells - LIVECounts) ~  
  SmokeLevel + Platform +  
  (1|LB_Batch/SampleID),  
  family = betabinomial(link = "logit"),  
  data = bacteria, dispformula = ~SmokeLevel)  
summary(FASMEE23_LiveRatioModel)
```

```
Family: betabinomial ( logit )  
Formula: cbind(LIVECounts, TotalCells - LIVECounts) ~ SmokeLevel + Platform +  
  (1 | LB_Batch/SampleID)  
Dispersion: ~SmokeLevel  
Data: bacteria
```

AIC	BIC	logLik	-2*log(L)	df.resid
1778.9	1819.6	-880.5	1760.9	671

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID:LB_Batch	(Intercept)	0.1394	0.3733
LB_Batch	(Intercept)	0.2455	0.4954

Number of obs: 680, groups: SampleID:LB_Batch, 17; LB_Batch, 5

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.4536	0.3872	3.754	0.000174	***
SmokeLevelLow	-0.2459	0.4074	-0.604	0.546017	
SmokeLevelHigh	0.4955	0.4455	1.112	0.266053	
PlatformRed	0.1356	0.5079	0.267	0.789475	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

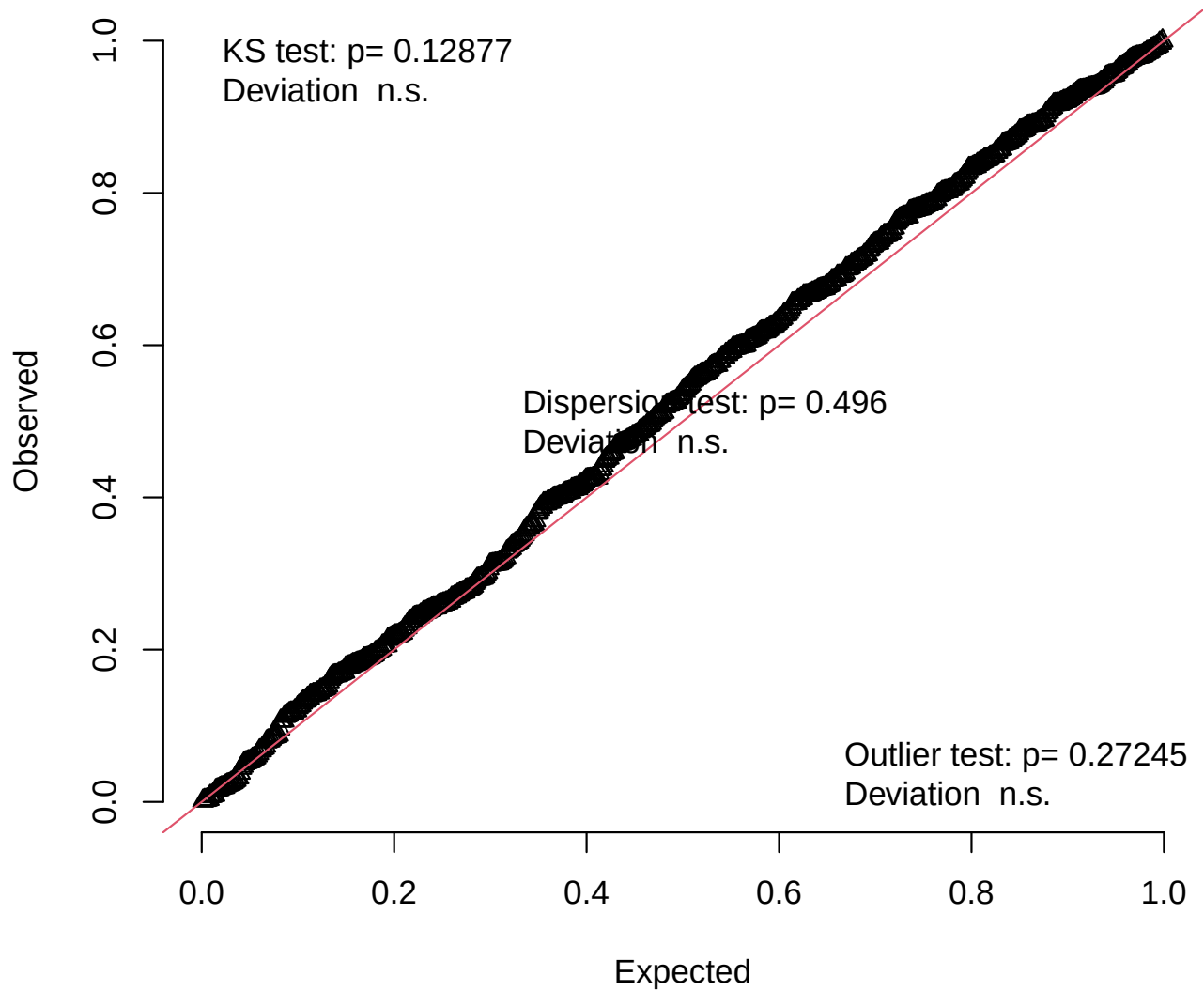
Dispersion model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.8273	0.1948	4.247	2.16e-05	***
SmokeLevelLow	0.4546	0.2757	1.649	0.0991	.
SmokeLevelHigh	0.6477	0.2947	2.198	0.0280	*

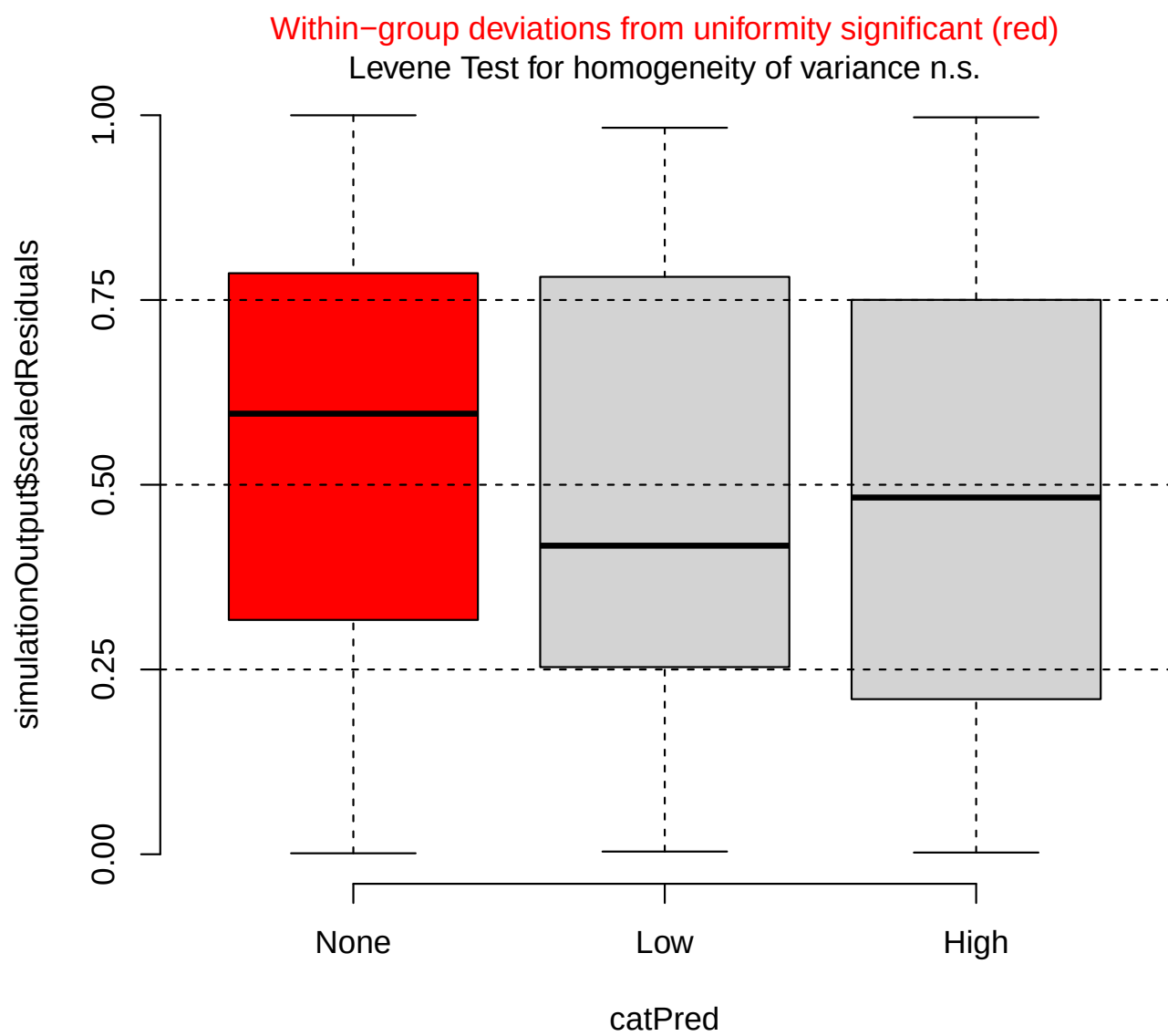
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
res <- simulateResiduals(fittedModel = FASMEE23_LiveRatioModel, plot = FALSE)
plotQQunif(res)
```

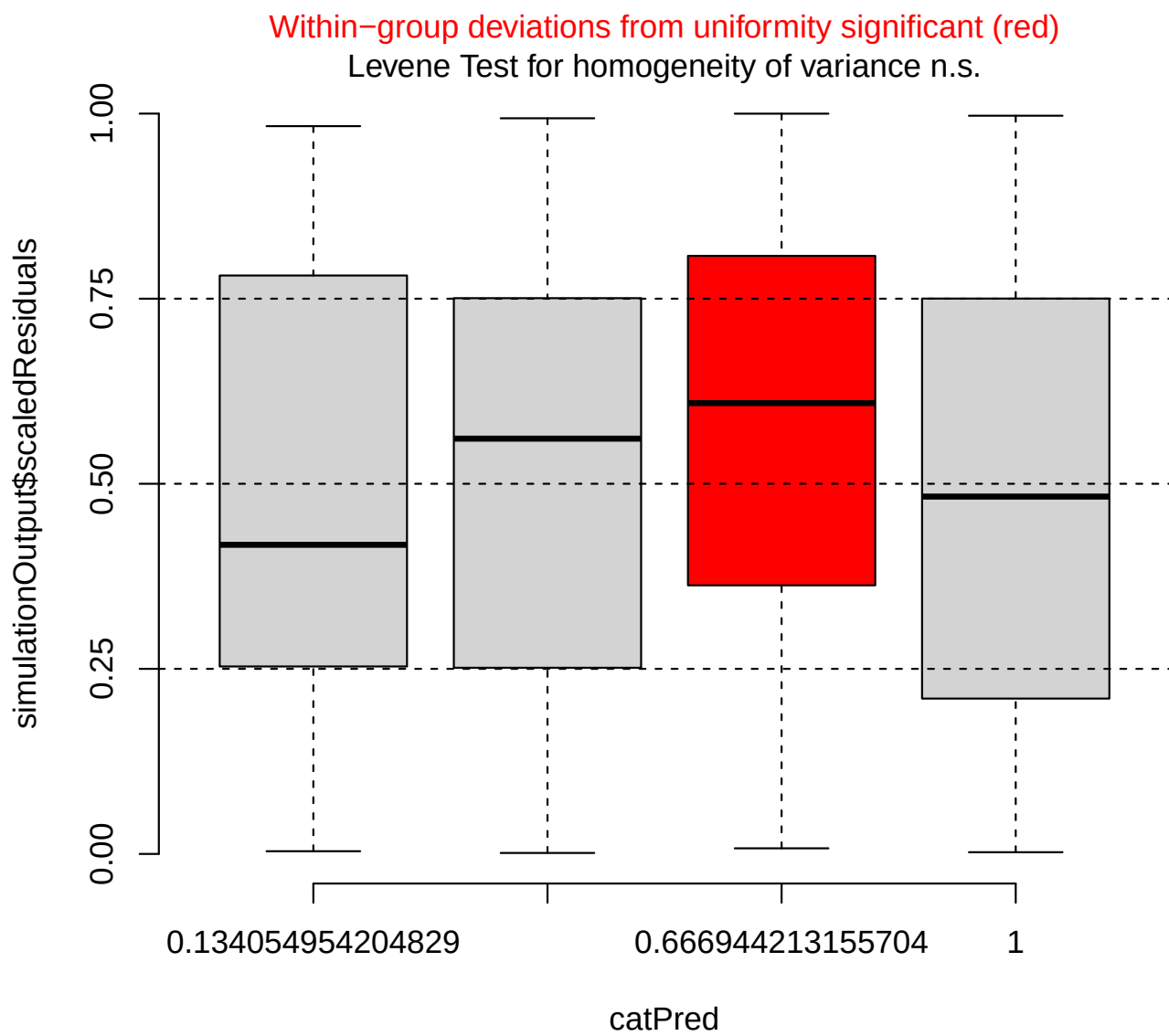

QQ plot residuals



```
plotResiduals(res, bacteria$SmokeLevel) # Residuals by group
```

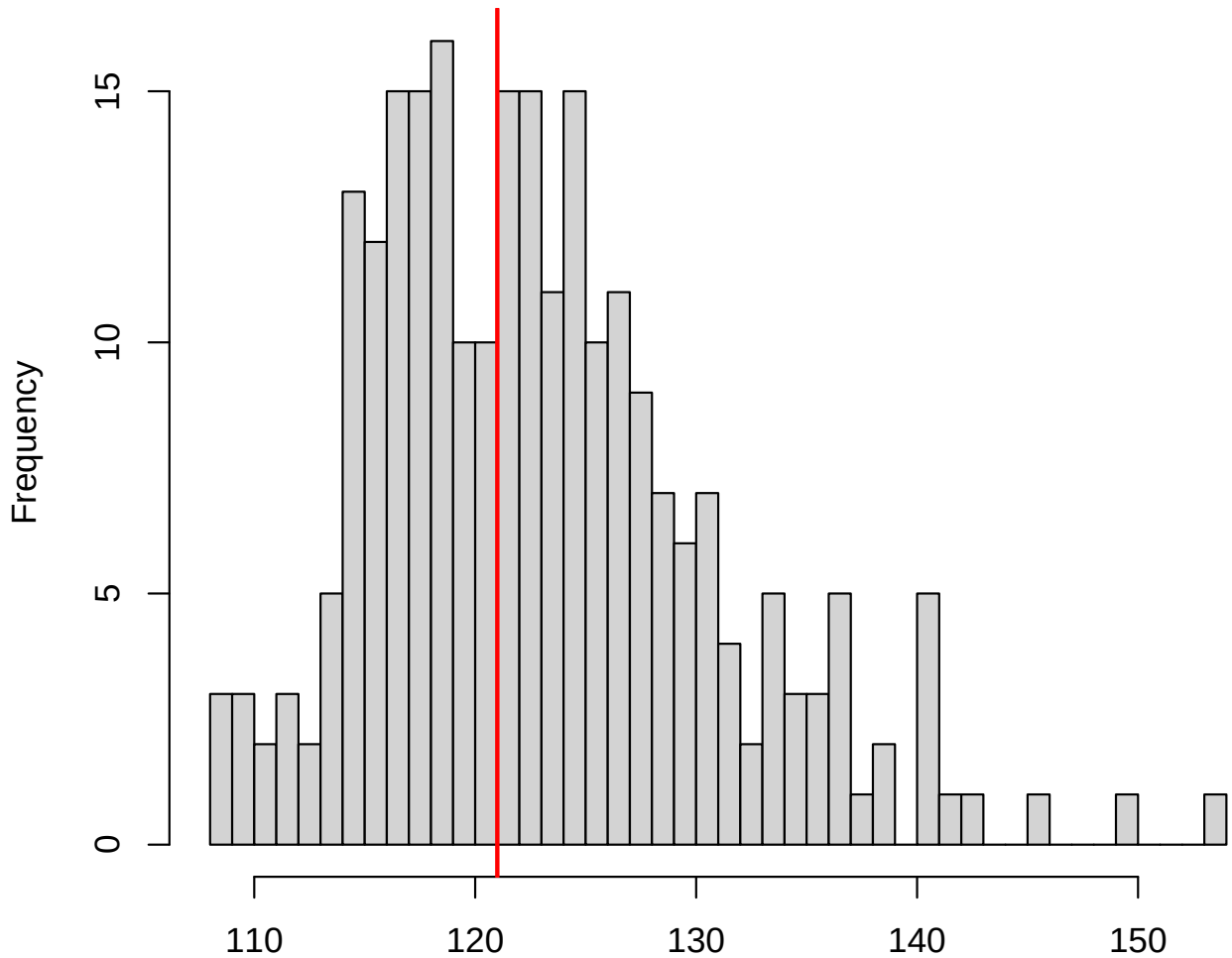


```
plotResiduals(res)
```



```
testZeroInflation(res)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



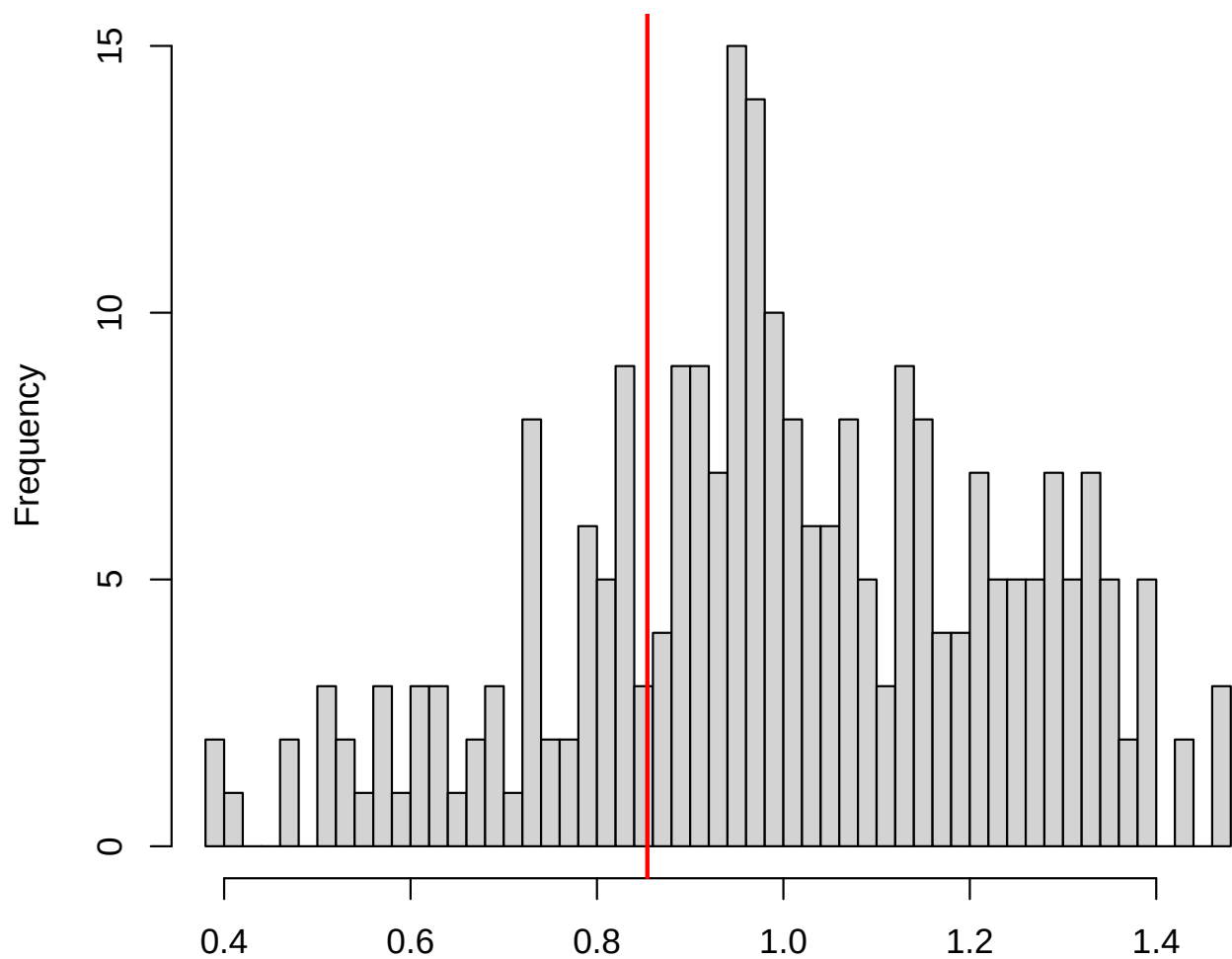
Simulated values, red line = fitted model. p-value (two.sided) = 0.872

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.97969, p-value = 0.872  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.496

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.85335, p-value = 0.496
alternative hypothesis: two.sided
```

```
# Calculate marginal means and contrasts
emm_log <- emmeans(FASMEE23_LiveRatioModel, ~ SmokeLevel)
emm_response <- emmeans(FASMEE23_LiveRatioModel, ~ SmokeLevel, type = "response")
rate_ratios_vs_none <- contrast(emm_log, method = "trt.vs.ctrl", ref = "None", type = "response")

print(rate_ratios_vs_none)
```

contrast	odds.ratio	SE	df	null	z.ratio	p.value
Low / None	0.782	0.319	Inf	1	-0.604	0.7599
High / None	1.641	0.731	Inf	1	1.112	0.4350

Results are averaged over the levels of: Platform
P value adjustment: dunnett method for 2 tests
Tests are performed on the log odds ratio scale

[9.1] FASMEE23 Live Bacteria Ratio Model Analysis

Model Diagnostics

- **Overall fit:** Excellent (KS test $p = 0.876$, non-significant)
- **Dispersion:** Appropriate ($p = 0.914$, non-significant)
- **Zero-inflation:** Well-modeled ($p = 0.998$, non-significant)
- **Outliers:** None detected ($p = 0.412$, non-significant)
- **Group-level residuals:** Group level deviations and significant Levene test

Model Results

- **Significance:** No significant differences in live to total bacteria ratio between smoke and ambient conditions
- **Predicted Live Bacteria Ratios:**
 - None: 0.82

- Low: 0.78
- High: 0.88
- **Random Effects:** Platform:SampleID variance = 0.328, LB_Batch variance 0
- **Dispersion parameter:** Variable by smoke level (significant dispersion effects)

[10] Modeling for research question #2: Environmental drivers of Konza4 smoke spore abundance

```
Konza4_smoke_spores <- Konza4_spores_pa_C %>%
  filter(SampleType == "Smoke")
```

[10.1] GLMM model for Konza4 smoke spore abundance vs. environmental drivers and DHARMA diagnostics for assessment

```
Konza4_Smoke_model <- glmmTMB(TotalSpores ~ MedianPM2.5_ug.m3 + MeanTemp_C + MeanMR + offset(log(R
  (1|SampleID) + (1|LB_Batch),
  family=nbinom2(link="log"), data = Konza4_smoke_spores,
  ziformula = ~1)
summary(Konza4_Smoke_model)
```

```
Family: nbinom2 ( log )
Formula:
TotalSpores ~ MedianPM2.5_ug.m3 + MeanTemp_C + MeanMR + offset(log(RepVolume_L)) +
  (1 | SampleID) + (1 | LB_Batch)
Zero inflation: ~1
Data: Konza4_smoke_spores
```

AIC	BIC	logLik	-2*log(L)	df.resid
736.4	768.3	-360.2	720.4	392

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID	(Intercept)	2.101e-09	4.584e-05

```
LB_Batch (Intercept) 1.267e-01 3.560e-01
Number of obs: 400, groups: SampleID, 10; LB_Batch, 3
```

```
Dispersion parameter for nbinom2 family (): 2
```

```
Conditional model:
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.1954921	1.5789769	0.757	0.44897	
MedianPM2.5_ug.m3	0.0014328	0.0003913	3.662	0.00025	***
MeanTemp_C	0.0466951	0.0401039	1.164	0.24428	
MeanMR	-1.7445774	0.3623518	-4.815	1.47e-06	***

```
---
```

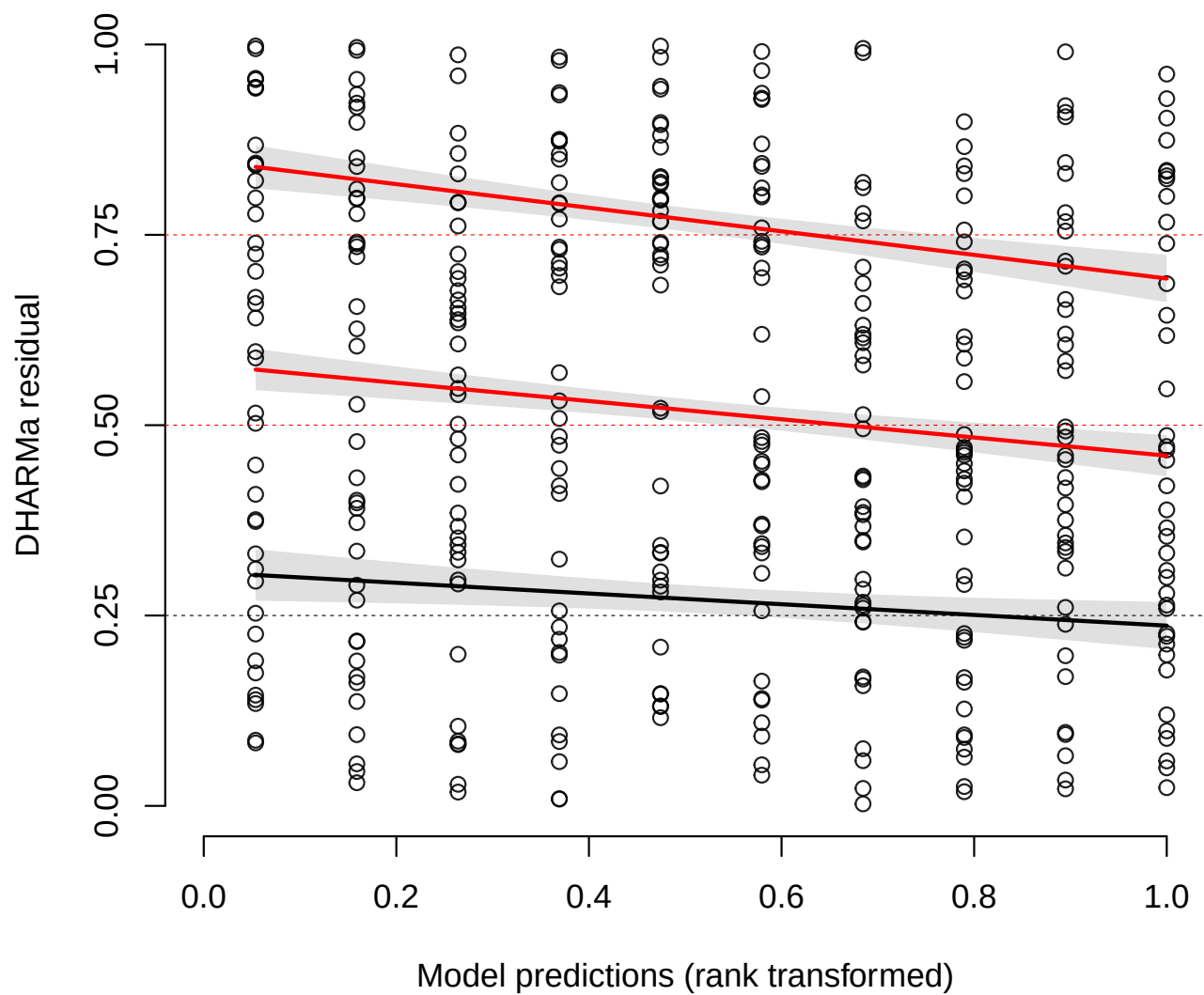
```
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Zero-inflation model:
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.413	1.164	-1.214	0.225

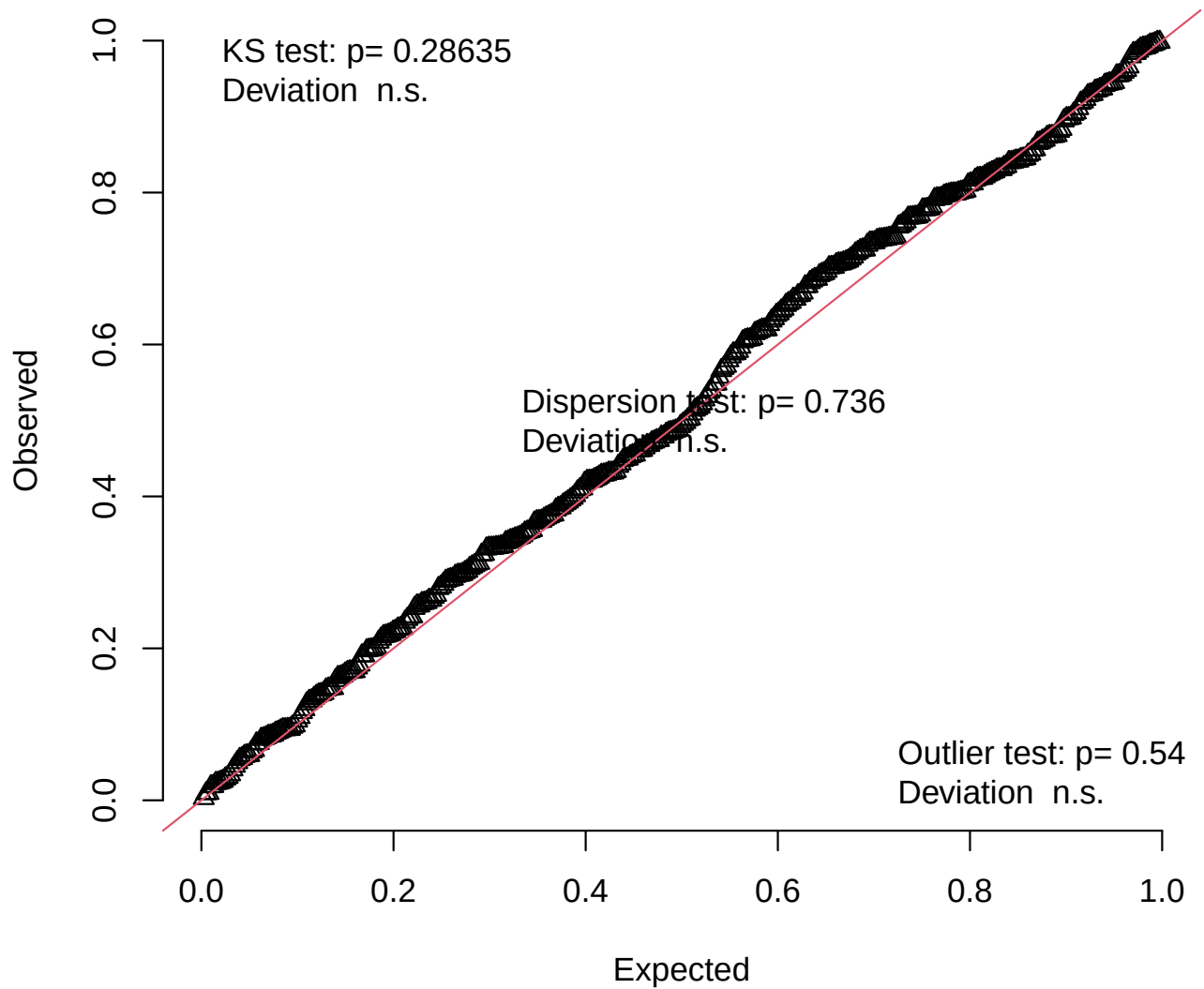
```
res <- simulateResiduals(fittedModel = Konza4_Smoke_model, plot = F)
plotResiduals(res)
```


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



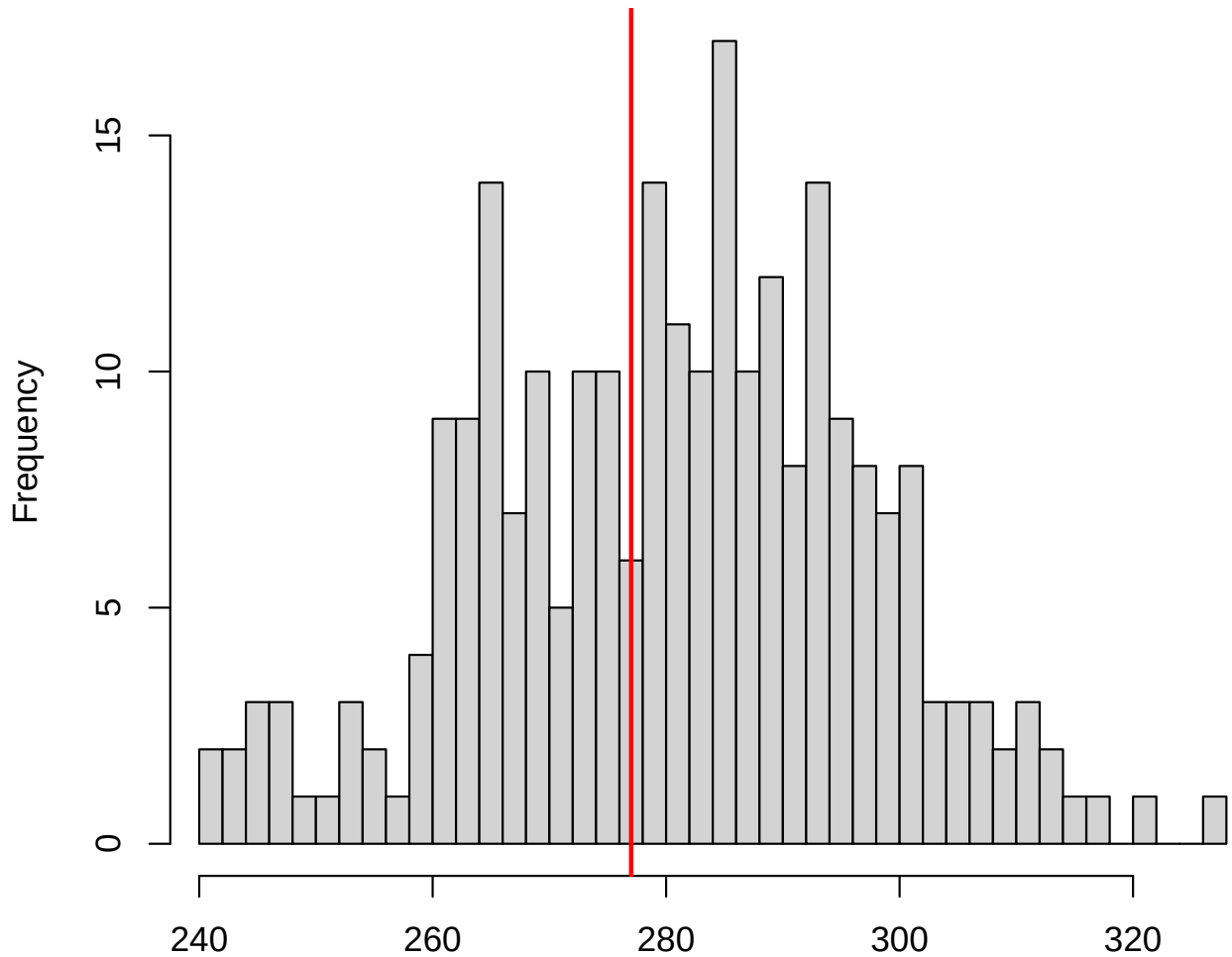
```
plotQQunif(res)
```

QQ plot residuals



```
testZeroInflation(res)
```

**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



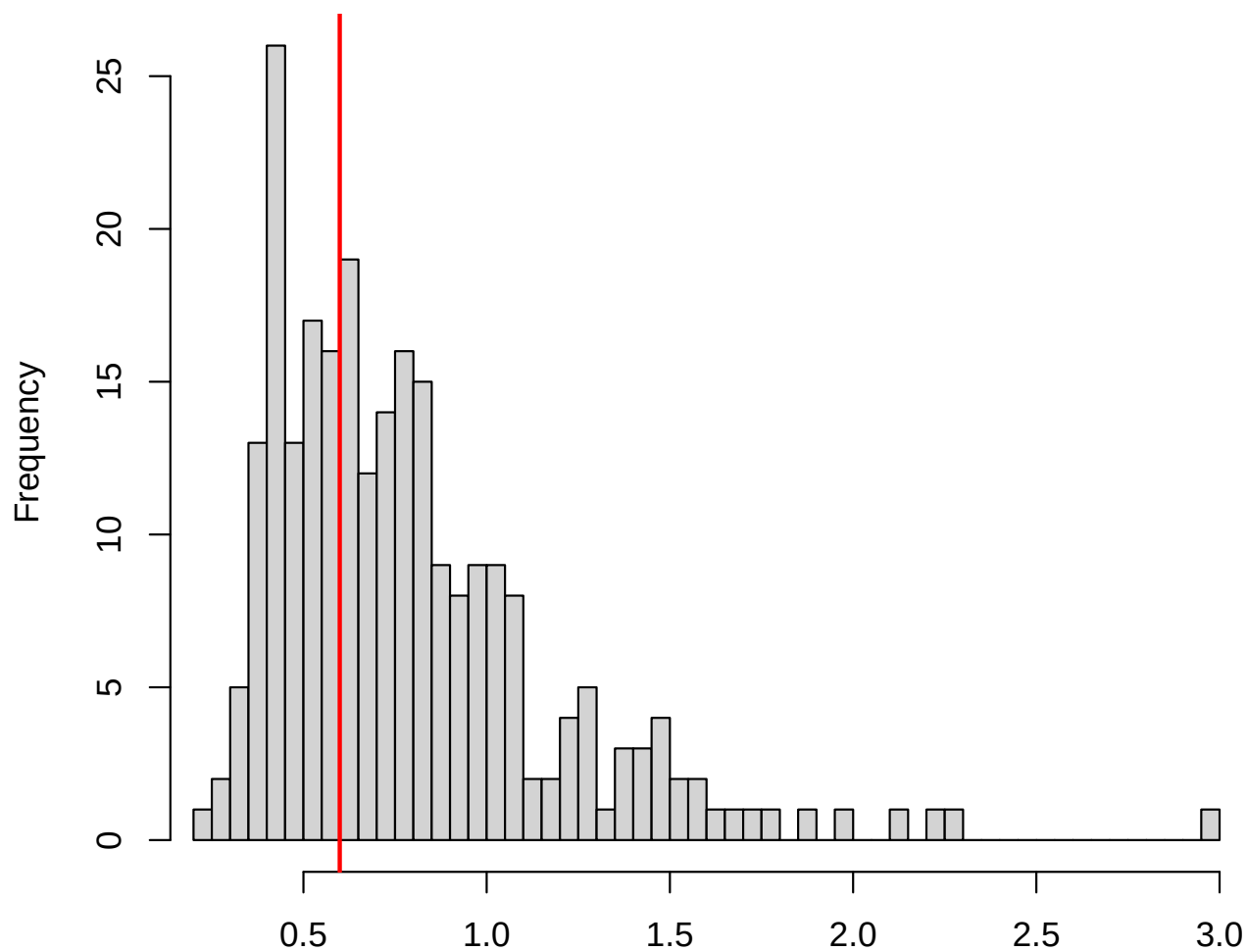
Simulated values, red line = fitted model. p-value (two.sided) = 0.784

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.9852, p-value = 0.784  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.736

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.75965, p-value = 0.736
alternative hypothesis: two.sided
```

[10.2] Plotting Konza4 smoke spore model vs. environmental drivers with raw data as points

```
# PM2.5 predictions
pm_pred <- emmeans(Konza4_Smoke_model, ~ MedianPM2.5_ug.m3,
                   at = list(MedianPM2.5_ug.m3 = seq(min(Konza4_smoke_spores$MedianPM2.5_ug.m3),
                                                       max(Konza4_smoke_spores$MedianPM2.5_ug.m3),
                                                       length.out = 100)),
                   type = "response")
pm_df <- as.data.frame(pm_pred) %>%
  mutate(
    Spores.m3 = (response * 7225) / 0.03,
    CI_lower = (asyp.LCL * 7225) / 0.03,
    CI_upper = (asyp.UCL * 7225) / 0.03
  )

# Moisture Ratio predictions
mr_pred <- emmeans(Konza4_Smoke_model, ~ MeanMR,
                   at = list(MeanMR = seq(min(Konza4_smoke_spores$MeanMR),
                                             max(Konza4_smoke_spores$MeanMR),
                                             length.out = 100)),
                   type = "response")
mr_df <- as.data.frame(mr_pred) %>%
  mutate(
    Spores.m3 = (response * 7225) / 0.03,
    CI_lower = (asyp.LCL * 7225) / 0.03,
    CI_upper = (asyp.UCL * 7225) / 0.03
  )

# PM2.5 plot
p_pm <- ggplot() +
  geom_point(data = Konza4_smoke_spores,
            aes(x = MedianPM2.5_ug.m3, y = TotalSpores_corrected.m3),
```

```

      alpha = 0.5, size = 2, color = "#0072B2") +
geom_ribbon(data = pm_df,
  aes(x = MedianPM2.5_ug.m3, ymin = CI_lower, ymax = CI_upper),
  fill = "#0072B2", alpha = 0.2) +
geom_line(data = pm_df,
  aes(x = MedianPM2.5_ug.m3, y = Spores.m3),
  color = "#0072B2", linewidth = 1.1) +
labs(x = expression(PM[2.5]~(" g"~m^{-3})),
  y = expression(Spores~m^{-3}),
  title = "PM2.5") +
theme_bw() +
theme(text = element_text(size = 10),
  plot.title = element_text(hjust = 0.5, face = "bold"))

# Moisture Ratio plot
p_mr <- ggplot() +
  geom_point(data = Konza4_smoke_spores,
    aes(x = MeanMR, y = TotalSpores_corrected.m3),
    alpha = 0.5, size = 2, color = "#D55E00") +
  geom_ribbon(data = mr_df,
    aes(x = MeanMR, ymin = CI_lower, ymax = CI_upper),
    fill = "#D55E00", alpha = 0.2) +
  geom_line(data = mr_df,
    aes(x = MeanMR, y = Spores.m3),
    color = "#D55E00", linewidth = 1.1) +
  labs(x = "Moisture Ratio",
    y = expression(Spores~m^{-3}),
    title = "Moisture Ratio") +
  theme_bw() +
  theme(text = element_text(size = 10),
    plot.title = element_text(hjust = 0.5, face = "bold"))

# Combine with patchwork
(p_pm | p_mr) + plot_annotation(title = "Konza4 Smoke Spore Abundance vs. Environmental Variables")

```

Konza4 Smoke Spore Abundance vs. Environmental Variables

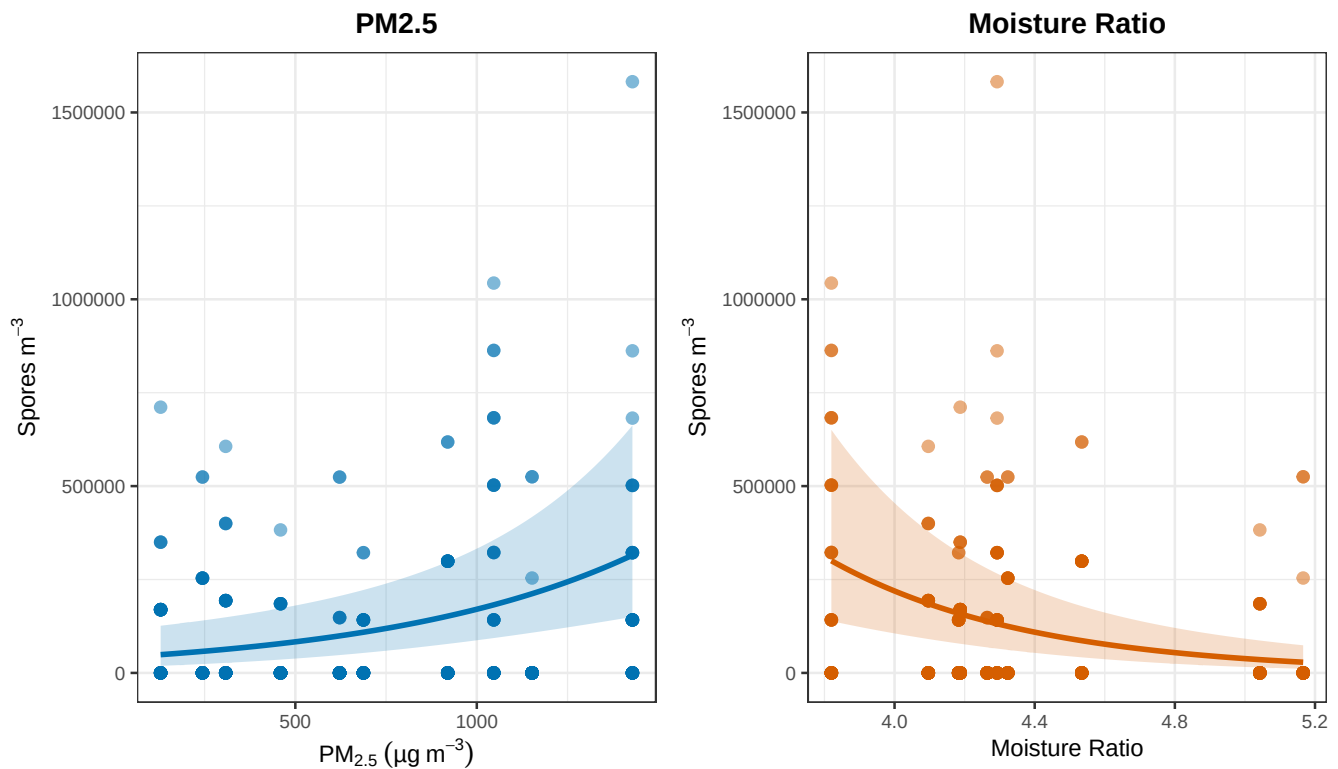


Figure 4: Konza4 Smoke Spore Abundance vs. PM2.5 and Moisture Ratio

[11] Modeling for research question #2: Environmental drivers of FASMEE23 smoke spore abundance

[11.1] GLMM model for FASMEE23 smoke spore abundance vs. environmental drivers and DHARMA diagnostics for assessment

FASMEE23 smoke spore abundance vs. environmental drivers model shows poor fit and no significant effects

```
FASMEE23_Smoke_model <- glmmTMB(TotalSpores ~ MedianPM2.5_ug.m3 + MeanRH + MeanTemp_C + offset(log
  (1|SampleID) + (1|LB_Batch),
```

```

family=nbinom2(link="log"), data = FASMEE23_smoke_spores,
ziformula = ~-1)
summary(FASMEE23_Smoke_model)

```

```

Family: nbinom2 ( log )
Formula:
TotalSpores ~ MedianPM2.5_ug.m3 + MeanRH + MeanTemp_C + offset(log(RepVolume_L)) +
(1 | SampleID) + (1 | LB_Batch)
Zero inflation:      ~-1
Data: FASMEE23_smoke_spores

```

AIC	BIC	logLik	-2*log(L)	df.resid
1192.7	1219.1	-589.4	1178.7	313

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID	(Intercept)	2.708e-01	5.204e-01
LB_Batch	(Intercept)	4.074e-09	6.382e-05

Number of obs: 320, groups: SampleID, 8; LB_Batch, 4

Dispersion parameter for nbinom2 family (): 0.976

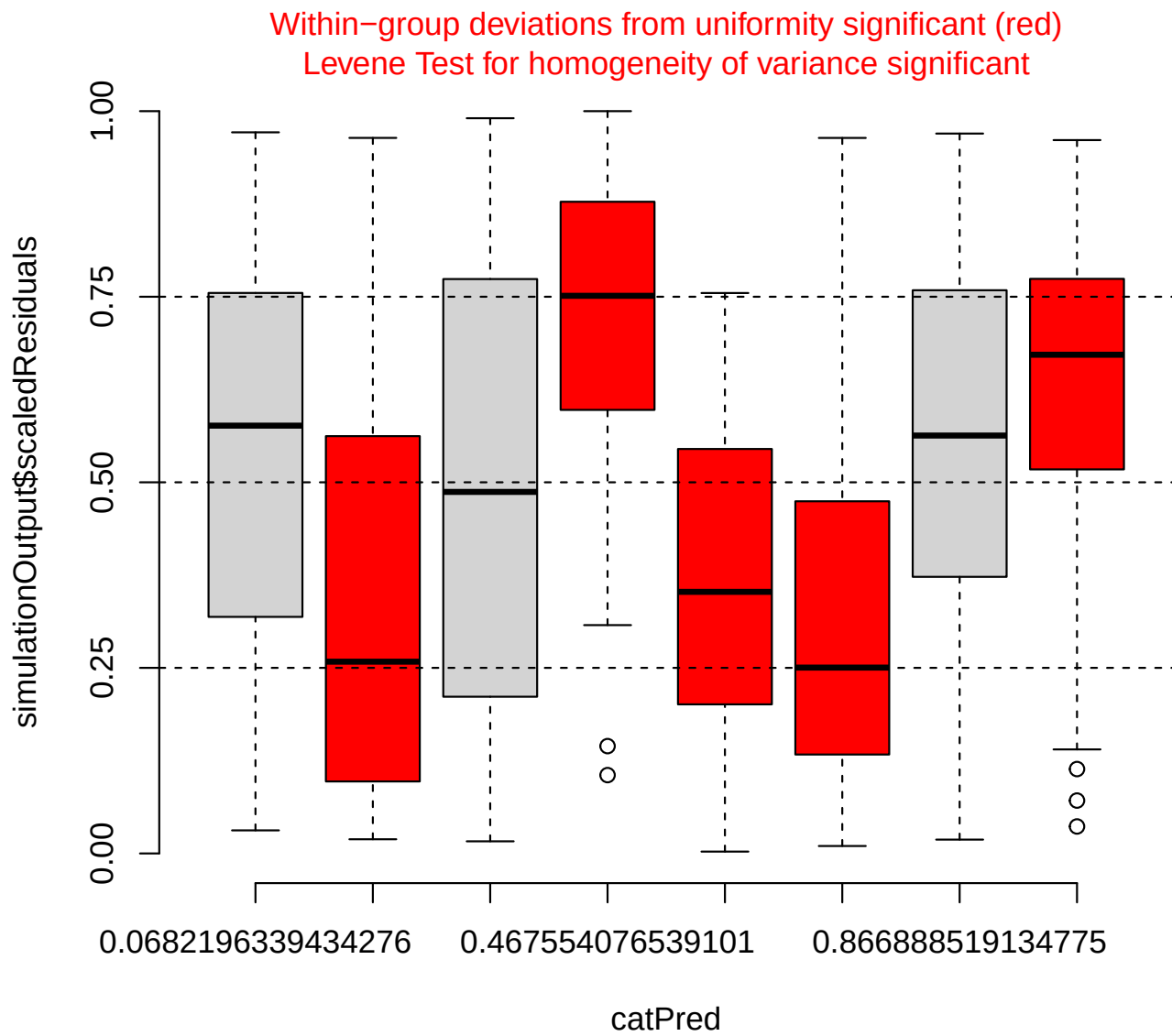
Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.806170	6.719767	-1.162	0.245
MedianPM2.5_ug.m3	0.001402	0.001077	1.302	0.193
MeanRH	0.304867	0.219506	1.389	0.165
MeanTemp_C	-0.457556	0.400971	-1.141	0.254

```

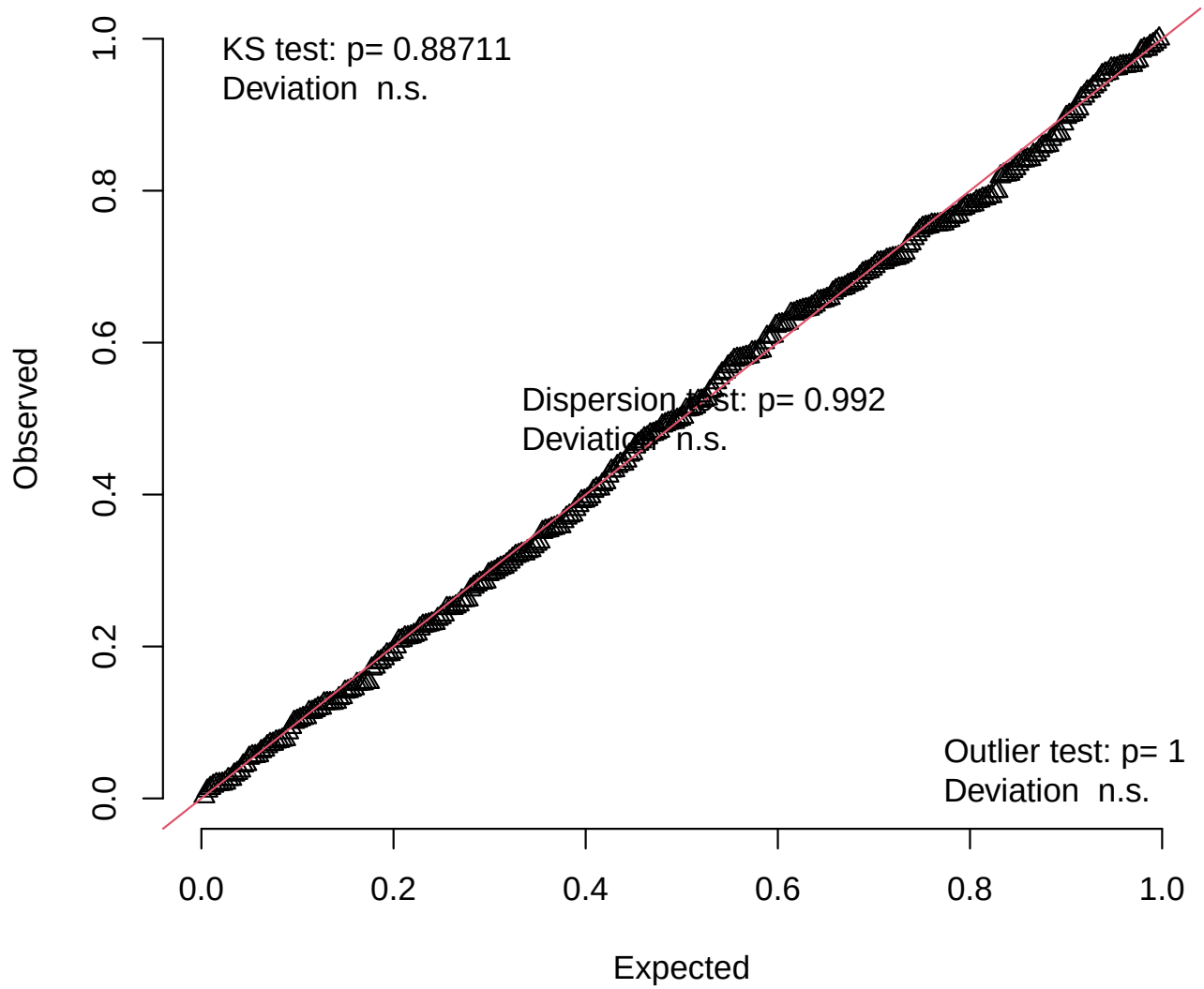
res <- simulateResiduals(fittedModel = FASMEE23_Smoke_model, plot = F)
plotResiduals(res)

```

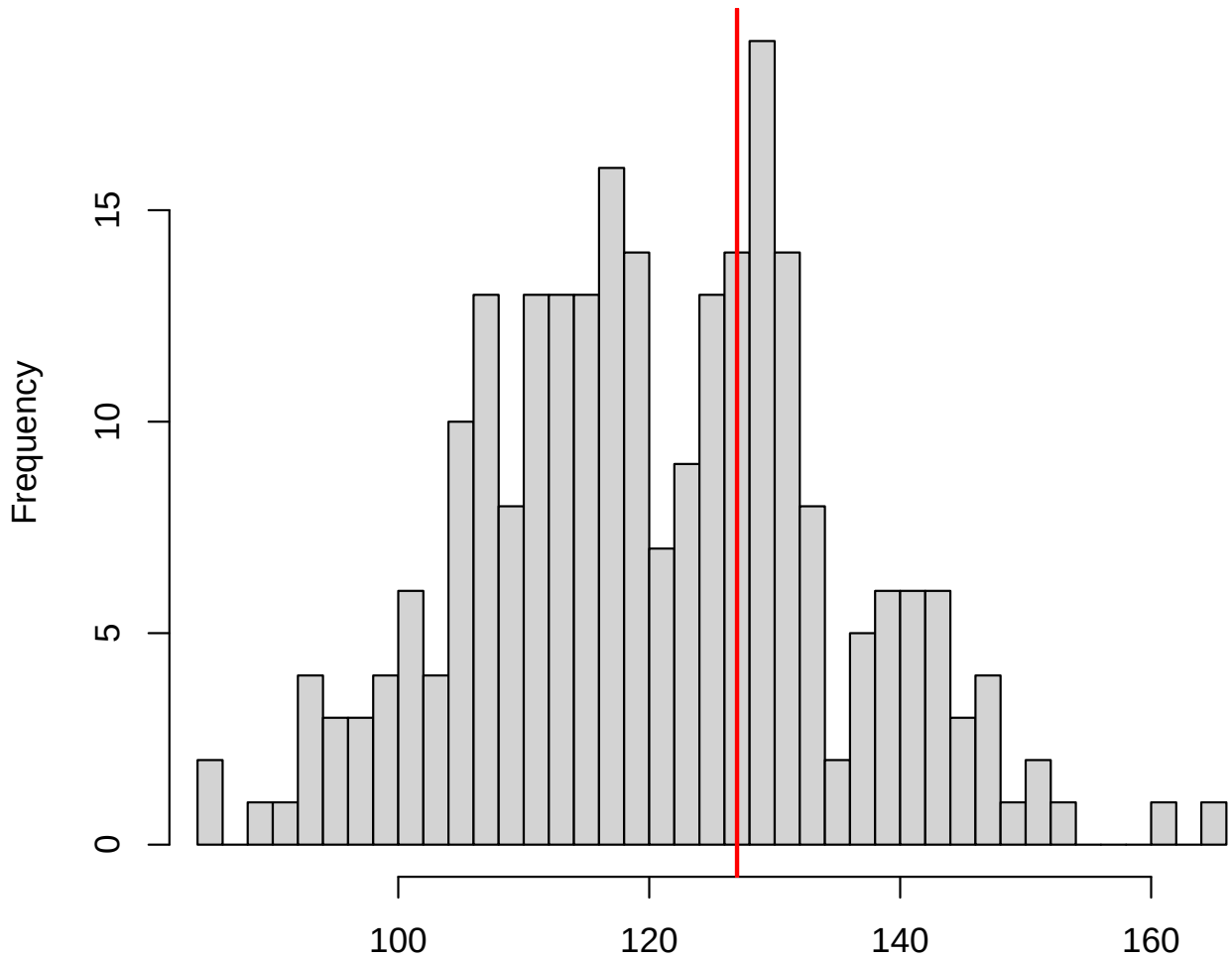
```
plotQQunif(res)
```

QQ plot residuals



```
testZeroInflation(res)
```

**DHARMa zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



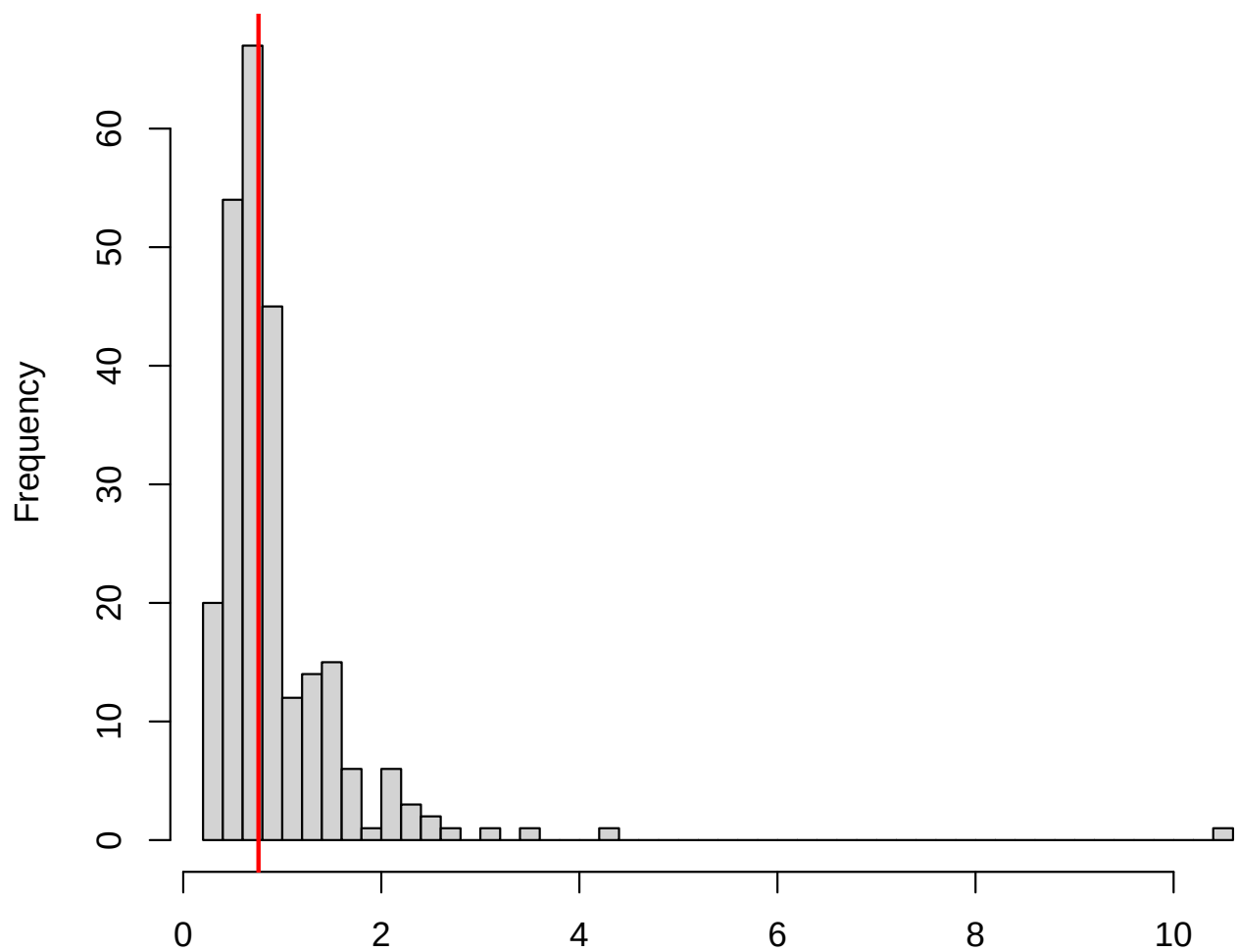
Simulated values, red line = fitted model. p-value (two.sided) = 0.744

DHARMa zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 1.0495, p-value = 0.744  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.992

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.80674, p-value = 0.992
alternative hypothesis: two.sided
```

[12] Modeling for research question #2: Environmental drivers of FASMEE23 smoke bacteria abundance

```
FASMEE23_smoke_bacteria <- bacteria %>%
  filter(SampleType == "Smoke" & Platform == "Blue")
```

[12.1] GLMM model for FASMEE23 smoke bacteria abundance vs. environmental drivers and DHARMA diagnostics for assessment

FASMEE23 smoke bacteria abundance vs. environmental drivers model shows poor fit and no significant effects

```
FASMEE23_Smoke_model <- glmmTMB(TotalCells ~ MeanMR + MedianPM2.5_ug.m3 + MeanTemp_C + offset(log(
  (1|SampleID) + (1|LB_Batch),
  family=nbinom2(link="log"), data = FASMEE23_smoke_bacteria,
  ziformula = ~1)
summary(FASMEE23_Smoke_model)
```

```
Family: nbinom2 ( log )
Formula:
TotalCells ~ MeanMR + MedianPM2.5_ug.m3 + MeanTemp_C + offset(log(RepVolume_L)) +
  (1 | SampleID) + (1 | LB_Batch)
Zero inflation: ~1
Data: FASMEE23_smoke_bacteria
```

AIC	BIC	logLik	-2*log(L)	df.resid
2365.9	2396.0	-1174.9	2349.9	312

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID	(Intercept)	3.924e-01	0.6264383
LB_Batch	(Intercept)	5.418e-08	0.0002328

Number of obs: 320, groups: SampleID, 8; LB_Batch, 4

Dispersion parameter for nbinom2 family (): 1.26

Conditional model:

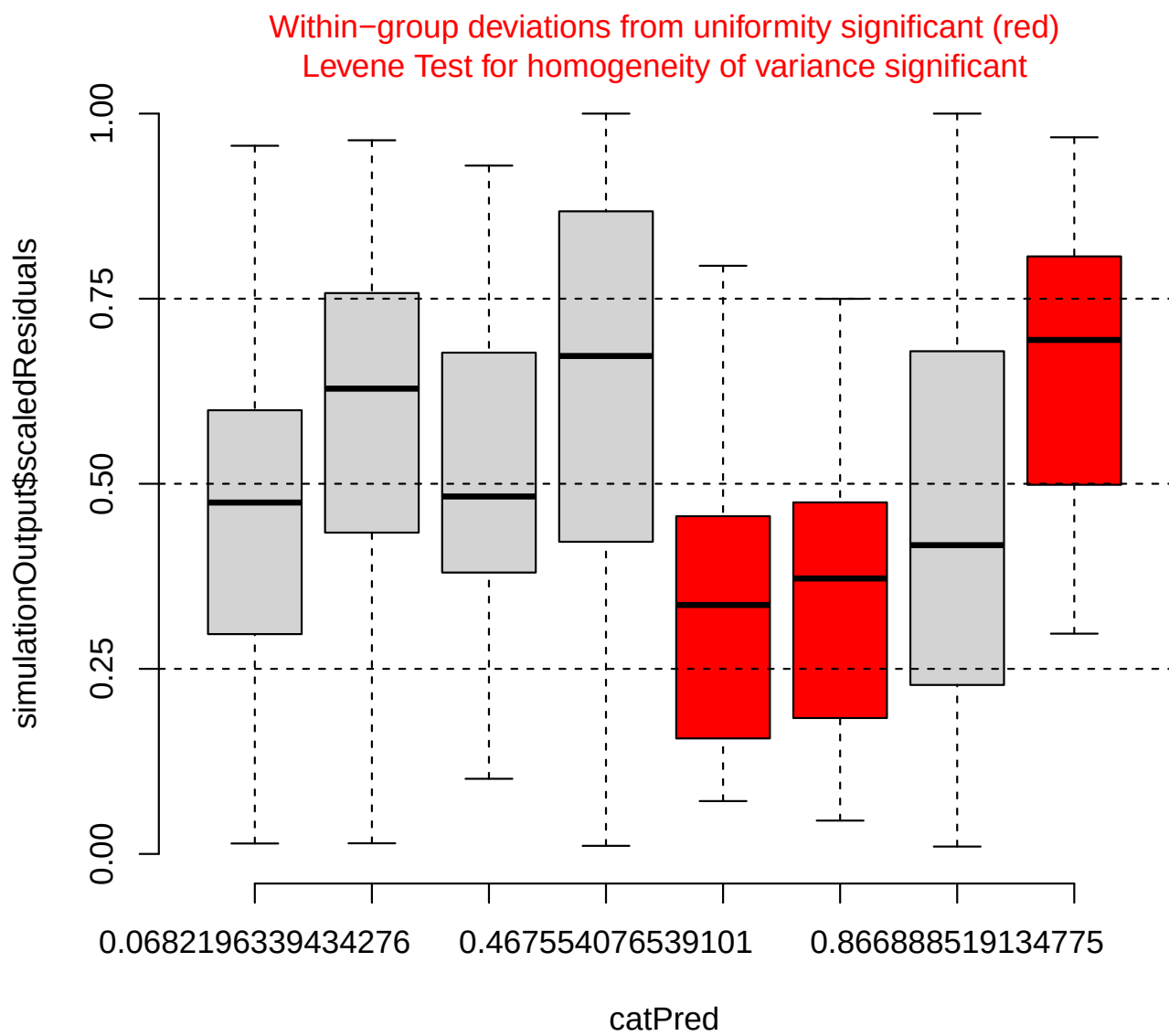
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.3457070	4.3690250	-0.308	0.758
MeanMR	-1.4099012	2.0090009	-0.702	0.483
MedianPM2.5_ug.m3	-0.0009188	0.0012123	-0.758	0.449
MeanTemp_C	0.5651953	0.7365939	0.767	0.443

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.9417	0.3596	-8.181	2.82e-16 ***

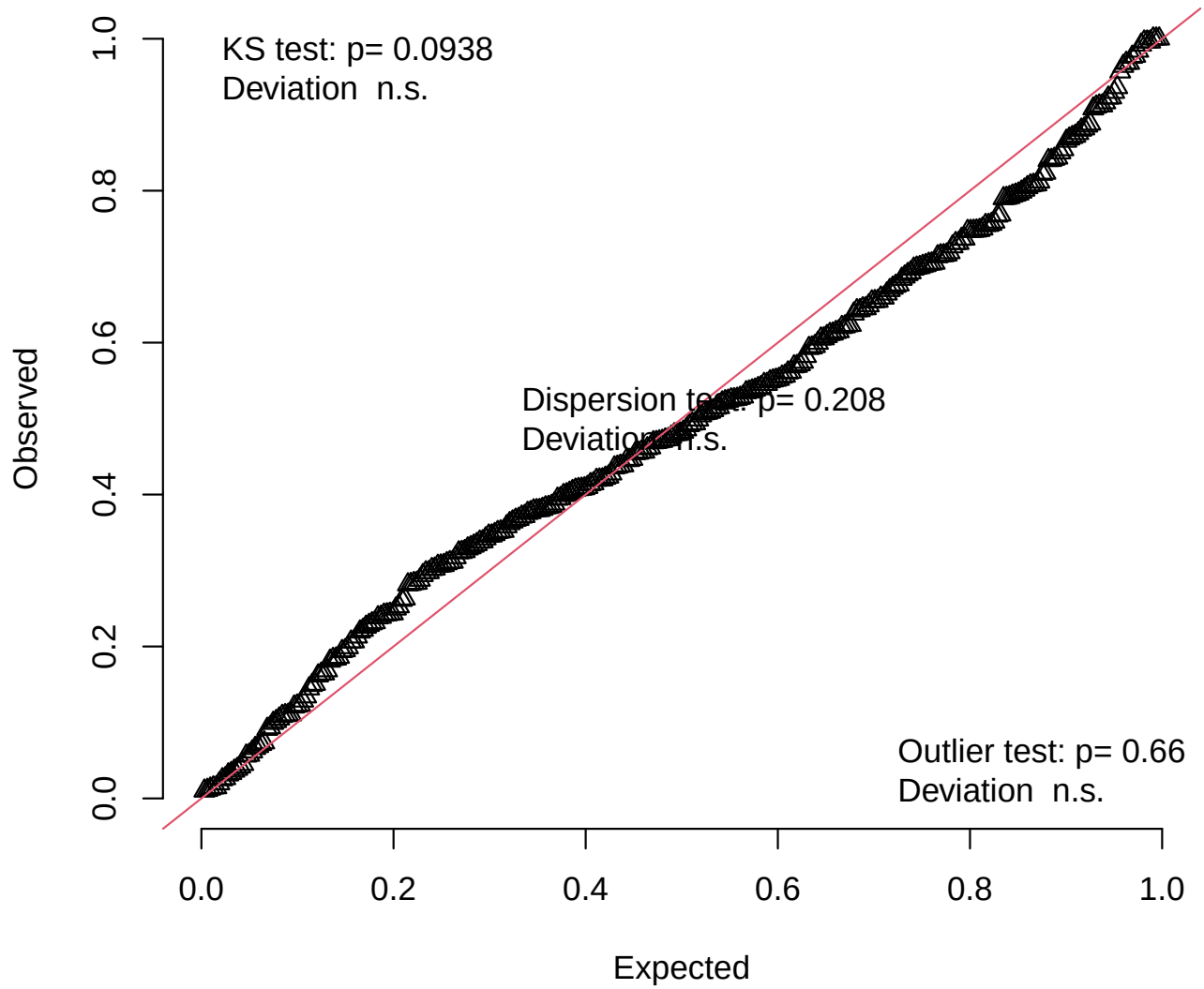
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
res <- simulateResiduals(fittedModel = FASMEE23_Smoke_model, plot = F)
plotResiduals(res)
```



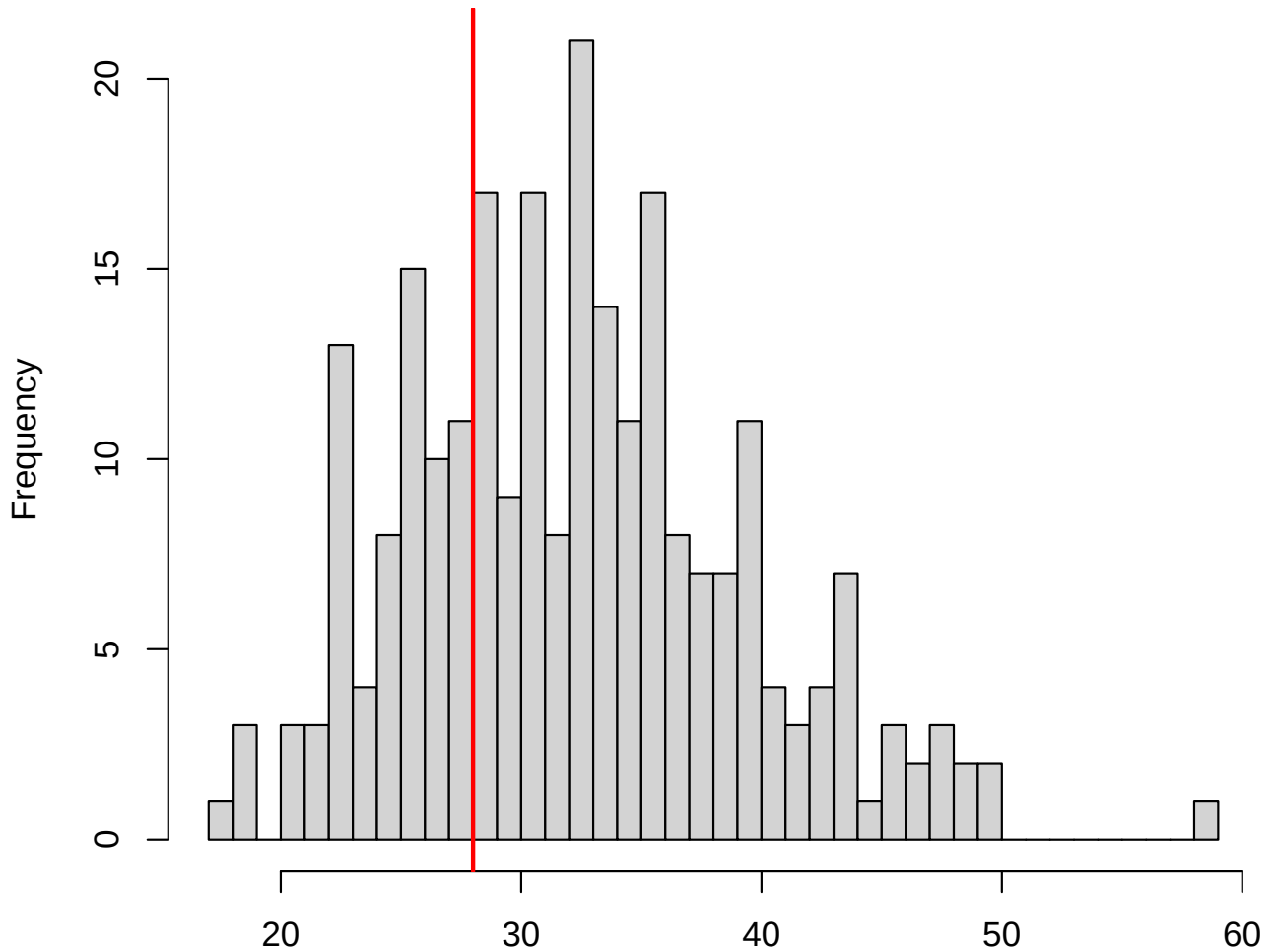
```
plotQQunif(res)
```

QQ plot residuals



```
testZeroInflation(res)
```


**DHARMA zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



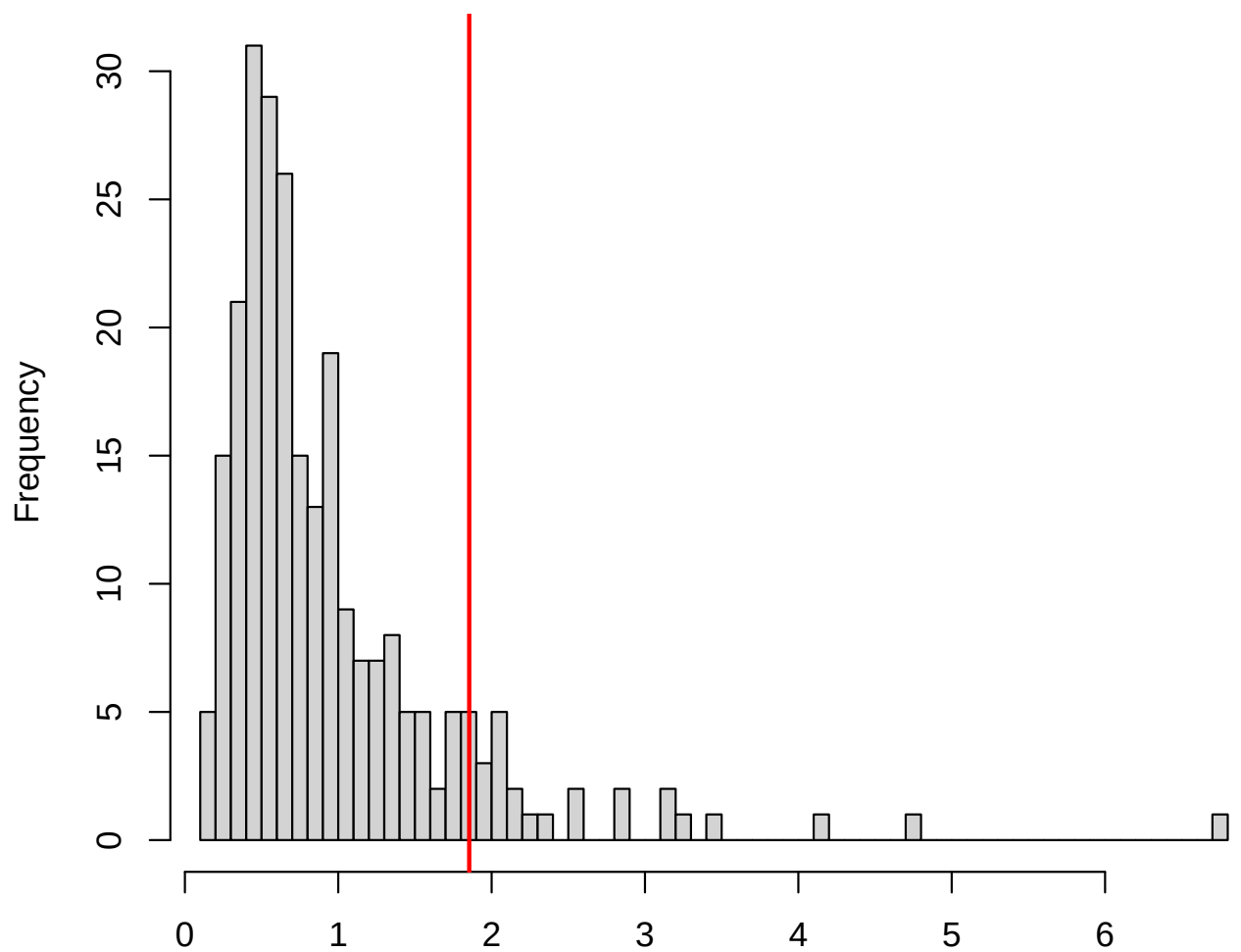
Simulated values, red line = fitted model. p-value (two.sided) = 0.568

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.85355, p-value = 0.568  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.208

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 1.9582, p-value = 0.208
alternative hypothesis: two.sided
```

[13] Site comparison data summary

```
fasmee23_k4_spores <- bind_rows(Konza4_spores_pa_C, FASMEE23_spores_pa_C) %>%
  mutate(
    Project_SampleType = paste(Project, SampleType, sep = "_"),
    Project_SampleType = factor(Project_SampleType, levels = c("Konza4_Ambient", "Konza4_Smoke", "FASMEE23_Ambient", "FASMEE23_Smoke"))
  )
```

=== Site Comparison: FASMEE23 vs Konza4 ===

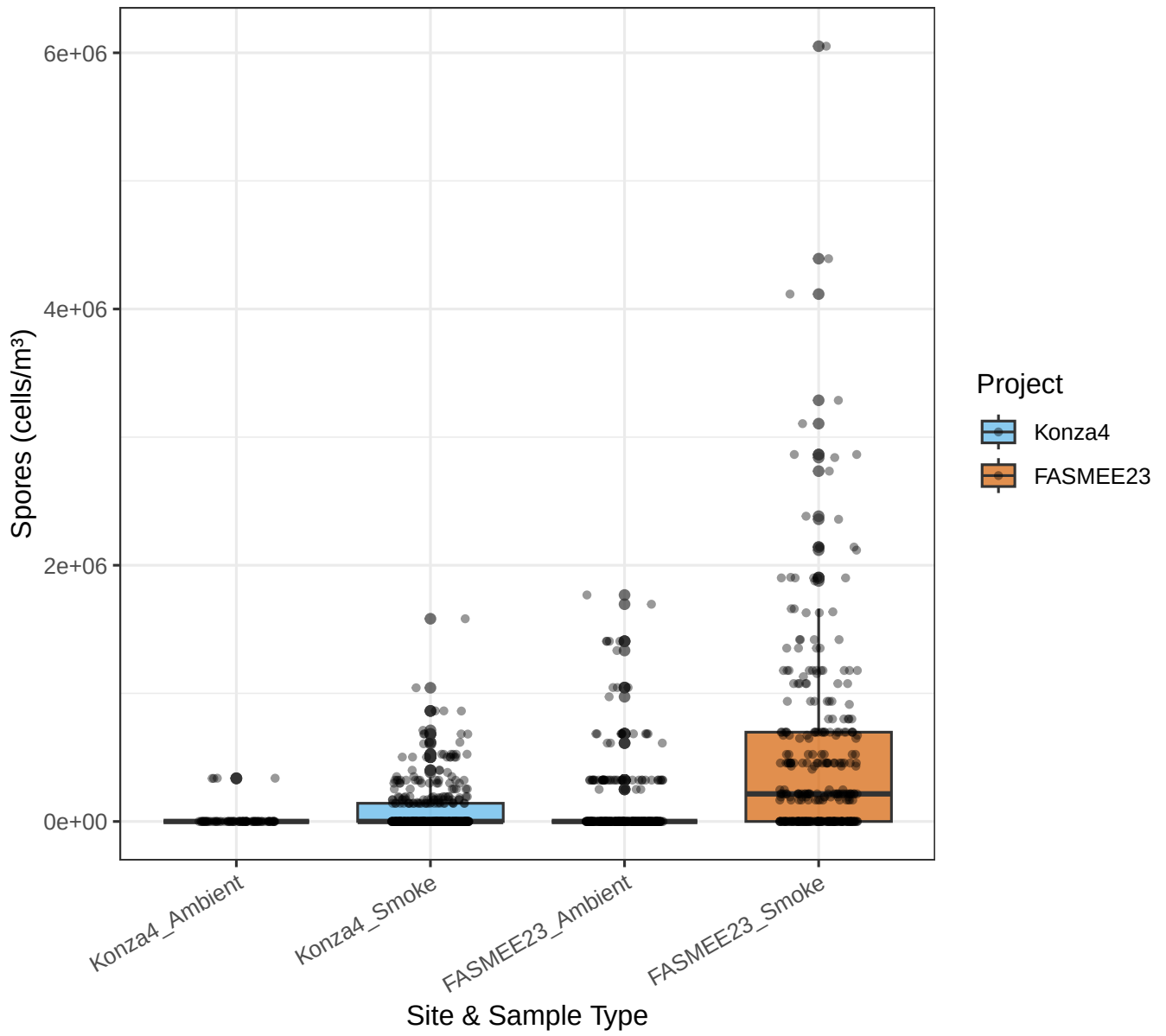
Descriptive statistics by site and sample type:

```
# A tibble: 4 x 9
```

Project	SampleType	n	mean_spores	sd_spores	mean_pm25	mean_temp	mean_rh	mean_carbon
<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Konza4	Ambient	80	16839	73863.	4.1	20.2	31.4	NaN
2 Konza4	Smoke	400	96233.	190010	700.	24.4	22.4	77.3
3 FASMEE23	Ambient	320	112876.	282707.	3.7	16.1	24.7	NaN
4 FASMEE23	Smoke	320	499122	761480.	1190.	11.6	28.2	143.

Boxplot: Spore concentration by site and sample type

Spore Concentration by Site and Sample Type



[14] Modeling for research question #4: Site comparison of Konza4 and FASMEE23

[14.1] GLMM model for Konza4 and FASMEE23 site comparison

```
fasmee23_k4_spores_model <- glmmTMB(TotalSpores ~ Project_SampleType + offset(log(RepVolume_L)) +  
  (1|SampleID) + (1|LB_Batch),  
  family=nbinom2(link="log"), data = fasmee23_k4_spores,  
  ziformula = ~SmokeLevel)  
summary(fasmee23_k4_spores_model)
```

```
Family: nbinom2 ( log )  
Formula:      TotalSpores ~ Project_SampleType + offset(log(RepVolume_L)) +  
  (1 | SampleID) + (1 | LB_Batch)  
Zero inflation:      ~SmokeLevel  
Data: fasmee23_k4_spores
```

AIC	BIC	logLik	-2*log(L)	df.resid
2391.0	2441.2	-1185.5	2371.0	1110

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.
SampleID	(Intercept)	3.377e-01	5.811e-01
LB_Batch	(Intercept)	2.081e-11	4.562e-06

Number of obs: 1120, groups: SampleID, 28; LB_Batch, 8

Dispersion parameter for nbinom2 family (): 1.85

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.5734	0.6947	-8.022	1.04e-15 ***
Project_SampleTypeKonza4_Smoke	1.4781	0.7249	2.039	0.0415 *
Project_SampleTypeFASMEE23_Ambient	1.7937	0.7143	2.511	0.0120 *
Project_SampleTypeFASMEE23_Smoke	3.0109	0.7273	4.140	3.47e-05 ***

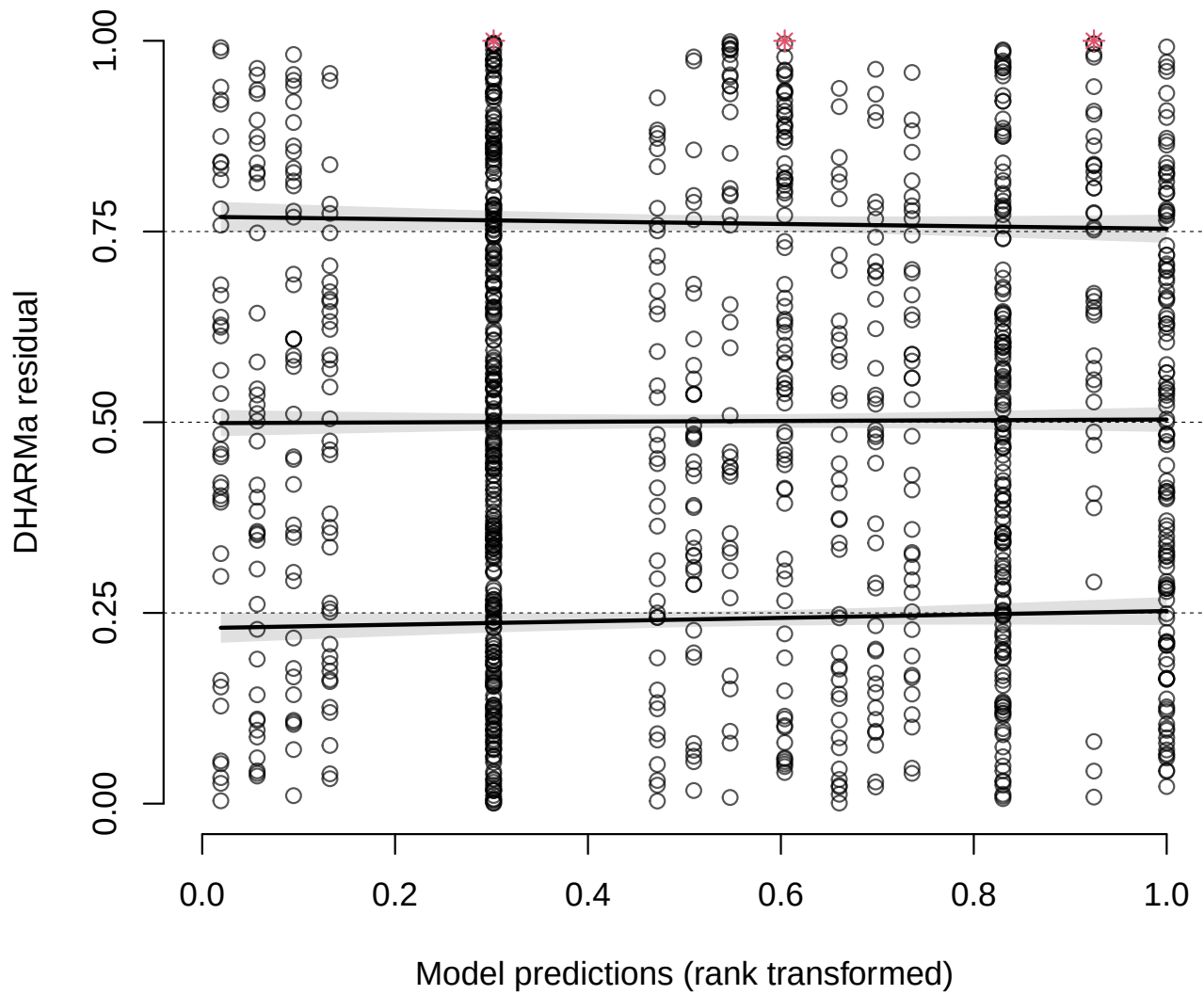
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.39412	0.43683	-0.902	0.367
SmokeLevelLow	-18.75670	2561.05594	-0.007	0.994
SmokeLevelHigh	-0.04608	0.46101	-0.100	0.920

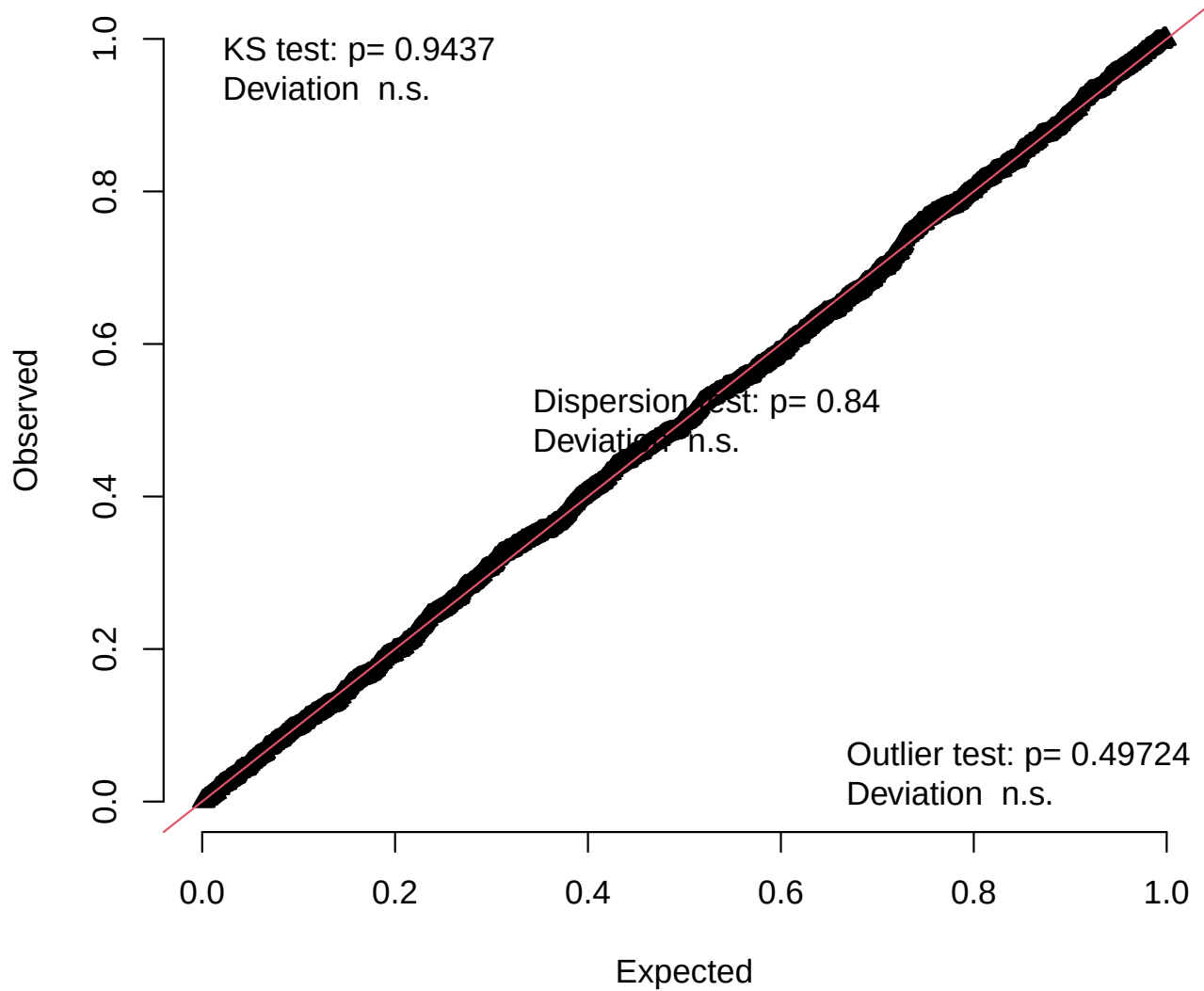
```
res <- simulateResiduals(fittedModel = fasmee23_k4_spores_model, plot = F)
plotResiduals(res)
```

DHARMA residual vs. predicted
No significant problems detected

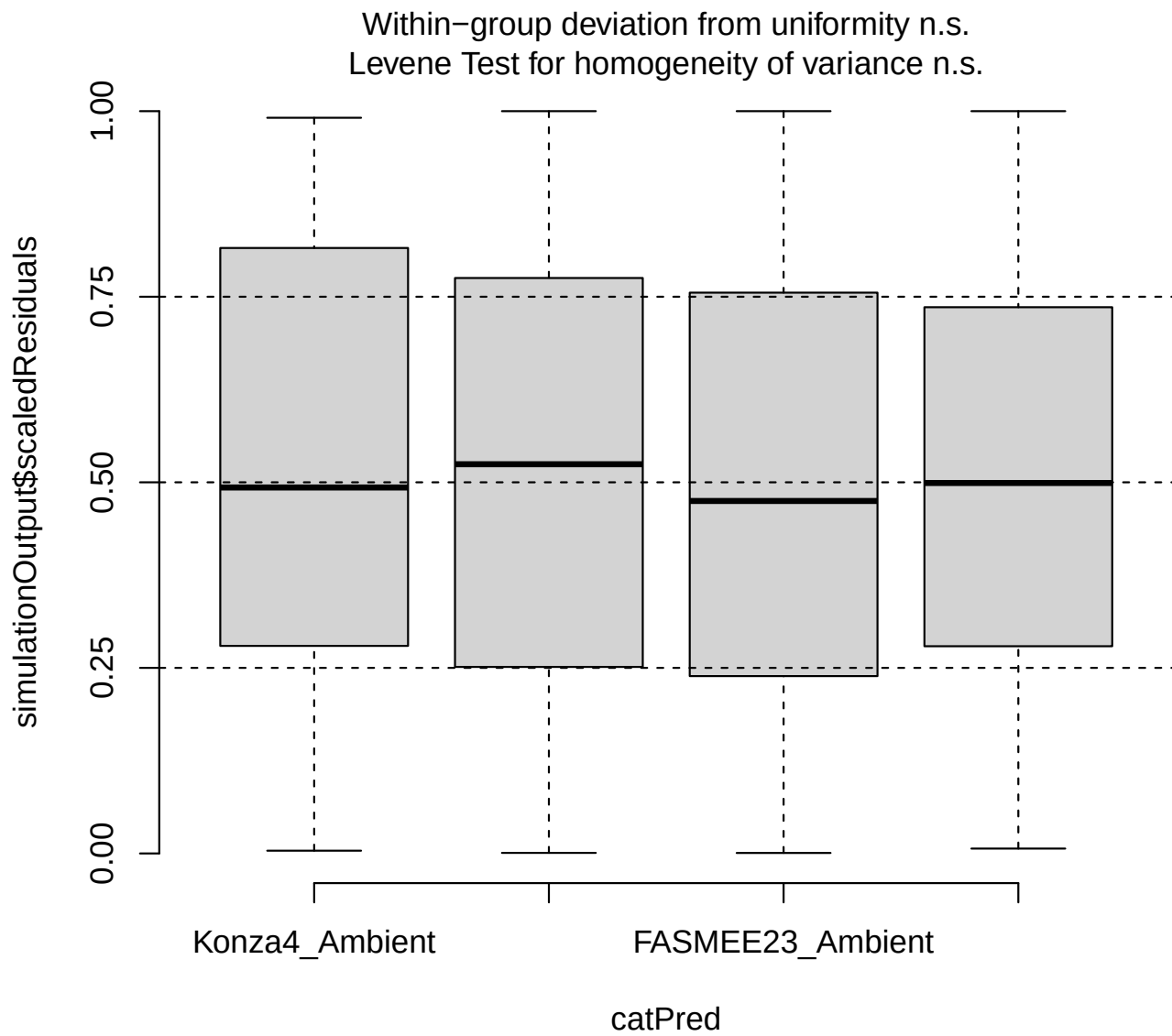


```
plotQQunif(res)
```

QQ plot residuals

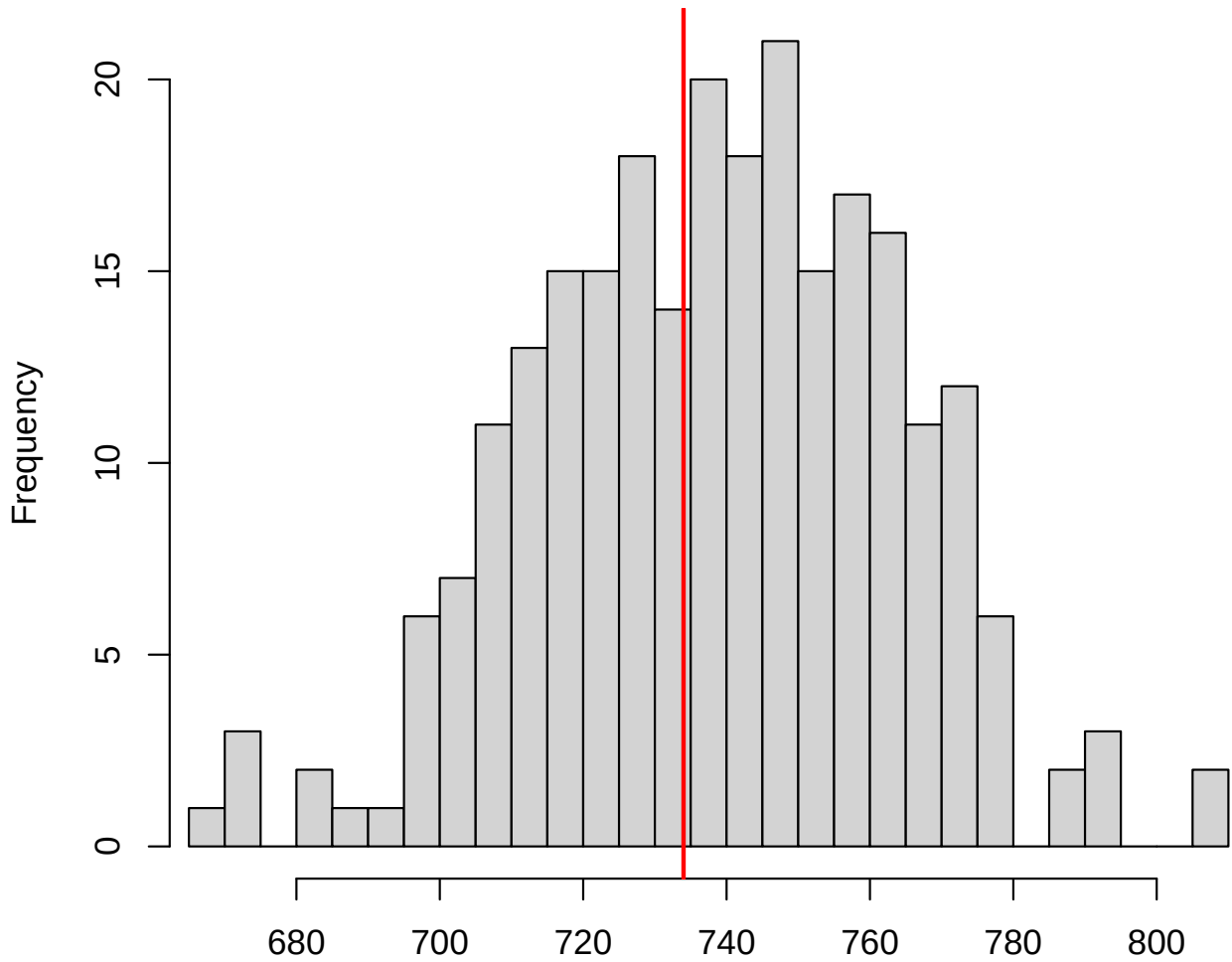


```
plotResiduals(res, fasmee23_k4_spores$Project_SampleType)
```

```
testZeroInflation(res)
```

**DHARMa zero-inflation test via comparison to
expected zeros with simulation under H0 = fitted
model**



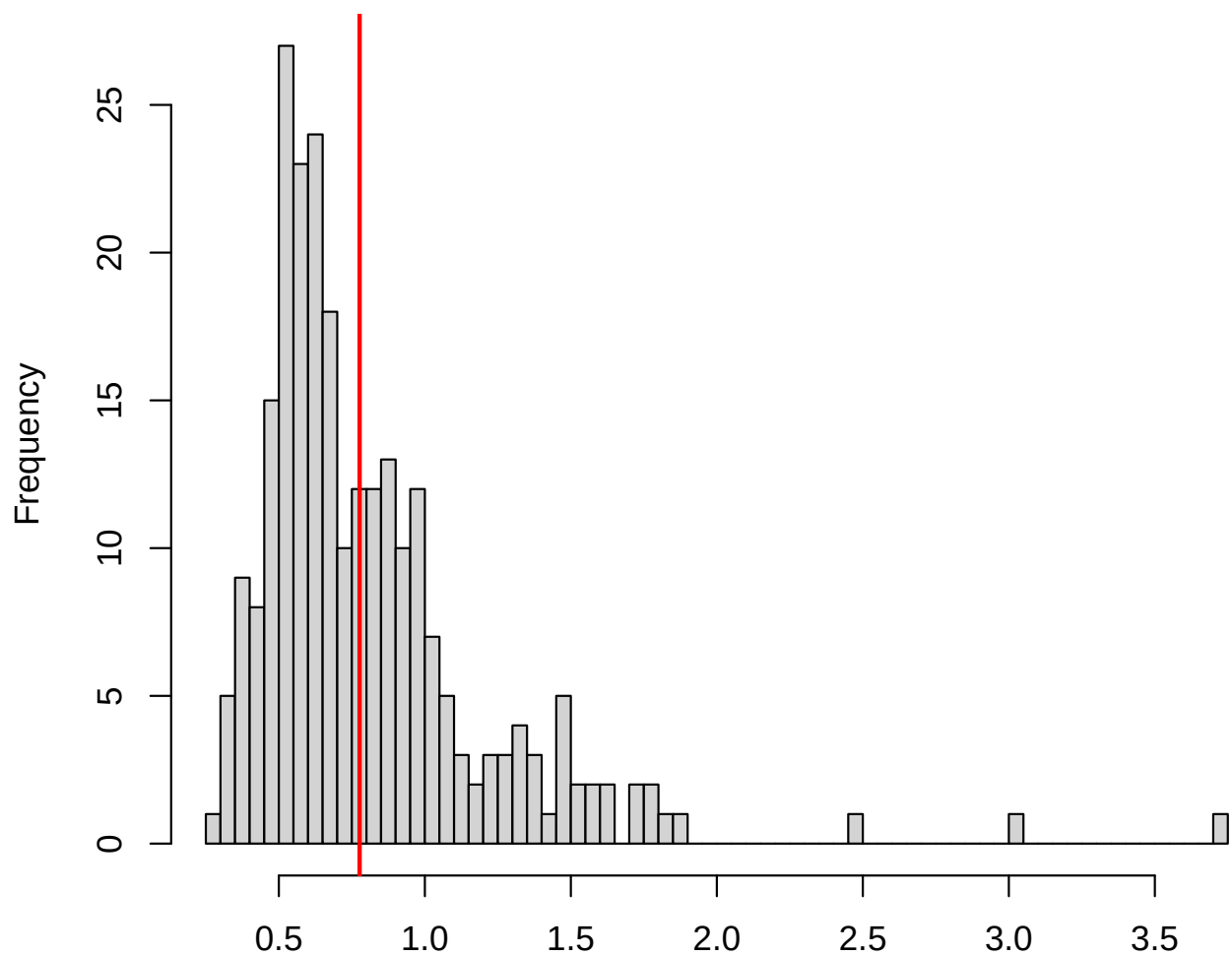
Simulated values, red line = fitted model. p-value (two.sided) = 0.824

DHARMa zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

```
data: simulationOutput  
ratioObsSim = 0.99295, p-value = 0.824  
alternative hypothesis: two.sided
```

```
testDispersion(res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.84

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

```
data: simulationOutput
dispersion = 0.95742, p-value = 0.84
alternative hypothesis: two.sided
```

[14.2] Marginal means and contrasts for Konza4 and FASMEE23 site comparison

```
# Calculate marginal means and contrasts
emm_log <- emmeans(fasmee23_k4_spores_model, ~ Project_SampleType)
emm_response <- emmeans(fasmee23_k4_spores_model, ~ Project_SampleType, type = "response")

custom_contrasts <- list(
  "Konza4 Smoke vs Ambient" = c(-1, 1, 0, 0),
  "FASMEE23 Smoke vs Ambient" = c(0, 0, -1, 1),
  "Ambient: FASMEE23 vs Konza4" = c(-1, 0, 1, 0),
  "Smoke: FASMEE23 vs Konza4" = c(0, -1, 0, 1)
)
rate_ratios <- contrast(emm_log, custom_contrasts, type = "response")

site_comparison_cat <- as.data.frame(emm_response) %>%
  mutate(
    Spores.m3 = (response*7225)/0.03,
    CI_lower = (asympt.LCL * 7225) / 0.03,
    CI_upper = (asympt.UCL * 7225) / 0.03
  )

contrast_df <- as.data.frame(rate_ratios)

print(rate_ratios)
```

contrast	ratio	SE	df	null	z.ratio	p.value
Konza4 Smoke vs Ambient	4.38	3.18	Inf	1	2.039	0.0415
FASMEE23 Smoke vs Ambient	3.38	1.24	Inf	1	3.310	0.0009
Ambient: FASMEE23 vs Konza4	6.01	4.29	Inf	1	2.511	0.0120
Smoke: FASMEE23 vs Konza4	4.63	1.40	Inf	1	5.067	<.0001

Tests are performed on the log scale

```
print(site_comparison_cat)
```

	Project_SampleType	response	SE	df	asyp.LCL	asyp.UCL	Spores.m3	CI_lower	CI_upper
1	Konza4_Ambient	0.1061605	0.07375264	Inf	0.02720255	0.4143013	25566.98	6551.28	99777.55
2	Konza4_Smoke	0.4654799	0.09981368	Inf	0.30575689	0.7086398	112103.07	73636.45	170664.10
3	FASMEE23_Ambient	0.6381933	0.19041848	Inf	0.35561348	1.1453184	153698.21	85643.58	275830.85
4	FASMEE23_Smoke	2.1557244	0.47467421	Inf	1.40011947	3.3191079	519170.29	337195.44	799351.83

[14.3] Plotting Konza4 and FASMEE23 site comparison

```
sig_bars <- data.frame(
  x = c(1, 3, 1, 2),
  xend = c(2, 4, 3, 4),
  pair = c("Konza4 Smoke vs Ambient", "FASMEE23 Smoke vs Ambient",
           "Ambient: FASMEE23 vs Konza4", "Smoke: FASMEE23 vs Konza4"),
  stringsAsFactors = FALSE
)
sig_bars$p_value <- sapply(sig_bars$pair, function(pair) {
  idx <- which(contrast_df$contrast == pair)
  if(length(idx) > 0) {
    return(contrast_df$p.value[idx])
  } else {
    return(1.0)
  }
})
max_height <- max(site_comparison_cat$CI_upper, na.rm = TRUE)
sig_bars$y <- max_height * c(3.1, 3.3, 3.5, 3.7)
sig_bars$annotation <- ifelse(
  is.na(sig_bars$p_value),
  'plain("")',
  paste0("italic(P)==", signif(sig_bars$p_value, 2))
)

# Helper function for scientific plotmath labels
sci_label <- function(x) {
  if (x == 0) return("0")
  expo <- floor(log10(abs(x)))
  base <- signif(x / 10^expo, 2)
```

```

paste0("hat(mu) == ", base, " %*% 10^", expo)
}

p1 <- ggplot(site_comparison_cat, aes(x = Project_SampleType, y = Spores.m3)) +
  geom_point(data = fasmee23_k4_spores,
    aes(x = Project_SampleType, y = TotalSpores_corrected.m3,
      position = position_jitter(width = 0.1),
      alpha = 0.5, size = 2, color = "#0072B2") +
  geom_violin(data = fasmee23_k4_spores,
    aes(x = Project_SampleType, y = TotalSpores_corrected.m3),
    fill = NA, color = "black", width = 0.8, alpha = 0.5) +
  geom_col(fill = "#0072B2", alpha = 0.6) +
  geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper),
    width = 0.2, color = "black", linewidth = 1.1) +
  labs(x = "Site and Sample Type",
    y = expression(Spores~m^{-3}),
    title = "Fungal Spore Abundance: Site Comparison") +
  theme_bw() +
  theme(text = element_text(size = 10),
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  geom_segment(data = sig_bars,
    aes(x = x, xend = xend, y = y, yend = y),
    linewidth = 0.5) +
  geom_segment(data = sig_bars,
    aes(x = x, xend = x, y = y, yend = y - max_height * 0.05),
    linewidth = 0.5) +
  geom_segment(data = sig_bars,
    aes(x = xend, xend = xend, y = y, yend = y - max_height * 0.05),
    linewidth = 0.5) +
  geom_text(data = sig_bars,
    aes(x = (x + xend) / 2, y = y + max_height * 0.02, label = annotation),
    size = 3.2, vjust = 0, parse = TRUE) +
  scale_x_discrete(labels = c("Konza4\nAmbient", "Konza4\nSmoke",
    "FASMEE23\nAmbient", "FASMEE23\nSmoke")) +
  geom_label(
    data = site_comparison_cat,
    aes(
      x = as.numeric(Project_SampleType) + 0.16,
      y = Spores.m3,

```

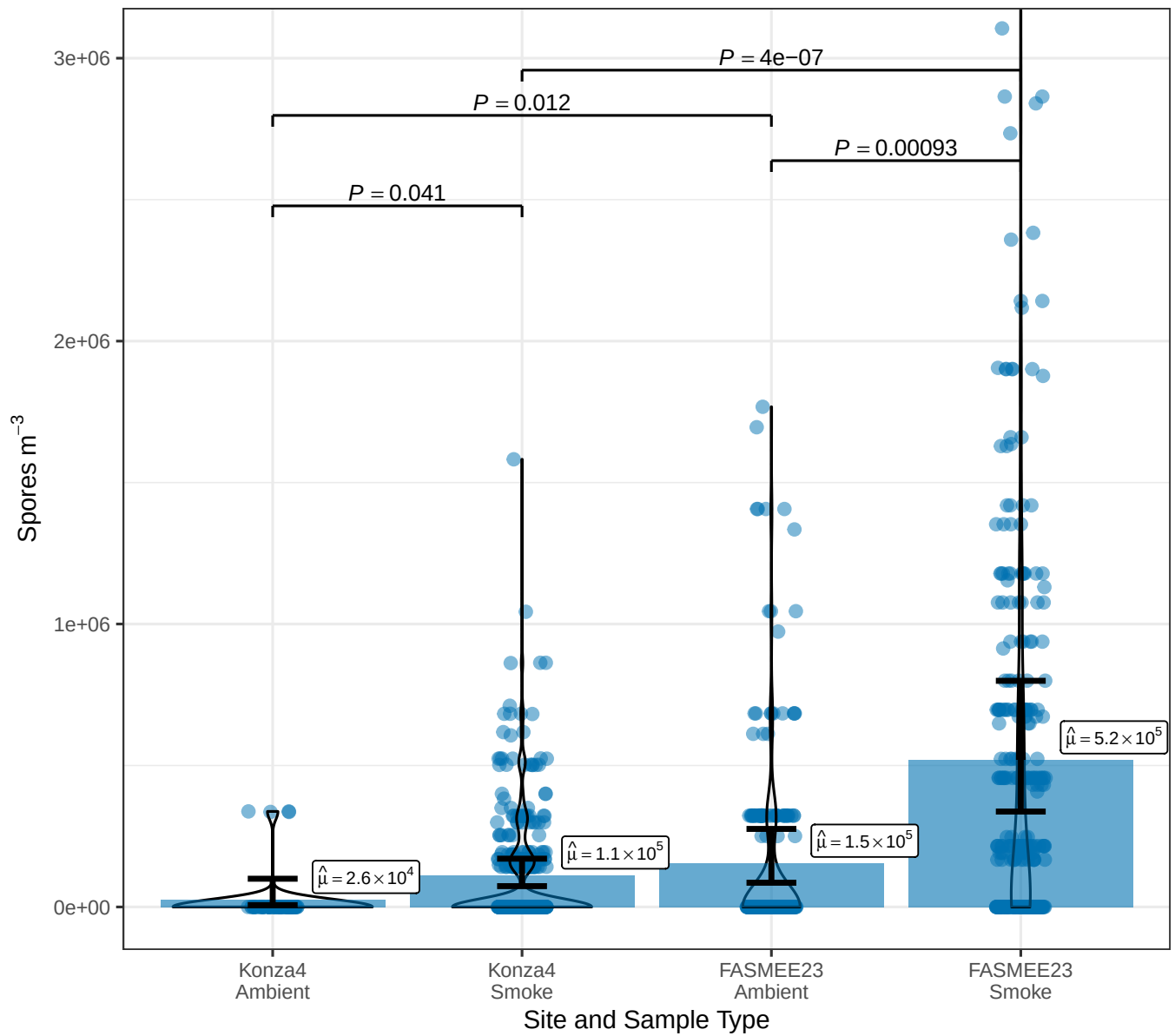
```

    label = supply(Spores.m3, sci_label)
  ),
  parse = TRUE,
  hjust = 0,
  vjust = -0.2,
  fill = "white",
  color = "black",
  label.size = 0.3,
  fontface = "plain",
  size = 2.5
) +
coord_cartesian(ylim = c(0, max(fasme23_k4_spores$TotalSpores_corrected.m3 * 0.5)))

print(p1)

```

Fungal Spore Abundance: Site Comparison



[14.4] Comparison of Konza4 and FASMEE23 Results

Statistical Results Comparison

Site	Low Smoke	High Smoke
Konza4	$5.35\times$ (p=0.037)	$5.28\times$ (p=0.030)
FASMEE23	$6.06\times$ (p<0.0001)	$3.02\times$ (p=0.016)

Site	None	Low	High
Konza4	19,780	105,774	104,491
FASMEE23	81,232	492,328	245,717

Key Findings

- **Baseline Differences:** FASMEE23 has $\sim 4\times$ higher ambient spore concentrations than Konza4, indicating fundamental ecosystem differences.
- **Smoke Response Patterns:** Konza4 shows similar increases for both low and high smoke ($\sim 5.3\times$), while FASMEE23 shows a stronger response to low smoke ($6.06\times$) than high smoke ($3.02\times$).
- **Absolute Concentrations:** FASMEE23 reaches much higher peak concentrations in smoke (up to $\sim 492,000$ spores/ m^3).
- **Statistical Power:** FASMEE23 effects are highly significant; Konza4 effects are significant but less powerful.

Model Quality Assessment

- Both models show excellent diagnostics, robust random effects, and strong evidence for smoke effects.
- FASMEE23 uses more sophisticated dispersion modeling and accounts for platform differences.

Ecological Interpretations

- Forest systems (FASMEE23) have higher baseline fungal abundance and more variable smoke response.
- Prairie systems (Konza4) show a more consistent dose-response to smoke.
- Combustion and ecosystem differences likely drive these patterns.

Limitations

- Point-in-time sampling; no seasonal/long-term trends captured.
- Localized drone sampling may not represent landscape-scale patterns.
- Some model overparameterization and unbalanced designs.
- Measurement uncertainty in spore viability and PM_{2.5}-based smoke classification.

Conclusions

Both GLMMs provide robust evidence for smoke effects on fungal spore concentrations, with forest fires producing higher absolute concentrations and prairie fires showing more consistent dose-response. The modeling approach effectively captures bioaerosol dynamics across fire regimes, though temporal and spatial limitations should be considered.